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Introduction to computational statistics

Contents

1 Markov chain Monte Carlo . . . 5
1.1 Introduction . . . 5
1.2 Key elements on Markov chains . . . 7
1.3 Metropolis–Hastings algorithm . . . 10
1.4 Variants of MH algorithms . . . 14
1.4.1 Metropolis–adjusted Langevin Algorithm . . . 14
1.4.2 Generalization of MH algorithms . . . 16
1.4.3 Pseudo-marginal Monte Carlo methods . . . 17
1.4.4 Hamiltonian Monte Carlo . . . 18

2 Variational inference and autoencoders . . . 23
2.1 Evidence lower bound . . . 23
2.2 Coordinate ascent variational inference . . . 24
2.3 Application to a mixture of Gaussian distributions . . . 27
2.4 Variational autoencoders . . . 31
2.5 Gradient-based optimization . . . 35
2.5.1 Pathwise gradient . . . 35
2.5.2 Score function gradient . . . 36

3 Sequential Monte Carlo Methods . . . 39
3.1 Importance sampling . . . 39
3.1.1 Convergence guarantees for the IS estimator . . . 40
3.1.2 Convergence guarantees for the self-normalized IS estimator . . . 42
3.2 Sequential importance sampling . . . 43
3.2.1 Hidden Markov models . . . 43
3.2.2 Sequential Monte Carlo methods for HMMs . . . 46

A M-estimation Z-estimation, maximum likelihood . . . 49
A.1 Method of moments . . . 49
A.2 Z-estimation . . . 50
A.3 Maximum likelihood . . . 50
A.4 M-estimation . . . 51
A.5 Consistency . . . 51

B Expectation Maximization algorithm . . . 55
B.1 Introduction . . . 55
B.2 Algorithm . . . 55
B.3 Convergence properties . . . 56
B.4 Application to latent data models . . . 58
B.5 Example: mixture of Gaussian distributions . . . 59

B.6 Monte Carlo EM in the curved exponential family setting. 61
References 63

Chapter 1

Markov chain Monte Carlo

1.1 Introduction

This chapter aims at designing algorithms to obtain samples from a complex distribution π defined on a measurable space $(\mathsf{X}, \mathcal{X})$. Such algorithms can be applied in many situations, and the target distribution can have several forms depending on the context. In many areas of statistics and machine learning, we are interested in drawing samples from π although π is known only up to a normalizing constant. Such situations arise naturally in a wide variety of contexts, for example, in Bayesian inference, where π represents a posterior distribution over parameters given data, or in energy-based models, where π is a probability distribution defined through an energy function.

Direct sampling from π is often infeasible, either because the distribution has a complex, high-dimensional structure or because computing the normalizing constant is intractable. Markov Chain Monte Carlo (MCMC) methods provide a powerful framework for addressing this challenge. The central idea is to construct a Markov chain whose stationary distribution is precisely π, and then to use the long-run behavior of the chain to generate approximate samples from π.

In most applications, samples are not an end in themselves: they are used to approximate integralsof the form

$$\pi(h) = \int_{\mathsf{X}} h(x)\, \pi(\mathrm{d}x),$$

for measurable real-valued functions h such that $\pi(|h|) < \infty$. If $X_1, \ldots, X_n$ are i.i.d. with distribution π, the estimator $\pi_n(h) = n^{-1} \sum_{k=1}^{n} h(X_k)$ is unbiased, converges almost surely to $\pi(h)$ by the strong law of large numbers and, if $\pi(h^2) < \infty$, the central limit theorem yields

$$\sqrt{n}\,\big(\pi_n(h) - \pi(h)\big) \overset{\mathcal{L}}{\Rightarrow} \mathcal{N}\big(0, \mathrm{Var}_\pi(h)\big),$$

with $\mathrm{Var}_\pi(h) = \pi(h^2) - \pi(h)^2$. In the context of MCMC methods, we use a similar estimator, $\pi_n(h) = n^{-1} \sum_{k=0}^{n-1} h(X_k)$, where $(X_k)_{k\in\mathbb{N}}$ is a Markov chain admitting π as invariant distribution. The samples are neither independent nor distributed according to π, and under suitable assumptions the law of large numbers and the central limit theorem still hold, with a different asymptotic variance.

Example 1.1 (Bayesian inference). In Bayesian inference, a prior belief is combined with data to obtain a posterior distribution on which statistical inference is based. The unknown parameter $\theta \in \Theta \subset \mathbb{R}^d$ is endowed with a *prior* distribution with density π_0 with respect to a measure λ (often the Lebesgue measure), and the *likelihood* $\theta \mapsto \mathrm{p}(y_{1:n} \mid \theta)$ is the density of the observations given θ. By Bayes' formula, the *posterior* density is

$$\theta \mapsto \pi(\theta) = \frac{\pi_0(\theta)\,\mathrm{p}(y_{1:n} \mid \theta)}{\int_\Theta \pi_0(\vartheta)\,\mathrm{p}(y_{1:n} \mid \vartheta)\lambda(\mathrm{d}\vartheta)} \propto \pi_0(\theta)\,\mathrm{p}(y_{1:n} \mid \theta).$$

The numerator is explicit while the denominator, the *evidence* or marginal likelihood, is an integral over Θ which is usually intractable.

- *A conjugate case.* Assume that conditionally on θ, $Y \sim \mathrm{Bin}(n, \theta)$ and let the prior be $\mathrm{Beta}(a, b)$, with $a, b > 0$: $\pi_0(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}\mathbf{1}_{(0,1)}(\theta)$. Then $\pi(\theta) \propto \theta^{y+a-1}(1-\theta)^{n-y+b-1}\mathbf{1}_{(0,1)}(\theta)$, i.e. the posterior is $\mathrm{Beta}(a+y, b+n-y)$ and no simulation algorithm is needed.
- *A non-conjugate case: Bayesian logistic regression.* Let $(x_i, y_i)_{1\le i\le n}$ with $x_i \in \mathbb{R}^d$, $y_i \in \{0,1\}$, and assume that, conditionally on θ, the $(Y_i)_{1\le i\le n}$ are independent with $\mathbb{P}(Y_i = 1 \mid \theta) = \sigma(x_i^\top \theta)$, where $\sigma(u) = 1/(1+\mathrm{e}^{-u})$. With a Gaussian prior $\mathcal{N}(0, \tau^2 I_d)$, the posterior density with respect to the Lebesgue measure satisfies

$$\log \pi(\theta) = -\frac{\|\theta\|^2}{2\tau^2} + \sum_{i=1}^n \left\{ y_i\, x_i^\top \theta - \log\left(1 + \mathrm{e}^{x_i^\top \theta}\right)\right\} + \text{const}, \tag{1.1}$$

  and

$$\nabla \log \pi(\theta) = -\frac{\theta}{\tau^2} + \sum_{i=1}^n \left(y_i - \sigma(x_i^\top \theta)\right) x_i.$$

  Both $\log \pi$ (up to an additive constant) and its gradient are cheap to evaluate, but the normalizing constant is a d-dimensional integral without closed form. Every algorithm presented in this chapter only requires evaluations of $\theta \mapsto \pi(\theta)$ up to a constant (and of $\nabla \log \pi$ for gradient-based methods).

Example 1.2. Energy-based models (EBM) are very flexible models which describe the target distribution using an unnormalized function, referred to as the energy function. These models are easier to design than models with a tractable likelihood such as autoregressive models, in particular in high-dimensional settings. As the energy function is not normalized, it can be easily parameterized with any nonlinear regression function. Using neural networks such as multi-layer perceptrons, or convolutional neural networks, it is straightforward to introduce energy functions with specific structures depending nonlinearly on the input. We can choose specific architectures tailored for every type of data, see [**?**] for a recent overview of EBM.

In a generic setting, the target random variable takes values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ and the target distribution (the target density with respect to the Lebesgue measure) is written:

$$x \mapsto \pi_\theta(x) \propto \exp\left(-\mathrm{E}_\theta(x)\right) = \frac{\exp\left(-\mathrm{E}_\theta(x)\right)}{\int \exp\left(-\mathrm{E}_\theta(u)\right)\mathrm{d}u},$$

where θ is an unknown parameter to estimate and E_θ is the energy function such that the normalizing constant is finite. The normalizing constant is often written Z_θ and referred to as the partition function:

$$Z_\theta = \int \exp\left(-\mathrm{E}_\theta(u)\right)\mathrm{d}u.$$

Since Z_θ is an intractable integral, evaluation and differentiation of $\theta \mapsto \log \pi_\theta(x)$ are not possible in usual settings. In order to estimate the unknown parameter θ using i.i.d. data, an appealing approach is to use gradient-based maximization procedures of the likelihood function. This means that we need to compute:

$$x \mapsto \nabla_\theta \log \pi_\theta(x) = -\nabla_\theta \mathrm{E}_\theta(x) - \nabla_\theta \log Z_\theta.$$

The first term can be evaluated easily as E_θ is known. For the second term, we can write, under regularity assumptions on the model:

$$\nabla_\theta \log Z_\theta = Z_\theta^{-1} \int \nabla_\theta \exp\left(-\mathrm{E}_\theta(u)\right) \mathrm{d}u$$
$$= \int \left\{-\nabla_\theta \mathrm{E}_\theta(u)\right\} Z_\theta^{-1} \exp\left(-\mathrm{E}_\theta(u)\right) \mathrm{d}u = \int \left\{-\nabla_\theta \mathrm{E}_\theta(u)\right\} \pi_\theta(u) \mathrm{d}u.$$

Therefore $\nabla_\theta \log Z_\theta = \mathbb{E}_{\pi_\theta}[-\nabla_\theta \mathrm{E}_\theta(X)]$ where $\mathbb{E}_\mu[f(X)]$ denotes the expectation of $f(X)$ when $X \sim \mu$. Given i.i.d. observations $X_1, \ldots, X_n$ with distribution π_θ, the gradient of the normalized log-likelihood is thus

$$\nabla_\theta \left(\frac{1}{n} \sum_{i=1}^n \log \pi_\theta(X_i) \right) = -\frac{1}{n} \sum_{i=1}^n \nabla_\theta \mathrm{E}_\theta(X_i) + \mathbb{E}_{\pi_\theta}[\nabla_\theta \mathrm{E}_\theta(X)],$$

which lowers the energy of the observed data and raises the energy of samples drawn from the model. It is therefore possible to train an EBM by providing a Monte Carlo estimate of $\nabla_\theta \log Z_\theta$, which requires samples from π_θ. However, this is not straightforward as π_θ is known only up to a multiplicative normalizing constant, as in the Bayesian setting of Example 1.1.

1.2 Key elements on Markov chains

Let $(\mathsf{X}, \mathcal{X})$ be a measurable space, i.e. $\mathcal{X}$ is a σ-algebra on X, and consider the following notations.

- $\mathsf{M}_+(\mathsf{X})$ is the set of non-negative measures on $(\mathsf{X}, \mathcal{X})$.
- $\mathsf{M}_1(\mathsf{X})$ is the set of probability measures on $(\mathsf{X}, \mathcal{X})$.
- $\mathsf{F}(\mathsf{X})$ is the set of real-valued measurable functions f on X and $\mathsf{F}_+(\mathsf{X})$ the set of non-negative measurable functions on X.
- For $\mu \in \mathsf{M}_+(\mathsf{X})$ and $h \in \mathsf{F}_+(\mathsf{X})$ (or h μ-integrable), $\mu(h) = \int_{\mathsf{X}} h(x)\mu(\mathrm{d}x)$.
- If $k \leq \ell$, $(u_k, \ldots, u_\ell)$ is denoted by $u_{k:\ell}$ and $(u_{k+\ell})_{\ell \in \mathbb{N}}$ by $u_{k:\infty}$.

Definition 1.3. We say that $P : \mathsf{X} \times \mathcal{X} \to \mathbb{R}^+$ is a Markov kernel if,

- for all $A \in \mathcal{X}$, $\mathsf{X} \ni y \mapsto P(y, A)$ is $\mathcal{X}/\mathcal{B}(\mathbb{R}^+)$ measurable,
- for all $x \in \mathsf{X}$, $\mathcal{X} \ni B \mapsto P(x, B)$ is a probability measure on $(\mathsf{X}, \mathcal{X})$.

For all $(x, A) \in \mathsf{X} \times \mathcal{X}$, as a function of the first component only, $P(\cdot, A)$ is measurable and as a function of the second component only, $P(x, \cdot)$ is a probability measure. In particular, $P(x, \mathsf{X}) = 1$ for all $x \in \mathsf{X}$. Since $P(x, \cdot)$ is a measure, we also use the infinitesimal notation: $P(x, \mathrm{d}y)$. For example,

$$P(x, A) = \int_{\mathsf{X}} \mathbf{1}_A(y) P(x, \mathrm{d}y) = \int_A P(x, \mathrm{d}y).$$

Example 1.4 (Three basic Markov kernels).

i) *Finite state space.* If $\mathsf{X} = \{1, \ldots, m\}$ and $\mathcal{X} = \mathcal{P}(\mathsf{X})$, a Markov kernel is characterized by the stochastic matrix $(P_{xy})_{1 \leq x,y \leq m}$, $P_{xy} = P(x, \{y\})$, with non-negative entries and rows summing to one.

ii) *Kernel with a density.* If $\lambda \in \mathsf{M}_+(\mathsf{X})$ is σ-finite and $p : \mathsf{X} \times \mathsf{X} \to \mathbb{R}^+$ is measurable with $\int p(x, y)\lambda(\mathrm{d}y) = 1$ for all $x \in \mathsf{X}$, then $P(x, \mathrm{d}y) = p(x, y)\lambda(\mathrm{d}y)$ is a Markov kernel. For instance, on $\mathbb{R}$, the *Gaussian autoregressive kernel* is $P(x, \cdot) = \mathcal{N}(c + \phi x, \sigma^2)$, $c \in \mathbb{R}$, $\phi \in \mathbb{R}$, $\sigma > 0$.

iii) *Random iterative functions.* Let $(\mathsf{U}, \mathcal{U})$ be a measurable space, $\Phi : \mathsf{X} \times \mathsf{U} \to \mathsf{X}$ be measurable and $\Gamma \in \mathsf{M}_1(\mathsf{U})$. Then $P(x, A) = \int_{\mathsf{U}} \mathbf{1}_A(\Phi(x, u))\Gamma(\mathrm{d}u)$ is a Markov kernel: $P(x, \cdot)$ is the distribution of $\Phi(x, U)$ when $U \sim \Gamma$. The random walk $\Phi(x, u) = x + u$ on $\mathbb{R}^d$ and the AR(1) kernel ($\Phi(x, u) = c + \phi x + \sigma u$, $\Gamma = \mathcal{N}(0, 1)$) are of this form. All algorithms of this chapter are written in this way.

For all $\mu \in \mathsf{M}_+(\mathsf{X})$, all Markov kernels P, Q on $\mathsf{X} \times \mathcal{X}$, and all measurable non-negative or bounded functions h on X, we use the following conventions and notations.

- μP is the non-negative measure: $\mathcal{X} \ni A \mapsto \mu P(A) = \int \mu(\mathrm{d}x) P(x, A)$,
- PQ is the Markov kernel: $(x, A) \mapsto \int_{\mathsf{X}} P(x, \mathrm{d}y) Q(y, A)$,
- Ph is the measurable function $x \mapsto \int_{\mathsf{X}} P(x, \mathrm{d}y) h(y)$.

It is easy to check that if μ is a probability measure, then μP is also a probability measure (since $\mu P(\mathsf{X}) = \int_{\mathsf{X}} \mu(\mathrm{d}x) P(x, \mathsf{X}) = \int_{\mathsf{X}} \mu(\mathrm{d}x) = 1$). With this notation, using the Fubini–Tonelli theorem,

$$\begin{aligned}\mu(P(Qh)) &= (\mu P)(Qh) = (\mu(PQ))h \\ &= \mu((PQ)h) = \int_{\mathsf{X}^3} \mu(\mathrm{d}x) P(x, \mathrm{d}y) Q(y, \mathrm{d}z) h(z).\end{aligned}$$

Therefore, all these parenthesis can be discarded and we can write μPQh without any ambiguity. For a given Markov kernel P on $\mathsf{X} \times \mathcal{X}$, define $P^0 = \mathrm{Id}$ where Id is the identity kernel: $(x, A) \mapsto \mathbf{1}_A(x)$, and set for $k \geq 0$, $P^{k+1} = P^k P$.

Definition 1.5. Let $\{X_k : k \in \mathbb{N}\}$ be a sequence of random variables defined on the same probability space $(\Omega, \mathcal{G}, \mathbb{P})$ and taking values in X. We say that $\{X_k : k \in \mathbb{N}\}$ is a Markov chain with Markov kernel P and initial distribution $\nu \in \mathsf{M}_1(\mathsf{X})$ if and only if the two following statements hold.

i) For all $(k, A) \in \mathbb{N} \times \mathcal{X}$, $\mathbb{P}(X_{k+1} \in A | X_{0:k}) = P(X_k, A)$, $\mathbb{P}$-a.s.
ii) For all $A \in \mathcal{X}$, $\mathbb{P}(X_0 \in A) = \nu(A)$.

Note that, in this definition, we consider $\mathbb{P}(X_{k+1} \in A | X_{0:k})$, that is, the conditional probability is with respect to the σ-field $\sigma(X_{0:k})$. More generally, if $(\mathcal{F}_k)_{k \geq 0}$ is a filtration such that $(X_k)_{k \geq 0}$ is $(\mathcal{F}_k)_{k \geq 0}$-adapted, we say that $(X_k)_{k \geq 0}$ is a Markov chain with respect to $(\mathcal{F}_k)_{k \geq 0}$ if $\mathbb{P}(X_{k+1} \in A | \mathcal{F}_k) = P(X_k, A)$, $\mathbb{P}$-a.s., for all $(k, A) \in \mathbb{N} \times \mathcal{X}$. Since $\sigma(X_{0:k}) \subset \mathcal{F}_k$, the tower property shows that a Markov chain with respect to $(\mathcal{F}_k)_{k \geq 0}$ is a Markov chain in the sense of the definition above. This is the property which is mostly used to analyze the algorithms below, where $\mathcal{F}_k$ contains all the auxiliary random variables simulated up to iteration k.

Proposition 1.6. *Let $(X_k)_{k \in \mathbb{N}}$ be a Markov chain with Markov kernel P and initial distribution ν. Then, for all $n \in \mathbb{N}$ and all $h \in \mathsf{F}_+(\mathsf{X}^{n+1})$,*

$$\mathbb{E}\left[h(X_{0:n})\right] = \int_{\mathsf{X}^{n+1}} \nu(\mathrm{d}x_0) P(x_0, \mathrm{d}x_1) \cdots P(x_{n-1}, \mathrm{d}x_n)\, h(x_{0:n}). \tag{1.2}$$

In particular, $X_n \sim \nu P^n$ and, for all $k, n \in \mathbb{N}$ and $h \in \mathsf{F}_+(\mathsf{X})$, $\mathbb{E}[h(X_{n+k}) | X_{0:n}] = P^k h(X_n)$, $\mathbb{P}$-a.s.

Proof. By a monotone class argument, it is enough to consider functions of the form $h(x_{0:n}) = h_0(x_0) \cdots h_n(x_n)$ with $h_j \in \mathsf{F}_+(\mathsf{X})$ for all $0 \leq j \leq n$. The case $n = 0$ is the definition of ν. If (1.2) holds for n, then, since $\mathbb{E}[h_{n+1}(X_{n+1}) | X_{0:n}] = P h_{n+1}(X_n)$,

$$\mathbb{E}\left[\prod_{j=0}^{n+1} h_j(X_j)\right] = \mathbb{E}\left[\prod_{j=0}^{n} h_j(X_j)\, P h_{n+1}(X_n)\right],$$

and the induction hypothesis yields (1.2) for $n + 1$. □

The next lemma is the tool used to identify the Markov kernel of all the algorithms of this chapter.

Lemma 1.7. *Let $X_0 \sim \nu$ and $(U_k)_{k \geq 1}$ be i.i.d. with distribution Γ, independent of X_0. Let $\Phi : \mathsf{X} \times \mathsf{U} \to \mathsf{X}$ be measurable and define recursively $X_{k+1} = \Phi(X_k, U_{k+1})$. Then $(X_k)_{k \in \mathbb{N}}$ is a Markov chain with respect to $\mathcal{F}_k = \sigma(X_0, U_{1:k})$ with initial distribution ν and Markov kernel given, for all $(x, A) \in \mathsf{X} \times \mathcal{X}$, by $P(x, A) = \mathbb{P}(\Phi(x, U_1) \in A)$.*

Proof. X_k is $\mathcal{F}_k$-measurable and U_{k+1} is independent of $\mathcal{F}_k$. Therefore, $\mathbb{P}(X_{k+1} \in A|\mathcal{F}_k) = \psi(X_k)$ with $\psi(x) = \mathbb{P}(\Phi(x, U_{k+1}) \in A) = P(x, A)$. □

Definition 1.8. We say that $\pi \in \mathsf{M}_1(\mathsf{X})$ is an invariant probability measure for the Markov kernel P on $\mathsf{X} \times \mathcal{X}$ if $\pi P = \pi$.

If $(X_k)_{k\in\mathbb{N}}$ is a Markov chain with Markov kernel P and if $X_0 \sim \pi$ where π is invariant for P, then, by induction, $\pi P^k = \pi P^{k-1} P = \pi P = \pi$ for all $k \geq 1$, so that, by Proposition 1.6, $X_k \sim \pi$ for all $k \in \mathbb{N}$. More precisely, by (1.2), the sequence of random variables $\{X_k : k \in \mathbb{N}\}$ is then a *strongly stationary sequence* in the sense that for all $n, p \in \mathbb{N}^*$ and all $k_{1:n} \in \mathbb{N}^n$, the random vector $(X_{k_1}, \ldots, X_{k_n})$ has the same distribution as $(X_{k_1+p}, \ldots, X_{k_n+p})$. Invariant probability measures are also called *stationary distributions*.

Example 1.9 (Invariant distributions).

i) *Two-state chain.* Let $\mathsf{X} = \{1, 2\}$ and, for $a, b \in (0, 1)$,

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}.$$

Solving $\pi P = \pi$ with $\pi(1) + \pi(2) = 1$ yields the unique invariant probability $\pi = \big(b/(a + b), a/(a + b)\big)$.

ii) *Gaussian AR(1) process.* Let $X_k = c + \phi X_{k-1} + \sigma Z_k$ where $(Z_k)_{k\geq 1}$ are i.i.d. $\mathcal{N}(0, 1)$, independent of X_0, $|\phi| < 1$ and $\sigma > 0$. If $X_{k-1} \sim \mathcal{N}(m, s^2)$, then $X_k \sim \mathcal{N}(c + \phi m, \phi^2 s^2 + \sigma^2)$. Therefore $\mathcal{N}(m, s^2)$ is invariant if and only if $m = c/(1 - \phi)$ and $s^2 = \sigma^2/(1 - \phi^2)$. Figure 1.2 shows that the histogram of a single trajectory is close to this invariant distribution: the chain "forgets" its initial condition $X_0 = 0$ after a few dozens of iterations. When $|\phi| \geq 1$, there is no invariant probability measure.

iii) *Random walk.* If $X_k = X_{k-1} + Z_k$ with $Z_k \sim \mathcal{N}(0, 1)$ on $\mathbb{R}$, the Lebesgue measure Leb satisfies $\mathrm{Leb}\, P = \mathrm{Leb}$ (by Fubini's theorem and translation invariance), but P admits no invariant *probability* measure: if $\pi P = \pi$, then the characteristic function of π would satisfy $\hat{\pi}(t) = \hat{\pi}(t)\mathrm{e}^{-t^2/2}$, so that $\hat{\pi}(t) = 0$ for $t \neq 0$, which contradicts the continuity of $\hat{\pi}$ at 0.

Definition 1.10. Let $\pi \in \mathsf{M}_1(\mathsf{X})$ and P be a Markov kernel on $\mathsf{X} \times \mathcal{X}$. We say that P is π-reversible if and only if for all measurable bounded or non-negative functions h on $\big(\mathsf{X}^2, \mathcal{X}^{\otimes 2}\big)$,

$$\iint_{\mathsf{X}^2} h(x, y)\pi(\mathrm{d}x)P(x, \mathrm{d}y) = \iint_{\mathsf{X}^2} h(x, y)\pi(\mathrm{d}y)P(y, \mathrm{d}x). \tag{1.3}$$

A Markov kernel P is π-reversible if and only if the probability measure $\pi(\mathrm{d}x)P(x, \mathrm{d}y)$ is symmetric with respect to (x, y). We often write, with infinitesimal notation,

$$\pi(\mathrm{d}x)P(x, \mathrm{d}y) = \pi(\mathrm{d}y)P(y, \mathrm{d}x). \tag{1.4}$$

Proposition 1.11. *Let P be a Markov kernel on $\mathsf{X} \times \mathcal{X}$ and $\pi \in \mathsf{M}_1(\mathsf{X})$ such that P is π-reversible. Then π is invariant for P.*

Proof. For any $A \in \mathcal{X}$, applying (1.3) with $h(x, y) = \mathbf{1}_A(y)$,

$$\pi P(A) = \iint_{\mathsf{X}^2} \mathbf{1}_A(y)\pi(\mathrm{d}x)P(x, \mathrm{d}y) = \iint_{\mathsf{X}^2} \mathbf{1}_A(y)\pi(\mathrm{d}y)P(y, \mathrm{d}x) = \int_A \pi(\mathrm{d}y)P(y, \mathsf{X}) = \pi(A),$$

which completes the proof. □

Therefore, if we want to check easily that π is invariant for a kernel P, it is sufficient to check that P is π-reversible. Reversibility is a "local" condition involving pairs of points, whereas invariance is a "global" condition involving an integral over X.

Example 1.12 (Reversible and non-reversible kernels).

i) The two-state chain of Example 1.9 is reversible: $\pi(1)P_{12} = ab/(a+b) = \pi(2)P_{21}$.
ii) The stationary AR(1) chain is reversible: if $X_0 \sim \mathcal{N}(m, s^2)$ with $m = c/(1-\phi)$ and $s^2 = \sigma^2/(1-\phi^2)$, (X_0, X_1) is a Gaussian vector whose marginals are both $\mathcal{N}(m, s^2)$ and whose covariance matrix is symmetric, so $(X_0, X_1) \overset{\mathcal{L}}{=} (X_1, X_0)$.
iii) Reversibility is sufficient but not necessary for invariance. On $\mathsf{X} = \{0, 1, 2\}$, let $P(x, \{x + 1 \bmod 3\}) = 1 - \varepsilon$ and $P(x, \{x - 1 \bmod 3\}) = \varepsilon$, with $\varepsilon \in [0, 1/2)$. Each column of P sums to one so the uniform distribution is invariant, but $\pi(0)P_{01} = (1-\varepsilon)/3 \neq \varepsilon/3 = \pi(1)P_{10}$. Non-reversible kernels also appear naturally when composing reversible kernels.

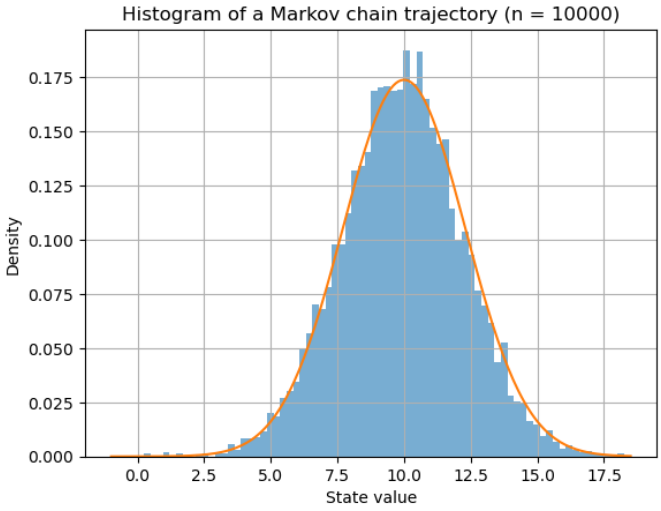


Fig. 1.1 Histogram of a Gaussian AR(1) process, $X_k = \mu + \phi X_{k-1} + \sigma Z_k$, where $(Z_k)_{k\in\mathbb{N}}$ is an iid sequence of standard Gaussian random variables, independent of X_0 with $|\phi| < 1$. The Gaussian distribution with mean $\mu/(1-\phi)$ and variance $\sigma^2/(1-\phi^2)$ is a stationary distribution of the Markov chain. Illustration of the histogram and of the stationary distribution with $\mu = 1$, $\phi = 0.9$ and $\sigma = 1$.

1.3 Metropolis–Hastings algorithm

In this section, we are given a probability measure $\pi \in \mathsf{M}_1(\mathsf{X})$ and the idea now is to construct a Markov chain $\{X_k : k \in \mathbb{N}\}$ admitting π as invariant probability measure, in which case we say that π is a target distribution. In other words, we try to find a Markov kernel P on $\mathsf{X} \times \mathcal{X}$ such that P is π-invariant.

For simplicity we now assume that π has a density with respect to some dominating σ-finite measure μ and by abuse of notation, we also denote by π this density, that is we write $\pi(\mathrm{d}x) = \pi(x)\mu(\mathrm{d}x)$ and we assume that this density π is positive. Moreover, let Q be a Markov kernel on $\mathsf{X} \times \mathcal{X}$ such that $Q(x, \mathrm{d}y) = q(x, y)\mu(\mathrm{d}y)$, that is, for any $x \in \mathsf{X}$, $Q(x, \cdot)$ is also dominated by μ and denoting by $q(x, \cdot)$ this density, we assume for simplicity that $q(x, y)$ is positive for all $x, y \in \mathsf{X}$. The kernel Q is called the *proposal kernel*.

For a given measurable function $\alpha : \mathsf{X}^2 \to [0, 1]$, Algorithm 1 describes the Metropolis–Hastings algorithm. The Markov kernel Q is used to propose a candidate for the next value of the Markov chain $(X_k)_{k\in\mathbb{N}}$ and this candidate is accepted or rejected with a probability that depends on

Input : Initial distribution ν, number of iterations n.
Output: Markov chain $X_0, \dots, X_n$.

At iteration $k = 0$, draw $X_0 \sim \nu$.
for $k = 0$ **to** $n - 1$ **do**
- Draw $Y_{k+1} \sim Q(X_k, \cdot)$ and $U_{k+1} \sim \mathrm{U}(0,1)$, independently of each other and of the past.
- Set $X_{k+1} = \begin{cases} Y_{k+1} & \text{if } U_{k+1} \le \alpha(X_k, Y_{k+1}), \\ X_k & \text{otherwise.} \end{cases}$

end

Algorithm 1: The Metropolis–Hastings algorithm

the function α. We now choose conveniently α in such a way that $(X_k)_{k\in\mathbb{N}}$ is a Markov chain with invariant probability measure π. First, we write down the Markov kernel associated with $(X_k)_{k\in\mathbb{N}}$. Write $\mathcal{F}_k = \sigma(X_0, U_{1:k}, Y_{1:k})$ and note that $(X_k)_{k\in\mathbb{N}}$ is adapted to the filtration $(\mathcal{F}_k)_{k\in\mathbb{N}}$ (which implies $\sigma(X_{0:k}) \subset \mathcal{F}_k$). Then, setting $\bar{\alpha}(x) = 1 - \int_{\mathsf{X}} Q(x, \mathrm{d}y)\alpha(x, y)$, we have for any bounded or non-negative measurable function h on X and any $k \in \mathbb{N}$, since conditionally on $\mathcal{F}_k$, $Y_{k+1} \sim Q(X_k, \cdot)$ and $U_{k+1} \sim \mathrm{U}(0,1)$ are independent, and $\mathbb{P}(U_{k+1} \le a) = a$ for $a \in [0,1]$,

$$\begin{aligned}
\mathbb{E}\left[h(X_{k+1})|\mathcal{F}_k\right] &= \mathbb{E}\left[\mathbf{1}_{\{U_{k+1}<\alpha(X_k,Y_{k+1})\}} h(Y_{k+1})|\mathcal{F}_t\right] + \mathbb{E}\left[\mathbf{1}_{\{U_{k+1}\ge\alpha(X_k,Y_{k+1})\}}|\mathcal{F}_k\right] h(X_k) \\
&= \int_{\mathsf{X}} Q(X_k, \mathrm{d}y)\alpha(X_k, y)h(y) + \bar{\alpha}(X_k)h(X_k) \\
&= \int_{\mathsf{X}} \left[Q(X_k, \mathrm{d}y)\alpha(X_k, y) + \bar{\alpha}(X_k)\delta_{X_k}(\mathrm{d}y)\right] h(y) = P^{MH}_{\langle\pi,Q\rangle} h(X_k).
\end{aligned}$$

Therefore, $\{X_k : k \in \mathbb{N}\}$ is a Markov chain (with respect to $(\mathcal{F}_k)_{k\in\mathbb{N}}$, hence also with respect to its natural filtration) with Markov kernel

$$P^{MH}_{\langle\pi,Q\rangle}(x, \mathrm{d}y) = Q(x, \mathrm{d}y)\alpha(x, y) + \bar{\alpha}(x)\delta_x(\mathrm{d}y). \tag{1.5}$$

The function $\bar{\alpha}(x)$ is the probability of rejecting the candidate when the chain is at x. Although this kernel depends on α, we keep the notation $P^{MH}_{\langle\pi,Q\rangle}$, as α will be chosen as a function of π and Q below.

Lemma 1.13. *The Markov kernel $P^{MH}_{\langle\pi,Q\rangle}$ is π-reversible if and only if for all measurable bounded or non-negative functions h on $\left(\mathsf{X}^2, \mathcal{X}^{\otimes 2}\right)$,*

$$\int_{\mathsf{X}^2} h(x, y)\pi(\mathrm{d}x)Q(x, \mathrm{d}y)\alpha(x, y) = \int_{\mathsf{X}^2} h(x, y)\pi(\mathrm{d}y)Q(y, \mathrm{d}x)\alpha(y, x). \tag{1.6}$$

Equation (1.6) is often called the detailed balance condition.

Proof. First, note that

$$\pi(\mathrm{d}x)\bar{\alpha}(x)\delta_x(\mathrm{d}y) = \pi(\mathrm{d}y)\bar{\alpha}(y)\delta_y(\mathrm{d}x). \tag{1.7}$$

Indeed, for any non-negative measurable function h on X^2, we have

$$\begin{aligned}
\iint_{\mathsf{X}^2} h(x, y)\pi(\mathrm{d}x)\bar{\alpha}(x)\delta_x\left(\mathrm{d}y\right) &= \int_{\mathsf{X}} h(x, x)\pi(\mathrm{d}x)\bar{\alpha}(x) \\
&= \int_{\mathsf{X}} h(y, y)\pi(\mathrm{d}y)\bar{\alpha}(y) = \iint_{\mathsf{X}^2} h(x, y)\pi(\mathrm{d}y)\bar{\alpha}(y)\delta_y(\mathrm{d}x).
\end{aligned}$$

Combining (1.5) with (1.7), we obtain that $P^{MH}_{\langle\pi,Q\rangle}$ is π-reversible if and only if the detailed balance condition (1.6) is satisfied. This completes the proof. □

Since $\pi(\mathrm{d}x)Q(x, \mathrm{d}y)\alpha(x, y) = \pi(x)q(x, y)\alpha(x, y)\,\mu(\mathrm{d}x)\mu(\mathrm{d}y)$, the detailed balance condition (1.6) holds if and only if

$$\pi(x)q(x,y)\alpha(x,y) = \pi(y)q(y,x)\alpha(y,x) \quad \text{for } \mu^{\otimes 2}\text{-almost all } (x,y) \in \mathsf{X}^2. \tag{1.8}$$

We now provide explicit expressions of the acceptance probability α.

Lemma 1.14. *Define*

$$\alpha^{MH}(x,y) = \min\left(\frac{\pi(y)q(y,x)}{\pi(x)q(x,y)}, 1\right)$$

and

$$\alpha^{b}(x,y) = \frac{\pi(y)q(y,x)}{\pi(x)q(x,y) + \pi(y)q(y,x)}.$$

Then, α^{MH} and α^b satisfy the detailed balance condition (1.6). Moreover, any function $\alpha : \mathsf{X}^2 \to [0,1]$ satisfying (1.8) is such that $\alpha \le \alpha^{MH}$, $\mu^{\otimes 2}$-a.e.

Proof. By (1.8), it is enough to note that

$$\pi(x)q(x,y)\alpha^{MH}(x,y) = \min\big(\pi(y)q(y,x), \pi(x)q(x,y)\big)$$

and

$$\pi(x)q(x,y)\alpha^{b}(x,y) = \frac{\pi(x)q(x,y)\pi(y)q(y,x)}{\pi(x)q(x,y) + \pi(y)q(y,x)}$$

are both symmetric functions of (x,y). Assume now that α satisfies (1.8). Since $\alpha(y,x) \le 1$, $\alpha(x,y) \le \pi(y)q(y,x)/(\pi(x)q(x,y))$ and, as $\alpha(x,y) \le 1$, $\alpha(x,y) \le \alpha^{MH}(x,y)$. □

Remark 1.15. i) *Unknown normalizing constants.* The acceptance probability only depends on π through the ratio $\pi(y)/\pi(x)$. If $\pi = \tilde{\pi}/Z$ with $\tilde{\pi}$ explicit and Z unknown, α^{MH} can be computed with $\tilde{\pi}$: this is the key property which makes MH algorithms applicable in many settings.

ii) *Numerical implementation.* The ratio is usually computed on the log scale: the candidate is accepted if $\log U_{k+1} \le \log\tilde{\pi}(Y_{k+1}) + \log q(Y_{k+1}, X_k) - \log\tilde{\pi}(X_k) - \log q(X_k, Y_{k+1})$, which avoids overflows and underflows.

Example 1.16 (The random walk MH sampler). Let $\mathsf{X} = \mathbb{R}^d$, $\mu = \lambda$ be the Lebesgue measure and let the proposal kernel be $Q(x, \mathrm{d}y) = g(y-x)\lambda(\mathrm{d}y)$ where g is a symmetric probability density on $\mathbb{R}^d$ (by symmetric, we mean that $g(u) = g(-u)$ for all $u \in \mathbb{R}^d$). Then at each time step in the MH algorithm, we draw a candidate $Y_{k+1} \sim g(y - X_k)\lambda(\mathrm{d}y)$, or equivalently $Y_{k+1} = X_k + \eta_{k+1}$ where $\eta_{k+1} \sim g$ is independent of the past. Since $q(x,y) = g(y-x) = g(x-y) = q(y,x)$, the acceptance probability is $\alpha^{MH}(x,y) = \min(\pi(y)/\pi(x), 1)$: a move towards a region of higher density is always accepted. The associated algorithm is called the *(symmetric) random walk Metropolis–Hastings* algorithm (RWMH).

Choice of the scale. With Gaussian increments $\eta_{k+1} \sim \mathcal{N}(0, s^2 I_d)$, the scale s governs a trade-off illustrated in Figure 1.2: if s is too small, almost all candidates are accepted but the chain moves by tiny steps; if s is too large, the candidates fall in regions of low probability and are almost always rejected, so that the chain stays still for long periods. In both cases, the autocorrelations decay slowly. For targets of product form $\pi(x) = \prod_{i=1}^d f(x_i)$, [?] showed that the optimal scale is $s = \ell^\star/\sqrt{d}$, and that it corresponds to an average acceptance rate of approximately 0.234 as $d \to \infty$. In dimension one, for a Gaussian target, the optimal acceptance rate is close to 0.44. Because $s \propto d^{-1/2}$, the number of iterations needed to explore π grows linearly with the dimension.

Example 1.17 (Multimodality). Figure 1.3 illustrates the main weakness of local proposals. With $a = 3$, a RWMH with unit increments must cross the region around 0 where $\pi(0)/\pi(a) \approx 2\mathrm{e}^{-a^2/2} \approx 0.02$, which only happens through a sequence of unlikely accepted moves. The mean time between two switches grows very fast with a (heuristically like $\mathrm{e}^{a^2/2}$), so that, for instance with $a = 6$, a

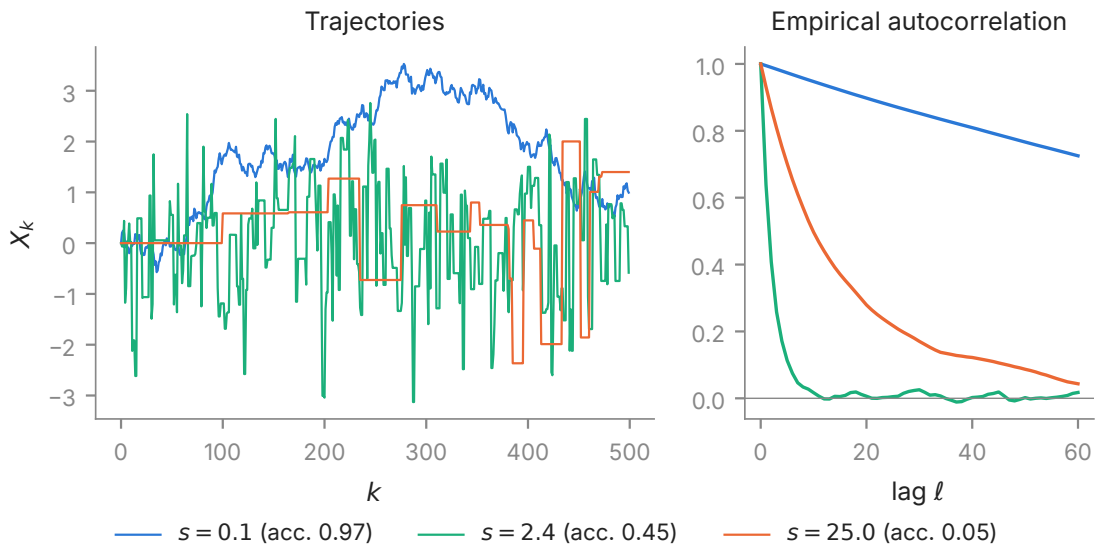


Fig. 1.2 RWMH with Gaussian increments $\mathcal{N}(0, s^2)$ targeting $\mathcal{N}(0, 1)$, for three values of the scale s (empirical acceptance rates in the legend). Left: first 500 iterations. Right: empirical autocorrelations $\hat{\rho}_\ell$ computed on 20 000 iterations. The intermediate scale yields the fastest decay of the autocorrelations, i.e. the smallest integrated autocorrelation time.

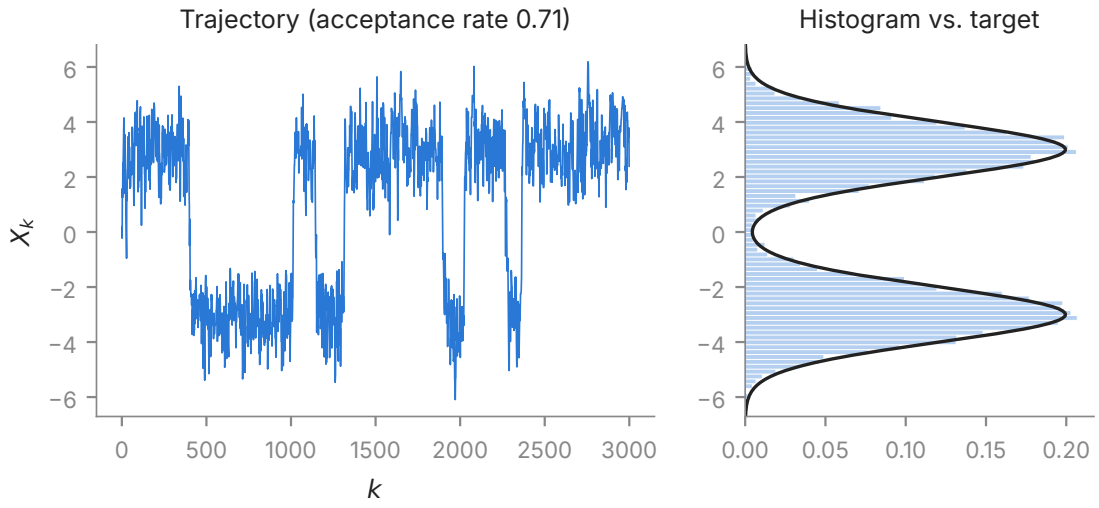


Fig. 1.3 Trajectory of a random walk MH sampler targeting the mixture $0.5\mathcal{N}(-a, 1) + 0.5\mathcal{N}(a, 1)$ with $a = 3$. At each iteration $k \geq 0$, the proposal distribution is a Gaussian distribution centered at the current state X_k with unit variance. The chain stays in one mode for hundreds of iterations before crossing the low-density region around 0.

chain of reasonable length started in one mode never visits the other one, while all the diagnostics computed on this single chain seem perfect. Running several chains from dispersed initial points, tempering strategies, or the sequential Monte Carlo methods of Chapter 3 are typical remedies.

Example 1.18 (Independent MH sampler). Let $Q(x, \mathrm{d}y) = g(y)\mu(\mathrm{d}y)$ where g is a positive probability density which does not depend on the current state. Writing $w = \pi/g$ for the importance weight function (see Chapter 3),

$$\alpha^{MH}(x, y) = \min\Big(\frac{\pi(y)g(x)}{\pi(x)g(y)}, 1\Big) = \min\Big(\frac{w(y)}{w(x)}, 1\Big).$$

Assume that $M = \sup_x w(x) < \infty$. Since $\min(1, w(y)/w(x)) \geq \min(1, w(y)/M) = w(y)/M$, for all $x \in \mathsf{X}$ and $A \in \mathcal{X}$,

$$P^{MH}_{\langle \pi, Q\rangle}(x, A) \geq \int_A g(y)\frac{w(y)}{M}\mu(\mathrm{d}y) = \frac{\pi(A)}{M}.$$

This minorization condition (Doeblin's condition) implies that $\|P^n(x, \cdot) - \pi\|_{\mathrm{TV}} \leq (1 - 1/M)^n$ for all x, i.e. the chain is uniformly ergodic. Conversely, if w is not bounded (g has lighter tails than π), the independent sampler is not even geometrically ergodic [Mengersen and Tweedie, 1996]: the chain gets stuck for long periods at points x where $w(x)$ is large. The same quantity M governs the efficiency of rejection sampling, which needs on average M proposals per accepted sample but, unlike the independent MH sampler, requires M to be known.

Example 1.19 (A discrete target). Let π be the Poisson distribution with parameter $\theta > 0$ on $\mathsf{X} = \mathbb{N}$, $\pi(x) = \mathrm{e}^{-\theta}\theta^x/x!$, and let the proposal be $Y_{k+1} = X_k \pm 1$ with probability $1/2$ each. With the convention $\pi(-1) = 0$ (a proposal at -1 is always rejected), q is symmetric and

$$\alpha^{MH}(x, x+1) = \min\Big(\frac{\theta}{x+1}, 1\Big), \qquad \alpha^{MH}(x, x-1) = \min\Big(\frac{x}{\theta}, 1\Big), \quad x \geq 1.$$

Neither $\mathrm{e}^{-\theta}$ nor the factorials need to be computed. Here q is not positive everywhere, but $P^{MH}_{\langle \pi, Q\rangle}$ is still π-irreducible since any integer can be reached from any other in finitely many steps with positive probability, and aperiodic since $P^{MH}_{\langle \pi, Q\rangle}(0, \{0\}) > 0$.

Example 1.20 (Bayesian logistic regression). For the posterior (1.1), a RWMH iteration only requires the evaluation of $\log \tilde{\pi}(Y_{k+1})$, i.e. $\mathcal{O}(nd)$ operations. In practice, the posterior may be strongly anisotropic (correlated covariates), and isotropic increments $\mathcal{N}(0, s^2 I_d)$ are inefficient. A standard improvement is to use $\mathcal{N}(0, s^2\Sigma)$ where Σ approximates the posterior covariance (e.g. the inverse Hessian of $-\log \pi$ at the posterior mode), with $s \approx 2.38/\sqrt{d}$ as suggested by the optimal scaling results.

1.4 Variants of MH algorithms

1.4.1 Metropolis–adjusted Langevin Algorithm

The Metropolis–adjusted Langevin Algorithm (MALA) combines a Langevin-type proposal with a Metropolis–Hastings correction. We assume here that $\mathsf{X} = \mathbb{R}^d$ and that π has a positive and continuously differentiable density with respect to the Lebesgue measure. The algorithm was in particular analyzed in [Roberts and Tweedie, 1996].

For all $k \geq 0$, given the current state X_k, a proposal Y_{k+1} is generated according to the Euler discretization of the overdamped Langevin dynamics,

$$Y_{k+1} = X_k + \frac{h^2}{2}\nabla \log \pi(X_k) + hZ_{k+1},$$

where $Z_{k+1} \sim \mathcal{N}(0, I_d)$ is independent of the past and $h > 0$ denotes the stepsize parameter. This defines a proposal kernel Q given by

$$Q(x, \cdot) = \mathcal{N}\Big(x + \frac{h^2}{2}\nabla \log \pi(x),\, h^2 I_d\Big),$$

i.e., with a constant $c_{h,d}$ which does not depend on (x, y),

$$q(x, y) = c_{h,d} \exp\Big(-\frac{1}{2h^2}\Big\|y - x - \frac{h^2}{2}\nabla \log \pi(x)\Big\|^2\Big).$$

This proposal is not symmetric, and the candidate is accepted with probability $\alpha^{MH}(X_k, Y_{k+1})$ where

$$\log \frac{\pi(y)q(y,x)}{\pi(x)q(x,y)} = \log \pi(y) - \log \pi(x) \\ - \frac{1}{2h^2}\Big\|x - y - \frac{h^2}{2}\nabla \log \pi(y)\Big\|^2 + \frac{1}{2h^2}\Big\|y - x - \frac{h^2}{2}\nabla \log \pi(x)\Big\|^2.$$

As for RWMH, only $\log \pi$ up to an additive constant and its gradient are needed. MALA can be interpreted as a Metropolis-corrected Euler discretization of the overdamped Langevin diffusion

$$\mathrm{d}X_t = \frac{1}{2}\nabla \log \pi(X_t)\,\mathrm{d}t + \mathrm{d}B_t,$$

where $(B_t)_{t\geq 0}$ is a standard d-dimensional Brownian motion; under mild assumptions on π, this diffusion has π as its unique invariant distribution. The Metropolis–Hastings step corrects for the discretization bias introduced by the Euler scheme, ensuring exact invariance of π. The efficiency of MALA strongly depends on the choice of the stepsize h. In high dimension, for target distributions of product form (and some extensions), optimal scaling is achieved when

$$h \asymp d^{-1/6},$$

which leads to a non-degenerate limiting acceptance rate (approximately 0.574 at the optimum), as established in [Roberts and Rosenthal, 1998]; compare with $s \asymp d^{-1/2}$ for RWMH. Using larger stepsizes results in low acceptance probabilities, while overly small stepsizes lead to slow exploration of the state space. Compared to random walk Metropolis algorithms, MALA leverages gradient information through $\nabla \log \pi$, yielding improved mixing and reduced random-walk behavior, particularly in moderately high dimensions. This improvement comes at the expense of an increased computational cost per iteration due to gradient evaluations, resulting in a trade-off between statistical efficiency and numerical cost. MALA can also behave badly in the tails: for targets with tails lighter than Gaussian, it may fail to be geometrically ergodic [Roberts and Tweedie, 1996].

Omitting the accept/reject step, i.e. setting $X_{k+1} = Y_{k+1}$, defines the *unadjusted Langevin algorithm* (ULA). ULA is cheaper and does not suffer from rejections, but its invariant distribution π_h (when it exists) differs from π. For strongly log-concave and smooth targets, non-asymptotic bounds on the distance between the law of X_k and π show that this bias is controlled by the stepsize [Dalalyan, 2017, Durmus and Moulines, 2017]; for targets with tails lighter than Gaussian, ULA may even be transient [Roberts and Tweedie, 1996]. ULA and its stochastic gradient variants are widely used in machine learning and in score-based generative models.

Example 1.21 (ULA versus MALA for a Gaussian target). Let $\pi = \mathcal{N}(0, 1)$, so that $\nabla \log \pi(x) = -x$, and set $\gamma = h^2$. ULA reads

$$X_{k+1} = \Big(1 - \frac{\gamma}{2}\Big)X_k + \sqrt{\gamma}\, Z_{k+1},$$

which is the AR(1) process of Example 1.9 with $c = 0$, $\phi = 1 - \gamma/2$ and $\sigma^2 = \gamma$. For $\gamma \in (0, 4)$, $|\phi| < 1$ and its invariant distribution is

$$\pi_h = \mathcal{N}\left(0, \frac{\gamma}{1 - (1 - \gamma/2)^2}\right) = \mathcal{N}\left(0, \frac{1}{1 - \gamma/4}\right),$$

while for $\gamma \geq 4$ the chain has no invariant probability. The bias $\mathrm{Var}_{\pi_h}(X) - 1 = \gamma/(4 - \gamma)$ is of order γ. MALA with the same stepsize is exactly π-invariant (Figure 1.4).

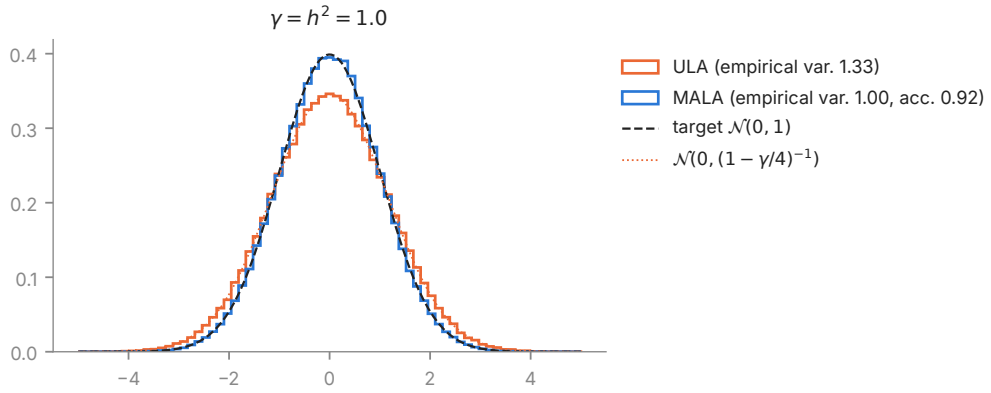


Fig. 1.4 Histograms of $2 \cdot 10^5$ iterations of ULA and MALA targeting $\mathcal{N}(0, 1)$ with $\gamma = h^2 = 1$. ULA samples from $\mathcal{N}(0, 4/3)$ (Example 1.21); MALA corrects this bias.

For the Bayesian logistic regression of Example 1.1, the gradient $\nabla \log \pi(\theta)$ is explicit, so that MALA and the Hamiltonian Monte Carlo algorithm below can be implemented at a cost of $\mathcal{O}(nd)$ operations per iteration.

1.4.2 Generalization of MH algorithms

Let $\pi \in \mathsf{M}_1(\mathsf{X})$ and let Q be a Markov kernel on $\mathsf{X} \times \mathcal{X}$. In this chapter, we have presented the Metropolis–Hastings algorithm when π and $Q(x, \cdot)$ both have densities with respect to a common dominating measure μ. In this section, we do not make such an assumption so that the expression of α^{MH} given in Lemma 1.14 is not available anymore and should be adapted. Instead, we will need the following assumption. Define

$$\mu_0(\mathrm{d}x\mathrm{d}y) = \pi(\mathrm{d}x)Q(x, \mathrm{d}y) \quad \text{and} \quad \mu_1(\mathrm{d}x\mathrm{d}y) = \pi(\mathrm{d}y)Q(y, \mathrm{d}x).$$

(B1) There exists a measurable function $(x, y) \mapsto r(x, y)$ such that $r(x, y) > 0$, μ_0-a.s. and for all $h \in \mathsf{F}_+(\mathsf{X}^2)$,

$$\int h(x, y)\mu_1(\mathrm{d}x\mathrm{d}y) = \int h(x, y)r(x, y)\mu_0(\mathrm{d}x\mathrm{d}y). \tag{1.9}$$

This equation shows that the measure μ_1 is dominated by μ_0 with a μ_0-a.s. positive density:

$$r(x, y) = \frac{\mathrm{d}\mu_1}{\mathrm{d}\mu_0}(x, y).$$

Then, by symmetry, $r(y, x) = 1/r(x, y)$, μ_0-a.s. Indeed, let $s(x, y) = (y, x)$, so that μ_1 is the image of μ_0 by s and conversely. For all $h \in \mathsf{F}_+(\mathsf{X}^2)$, applying (1.9) twice,

$$\int h \,\mathrm{d}\mu_0 = \int h \circ s \,\mathrm{d}\mu_1 = \int (h \circ s)\, r \,\mathrm{d}\mu_0 = \int h\,(r \circ s)\,\mathrm{d}\mu_1 = \int h\,(r \circ s)\, r \,\mathrm{d}\mu_0,$$

so that $r(x,y)r(y,x) = 1$, μ_0-a.s. Finally, the two measures μ_0 and μ_1 are equivalent (each one is dominated by the other). In this case, the generalized version of the Metropolis–Hastings kernel, where α^{MH} given in Lemma 1.14 is replaced by $\alpha(x,y) = r(x,y) \wedge 1$, is π-reversible, see [?]. When π and Q have densities, $r(x,y) = \pi(y)q(y,x)/(\pi(x)q(x,y))$ and we recover α^{MH}.

Lemma 1.22. *Assume (B1). Then, setting $\alpha(x,y) = r(x,y) \wedge 1$, the MH kernel:*

$$P^{MH}_{\langle \pi, Q\rangle}(x, \mathrm{d}y) = Q(x, \mathrm{d}y)\alpha(x,y) + \bar{\alpha}(x)\delta_x(\mathrm{d}y), \quad \textit{where} \quad \bar{\alpha}(x) = 1 - \int_{\mathsf{X}} Q(x, \mathrm{d}y)\alpha(x,y),$$

is π-reversible.

Proof. Similarly to Lemma 1.13, we only need to check the detailed balance condition. Let $h \in \mathsf{F}_+(\mathsf{X}^2)$. Then,

$$\begin{aligned}
&\int_{\mathsf{X}^2} \pi(\mathrm{d}x)Q(x,\mathrm{d}y)\alpha(x,y)h(x,y) = \int_{\mathsf{X}^2} \mu_0(\mathrm{d}x\mathrm{d}y)(r(x,y)\wedge 1)h(x,y) \\
&= \int_{\mathsf{X}^2} \mu_0(\mathrm{d}x\mathrm{d}y) r(x,y) \left(1 \wedge \frac{1}{r(x,y)}\right) h(x,y) = \int_{\mathsf{X}^2} \mu_1(\mathrm{d}x\mathrm{d}y) \left(1 \wedge \underbrace{1/r(x,y)}_{r(y,x)}\right) h(x,y) \\
&= \int_{\mathsf{X}^2} \pi(\mathrm{d}y)Q(y,\mathrm{d}x)\alpha(y,x)h(x,y).
\end{aligned}$$

Thus, the detailed balance condition is verified and the proof is completed. □

1.4.3 Pseudo-marginal Monte Carlo methods

Assume that π and Q are dominated by a common measure μ and write by abuse of notation, $\pi(\mathrm{d}x) = \pi(x)\mu(\mathrm{d}x)$ and $Q(x,\mathrm{d}y) = q(x,y)\mu(\mathrm{d}y)$. When considering a Metropolis–Hastings algorithm, we need an explicit expression of $\pi(x)$ for any $x \in \mathsf{X}$, up to a multiplicative constant. It may happen that we are not able to calculate $\pi(x)$ explicitly (even up to a multiplicative constant). Instead, assume that we are able to compute an unbiased estimator of $\pi(x)$. To obtain such an unbiased estimator, we draw $W \sim R(x,\mathrm{d}w)$ where R is a Markov kernel from X to $\mathbb{R}^+_\star$, that is, a Markov kernel on $\mathsf{X} \times \mathcal{B}(\mathbb{R}^+_\star)$ such that $\int_{\mathbb{R}^+_\star} wR(x,\mathrm{d}w) = \pi(x)$ (the *unbiasedness* condition). As for MH, it is enough that $\int wR(x,\mathrm{d}w) = C\pi(x)$ for a constant $C > 0$ which does not depend on x. The pseudo-marginal algorithm [Beaumont, 2003, Andrieu and Roberts, 2009] is described in Algorithm 2 below. We now justify the pseudo-marginal Monte Carlo algorithm by showing that

Input : Initial distribution μ, total number of samples n.
Output: $X_0, \ldots, X_n$

At $k = 0$, draw $X_0 \sim \nu$ and $W_0 \sim R(X_0, \cdot)$.
for $k \leftarrow 0$ **to** $n-1$ **do**
- Draw $\tilde{X}_{k+1} \sim Q(X_t, \cdot)$ and then $\tilde{W}_{k+1} \sim R(\tilde{X}_{k+1}, \cdot)$.
- Set $(X_{k+1}, W_{k+1}) = \begin{cases} (\tilde{X}_{k+1}, \tilde{W}_{k+1}) & \text{with probability } \frac{\tilde{W}_{k+1} q(\tilde{X}_{k+1}, X_k)}{W_k q(X_k, \tilde{X}_{k+1})} \wedge 1, \\ (X_k, W_k) & \text{with probability } 1 - \frac{\tilde{W}_{k+1} q(\tilde{X}_{k+1}, X_k)}{W_k q(X_k, \tilde{X}_{k+1})} \wedge 1. \end{cases}$

end

Algorithm 2: The Pseudo-Marginal MH Algorithm

it is actually a generalized MH algorithm (as described in Lemma 1.22) by considering extended Markov chain, $(\bar{X}_k)_{k\in\mathbb{N}} = (X_k, W_k)_{k\in\mathbb{N}}$ on an extended space and with an extended target. Define the extended target distribution $\bar{\pi}(\mathrm{d}\bar{x}) = \bar{\pi}(\mathrm{d}x\mathrm{d}w) = wR(x,\mathrm{d}w)\mu(\mathrm{d}x)$ (where we set $\bar{x} = (x,w)$). Note that $\bar{\pi}$ is indeed a probability measure on $\bar{\mathsf{X}} = \mathsf{X}\times\mathbb{R}_\star^+$, since

$$\int_{\mathsf{X}\times\mathbb{R}_\star^+} \bar{\pi}(\mathrm{d}x\mathrm{d}w) = \int_{\mathsf{X}} \left(\int_{\mathbb{R}_\star^+} wR(x,\mathrm{d}w) \right) \mu(\mathrm{d}x) = \int_{\mathsf{X}} \pi(x)\mu(\mathrm{d}x) = 1.$$

Moreover, in Algorithm 2, the candidate $(\tilde{X}_{k+1}, W_{k+1})$ is proposed according to $\bar{Q}$ where the proposal kernel $\bar{Q}$ is defined by $\bar{Q}(\bar{x}, \mathrm{d}\bar{x}') = Q(x,\mathrm{d}x')R(x',\mathrm{d}w')$. In order to check (B1), we first set

$$\begin{aligned}
\mu_0(\mathrm{d}\bar{x}\mathrm{d}\bar{x}') &= \mu_0(\mathrm{d}x\mathrm{d}w\mathrm{d}x'\mathrm{d}w') = wR(x,\mathrm{d}w)\mu(\mathrm{d}x)Q(x,\mathrm{d}x')R(x',\mathrm{d}w') \\
\mu_1(\mathrm{d}\bar{x}\mathrm{d}\bar{x}') &= \mu_1(\mathrm{d}x\mathrm{d}w\mathrm{d}x'\mathrm{d}w') = w'R(x',\mathrm{d}w')\mu(\mathrm{d}x')Q(x',\mathrm{d}x)R(x,\mathrm{d}w).
\end{aligned}$$

Then, writing $Q(x,\mathrm{d}y) = q(x,y)\mu(\mathrm{d}y)$, we obtain for all $h \in \mathsf{F}_+(\bar{\mathsf{X}}^2)$,

$$\begin{aligned}
\int_{\bar{\mathsf{X}}^2} h(\bar{x},\bar{x}')\mu_1(\mathrm{d}\bar{x}\mathrm{d}\bar{x}') &= \int_{\bar{\mathsf{X}}^2} h(\bar{x},\bar{x}')w'q(x',x)[R(x,\mathrm{d}w)R(x',\mathrm{d}w')\mu(\mathrm{d}x)\mu(\mathrm{d}x')] \\
&= \int_{\bar{\mathsf{X}}^2} h(\bar{x},\bar{x}')r(\bar{x},\bar{x}')\mu_0(\mathrm{d}\bar{x}\mathrm{d}\bar{x}'),
\end{aligned}$$

where

$$r(\bar{x},\bar{x}') = \frac{w'q(x',x)}{wq(x,x')}.$$

Since r is positive, we can apply Lemma 1.22 with $\alpha(\bar{x},\bar{x}') = r(\bar{x},\bar{x}') \wedge 1$ and we finally get that $P^{MH}_{\langle\bar{\pi},\bar{Q}\rangle}(\bar{x},\mathrm{d}\bar{x}')$ is $\bar{\pi}$-reversible. Since Algorithm 2 corresponds to applying the Markov kernel $P^{MH}_{\langle\bar{\pi},\bar{Q}\rangle}$, this completes the proof. Note that the extended target distribution $\bar{\pi}$ has the marginal π with respect to the first component:

$$\bar{\pi}(A\times\mathbb{R}_\star^+) = \int_A \int_{\mathbb{R}_\star^+} wR(x,\mathrm{d}w)\mu(\mathrm{d}x) = \int_A \pi(\mathrm{d}x) = \pi(A).$$

To sum up, $(\bar{X}_k)_{k\in\mathbb{N}} = (X_k, W_k)_{k\in\mathbb{N}}$ produced by Algorithm 2 is a generalized Metropolis–Hastings algorithm whose target distribution $\bar{\pi}$ admits π as marginal distribution on the first component. Note that $(X_k)_{k\in\mathbb{N}}$ alone is not a Markov chain anymore.

Example 1.23 (Latent variable models). Consider a Bayesian model with parameter x, prior density $p(x)$, latent variable Z and observation y, with complete-data likelihood $p(y,z \mid x)$. The posterior $\pi(x) \propto p(x)\int p(y,z \mid x)\,\mathrm{d}z$ involves an intractable integral, even up to a constant. Given an importance density g_x on the latent space (see Chapter 3), draw $Z_1,\ldots,Z_n$ i.i.d. with density g_x and set

$$W = p(x)\frac{1}{n}\sum_{i=1}^n \frac{p(y,Z_i \mid x)}{g_x(Z_i)}.$$

Then $\mathbb{E}[W] = p(x)p(y \mid x) \propto \pi(x)$, so that Algorithm 2 targets the exact posterior, whatever $n \geq 1$. The number n of samples only affects the efficiency: when the variance of W is large, an overestimated W_k makes the chain "stick" at X_k for many iterations.

1.4.4 Hamiltonian Monte Carlo

In Hamiltonian Monte Carlo (HMC) [Duane et al., 1987, Neal, 2011], we extend the target density and construct a Markov chain on an extended space. We assume that the state space is $\mathbb{R}^d$ and

that $\pi(q) \propto \mathrm{e}^{-U(q)}$ where U is continuously differentiable (in most of the HMC literature, the state variable x is replaced by q, which should not be confused with the proposal density of the previous sections). A classical setting is to consider π as the marginal of the extended target

$$\bar{\pi}(q,p) \propto \exp\left\{-U(q) - p^\top p/2\right\}, \quad p,q \in \mathbb{R}^d. \tag{1.10}$$

In this simple setting, the extended target density can be written as the product of π and the standard Gaussian probability density of $\mathcal{N}(0, I_d)$: under $\bar{\pi}$, q and p are independent. In what follows, we use the following terminology, very classical in the HMC literature.

- $q \in \mathbb{R}^d$ is the position and $U(q)$ is called the *potential energy*.
- $p \in \mathbb{R}^d$ is the momentum and $K(p) = p^\top p/2$ is called the *kinetic energy*.
- $H(q,p) = U(q) + K(p)$ is called the *Hamiltonian*, so that $\bar{\pi}(q,p) \propto \mathrm{e}^{-H(q,p)}$.

Several versions of HMC exist. We consider in this course the leapfrog HMC which produces a Markov chain $(X_k)_{k\in\mathbb{N}} = (q_k, p_k)_{k\in\mathbb{N}}$ and where a transition of this algorithm can be decomposed into two different moves.

- The first transition sets $X^0_{k+1} = (q^0_{k+1}, p^0_{k+1})$ where the first component is frozen, i.e. $q^0_{k+1} = q_k$, while the second one is sampled as $p^0_{k+1} \sim \mathcal{N}(0, I_d)$. As the standard Gaussian distribution is the conditional distribution of p given q under (1.10), this is a Gibbs move, which means that the transition kernel associated with this move is $\bar{\pi}$-reversible.
- The second transition sets $X^L_{k+1} = (q^L_{k+1}, p^L_{k+1})$ deterministically from X^0_{k+1} using L steps of a numerical integrator, and then accepts or rejects the proposed candidate according to some well-chosen probability. Therefore, we need to establish how to choose the acceptance probability with deterministic moves to obtain a $\bar{\pi}$-reversible transition.

Since both moves leave $\bar{\pi}$ invariant, so does their composition, which is however not reversible in general.

MH with deterministic moves

A natural question is therefore to understand whether we can construct a MH algorithm with target $\bar{\pi}$ on $\mathbb{R}^{2d}$ and where the proposed candidate is deterministic: $Q(x, \mathrm{d}y) = \delta_{\varphi(x)}(\mathrm{d}y)$, where $\varphi : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ is a measurable bijection. Due to the Dirac mass, we are not in the standard framework but we can use the generalized version of the MH algorithm as described in Section 1.4.2. Set

$$\mu_0(\mathrm{d}x\mathrm{d}y) = \bar{\pi}(\mathrm{d}x)\delta_{\varphi(x)}(\mathrm{d}y) \quad \text{and} \quad \mu_1(\mathrm{d}x\mathrm{d}y) = \bar{\pi}(\mathrm{d}y)\delta_{\varphi(y)}(\mathrm{d}x),$$

denote by $\bar{\pi} \circ \varphi^{-1}$ the image measure of $\bar{\pi}$ by φ, assume that it is dominated by $\bar{\pi}$, and write for any non-negative measurable function h,

$$\begin{aligned}\int h(x,y)\mu_1(\mathrm{d}x\mathrm{d}y) &= \int \bar{\pi}(\mathrm{d}u) h(\underbrace{\varphi(u)}_{v}, \underbrace{u}_{\varphi^{-1}(v)}) \\ &= \int \bar{\pi} \circ \varphi^{-1}(\mathrm{d}v) h(v, \varphi^{-1}(v)) = \int \frac{\mathrm{d}\bar{\pi} \circ \varphi^{-1}}{\mathrm{d}\bar{\pi}}(v)\bar{\pi}(\mathrm{d}v) h(v, \varphi^{-1}(v)).\end{aligned}$$

If we want to let appear the integral of h with respect to $\mu_0(\mathrm{d}x\mathrm{d}y) = \bar{\pi}(\mathrm{d}x)\delta_{\varphi(x)}(\mathrm{d}y)$, we need to assume that $\varphi^{-1}(v) = \varphi(v)$, that is, φ is an involution, and in such a case:

$$\int h(x,y)\mu_1(\mathrm{d}x\mathrm{d}y) = \int h(x,y)\frac{\mathrm{d}\bar{\pi} \circ \varphi^{-1}}{\mathrm{d}\bar{\pi}}(x)\mu_0(\mathrm{d}x\mathrm{d}y)$$

and the acceptance probability is then

$$\alpha(x,y) = \frac{\mathrm{d}\mu_1}{\mathrm{d}\mu_0}(x,y) \wedge 1 = \frac{\mathrm{d}\bar{\pi} \circ \varphi^{-1}}{\mathrm{d}\bar{\pi}}(x) \wedge 1.$$

A first point is that if we only use the involution, then the chain alternates between x and $\varphi(x)$ (after two accepted steps we are back to the initial state), so that it cannot be irreducible. Therefore, this deterministic transition is combined with another move that is not deterministic (in HMC, the refreshment of the momentum). For any involution, we obtain a Metropolis–Hastings algorithm with a theoretical expression of the acceptance probability, but, in an ideal HMC, we can choose φ so that it is equal to 1. If $\bar{\pi}$ has a positive density with respect to the Lebesgue measure that we still denote $\bar{\pi}$ and if φ is a C^1-diffeomorphism, the change of variables formula yields

$$\frac{\mathrm{d}\bar{\pi} \circ \varphi^{-1}}{\mathrm{d}\bar{\pi}}(x) = \frac{\bar{\pi}(\varphi^{-1}(x))}{\bar{\pi}(x)} \left|\det J_{\varphi^{-1}}(x)\right|,$$

where $J_{\varphi^{-1}}(x)$ is the Jacobian matrix of the mapping $\varphi^{-1} = \varphi$ at x. To get 1 in the acceptance probability, we can impose that the two terms are equal to 1.

- The first term $\bar{\pi}(\varphi^{-1}(x))/\bar{\pi}(x)$ is one if the involution stays on the same level set of $\bar{\pi}$ (i.e. the moves according to φ do not change the value of $\bar{\pi}$).
- The second term $\left|\det J_{\varphi^{-1}}(x)\right|$ is one if the involution is volume-preserving. If the involution is only volume-preserving, then the Radon–Nikodym derivative simplifies to

$$\frac{\bar{\pi}(\varphi^{-1}(x))}{\bar{\pi}(x)} = \frac{\bar{\pi}(\varphi(x))}{\bar{\pi}(x)} \quad \text{since} \quad \varphi \circ \varphi = \mathrm{I},$$

  and the acceptance probability is $\alpha(x,\varphi(x)) = 1 \wedge \bar{\pi}(\varphi(x))/\bar{\pi}(x) = 1 \wedge \exp\{H(x) - H(\varphi(x))\}$.

Hamiltonian dynamics

In order to propose a deterministic mapping, assume that the state depends on a real parameter t and we impose to stay on a level set of H. We get:

$$\frac{\mathrm{d}H(q_t,p_t)}{\mathrm{d}t} = 0 = \sum_{i=1}^{d} \frac{\partial H(q_t,p_t)}{\partial q_{t,i}} \frac{\mathrm{d}q_{t,i}}{\mathrm{d}t} + \frac{\partial H(q_t,p_t)}{\partial p_{t,i}} \frac{\mathrm{d}p_{t,i}}{\mathrm{d}t}.$$

This motivates the use the following dynamics: for all $1 \leq i \leq d$,

$$\begin{aligned} \frac{\partial H}{\partial q_{t,i}}(q_t,p_t) &= \frac{\partial U(q_t)}{\partial q_{t,i}} = -\frac{\mathrm{d}p_{t,i}}{\mathrm{d}t} \\ \frac{\partial H}{\partial p_{t,i}}(q_t,p_t) &= \frac{\partial K(p_t)}{\partial p_{t,i}} = p_{t,i} = \frac{\mathrm{d}q_{t,i}}{\mathrm{d}t}. \end{aligned} \tag{1.11}$$

Denote by Φ_t the flow of (1.11), i.e. $\Phi_t(q_0,p_0) = (q_t,p_t)$, and by $s(q,p) = (q,-p)$ the momentum flip. The Hamiltonian flow has three fundamental properties.

i) *Conservation of energy:* $H \circ \Phi_t = H$, by construction.
ii) *Volume preservation (Liouville's theorem):* the vector field $F(q,p) = (\nabla_p H, -\nabla_q H)$ has zero divergence, $\sum_i \partial^2 H/\partial q_i \partial p_i - \partial^2 H/\partial p_i \partial q_i = 0$, so that $|\det J_{\Phi_t}| = 1$.
iii) *Time reversibility:* if $(q_t,p_t)_{0 \leq t \leq T}$ solves (1.11), so does $(\tilde{q}_t,\tilde{p}_t) = (q_{T-t},-p_{T-t})$. Indeed, $\mathrm{d}\tilde{q}_t/\mathrm{d}t = -\dot{q}_{T-t} = -p_{T-t} = \tilde{p}_t$ and $\mathrm{d}\tilde{p}_t/\mathrm{d}t = \dot{p}_{T-t} = -\nabla U(q_{T-t}) = -\nabla U(\tilde{q}_t)$. Since $(\tilde{q}_0,\tilde{p}_0) = s \circ \Phi_T(q_0,p_0)$ and $(\tilde{q}_T,\tilde{p}_T) = s(q_0,p_0)$, we get $\Phi_T \circ s \circ \Phi_T = s$, i.e. $s \circ \Phi_T$ is an involution.

The flow Φ_T itself is not an involution, but $\varphi = s \circ \Phi_T$ is an involution which is volume-preserving and preserves H (since $H \circ s = H$): the corresponding MH move is always accepted. Moreover,

since the momentum is refreshed at the next iteration and $\bar{\pi}$ is invariant by s, the flip does not need to be applied in practice. Unfortunately, it is not possible to compute (q_t, p_t) exactly in general, and we need to use a discretization scheme to approximate solutions to the Hamiltonian dynamics.

The leapfrog integrator

Recall that for all $1 \leq i \leq d$,

$$\frac{\partial U(q_t)}{\partial q_{t,i}} = -\frac{\mathrm{d}p_{t,i}}{\mathrm{d}t}, \quad \text{and} \quad p_{t,i} = \frac{\mathrm{d}q_{t,i}}{\mathrm{d}t}.$$

A first idea of discretization would be to choose a small stepsize h and a number of steps L and compute, for $0 \leq k \leq L-1$,

$$\begin{aligned} p^{k+1} &= p^k - h\nabla U(q^k) \\ q^{k+1} &= q^k + hp^k. \end{aligned}$$

Unfortunately, this discretization is associated with a mapping that is not volume-preserving. That is the absolute value of the Jacobian determinant of the mapping is not equal to one. This is the reason why we use in practice the leapfrog discretization, defined by the following scheme: for $0 \leq k \leq L-1$,

$$\begin{aligned} p^{k+1/2} &= p^k - (h/2)\nabla U(q^k) \\ q^{k+1} &= q^k + hp^{k+1/2} \\ p^{k+1} &= p^{k+1/2} - (h/2)\nabla U(q^{k+1}). \end{aligned} \tag{1.12}$$

To see that it is volume-preserving, just note that a leapfrog update can be decomposed into three mappings

$$(q^k, p^k) \xrightarrow{\varphi_1} (q^k, p^{k+1/2}) \xrightarrow{\varphi_2} (q^{k+1}, p^{k+1/2}) \xrightarrow{\varphi_1} (q^{k+1}, p^{k+1}), \tag{1.13}$$

where

$$\varphi_1(x, y) = (x, y - (h/2)\nabla U(x)) \quad \text{and} \quad \varphi_2(x, y) = (x + hy, y). \tag{1.14}$$

Each of these mappings is a shear whose Jacobian matrix is block triangular with identity diagonal blocks: it is volume-preserving and so is the leapfrog update $\Psi_h = \varphi_1 \circ \varphi_2 \circ \varphi_1$.

The symmetric structure $\varphi_1 \circ \varphi_2 \circ \varphi_1$ is what makes the leapfrog integrator reversible; the explicit Euler scheme is neither volume-preserving nor reversible. To sum up, the leapfrog update is an approximation of the Hamiltonian dynamics which, combined with a momentum flip, is a deterministic volume-preserving involution. By the previous paragraphs, the MH move with this deterministic proposal leaves $\bar{\pi}$ invariant when the candidate is accepted with probability

$$1 \wedge \exp\left\{H(q^0, p^0) - H(q^L, p^L)\right\},$$

where we used $H(q, -p) = H(q, p)$. Contrary to the exact flow, the leapfrog integrator does not preserve H exactly, so that some candidates are rejected; the energy error is of order h^2 over bounded time intervals for smooth U. The resulting algorithm is summarized in Algorithm 3.

Example 1.24 (Leapfrog for a Gaussian target). Let $d = 1$ and $U(q) = q^2/2$, i.e. $\pi = \mathcal{N}(0, 1)$. The exact flow is the rotation $(q_t, p_t) = (q_0 \cos t + p_0 \sin t, p_0 \cos t - q_0 \sin t)$, which preserves $H(q, p) = (q^2 + p^2)/2$. One leapfrog step is the linear map

$$\begin{pmatrix} q^{k+1} \\ p^{k+1} \end{pmatrix} = \begin{pmatrix} 1 - h^2/2 & h \\ -h(1 - h^2/4) & 1 - h^2/2 \end{pmatrix} \begin{pmatrix} q^k \\ p^k \end{pmatrix},$$

Input : Initial position q_0, stepsize h, number of leapfrog steps L, number of iterations n.
Output: $q_0, \dots, q_n$.

for $k = 0$ **to** $n-1$ **do**
- Set $q^0 = q_k$ and draw $p^0 \sim \mathcal{N}(0, I_d)$ independently of the past.
- **for** $\ell = 0$ **to** $L-1$ **do**
  $p^{\ell+1/2} = p^\ell - (h/2)\nabla U(q^\ell)$; $q^{\ell+1} = q^\ell + h p^{\ell+1/2}$; $p^{\ell+1} = p^{\ell+1/2} - (h/2)\nabla U(q^{\ell+1})$.
- **end**
- Draw $U_{k+1} \sim \mathrm{U}(0,1)$. If $\log U_{k+1} \le H(q^0,p^0) - H(q^L,p^L)$, set $q_{k+1} = q^L$; otherwise set $q_{k+1} = q_k$.

end

Algorithm 3: Hamiltonian Monte Carlo with leapfrog integrator

whose determinant is $(1 - h^2/2)^2 + h^2(1 - h^2/4) = 1$, and whose trace is $2 - h^2$. For $h < 2$, its eigenvalues are complex conjugate with modulus one: the leapfrog trajectories stay on ellipses close to the circles $\{H = \text{const}\}$ and the energy error remains bounded (and of order h^2) for all L. For $h > 2$, one eigenvalue has modulus larger than one, trajectories diverge exponentially and almost all candidates are rejected: the stepsize must be smaller than a stability threshold related to the smallest length scale of π. By contrast, the explicit Euler map has determinant $1 + h^2 > 1$ and H is multiplied by $1 + h^2$ at each step (Figure 1.5).

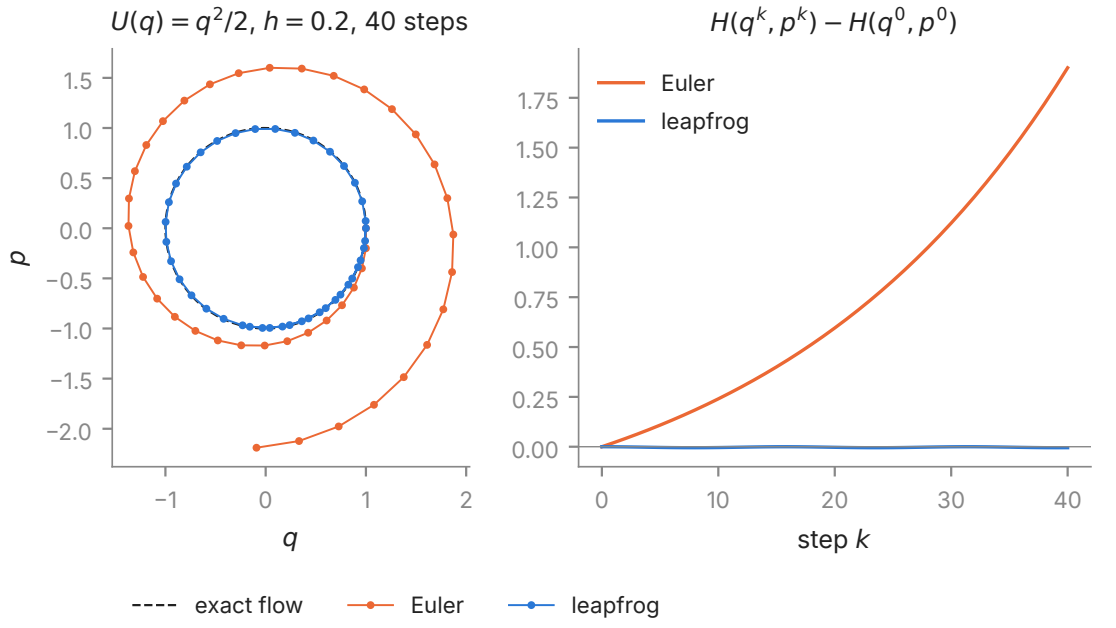


Fig. 1.5 Left: trajectories of the explicit Euler scheme and of the leapfrog integrator for $U(q) = q^2/2$, started at $(q, p) = (1, 0)$, with $h = 0.2$ and 40 steps; the exact flow is the unit circle. Right: evolution of the energy error. The leapfrog energy error oscillates with amplitude of order h^2, while the Euler scheme spirals out.

Chapter 2

Variational inference and autoencoders

2.1 Evidence lower bound

In this chapter, we consider models with latent (unobserved) data. Let (Z, X) be random variables in $\mathbb{R}^d \times \mathbb{R}^m$. We assume that the law of (Z, X) has a density $(z, x) \mapsto p(z, x)$ with respect to the Lebesgue measure (all the results of this chapter hold with a general dominating measure, replacing $\mathrm{d}z$ by $\lambda(\mathrm{d}z)$; for discrete latent variables, integrals are replaced by sums). In this setting, we write

$$(z, x) \mapsto p(z, x) = p(z)p(x|z)\,,$$

where $z \mapsto p(z)$ is a prior density for Z and $x \mapsto p(x|z)$ is the conditional density (likelihood) of X given Z. The conditional density of Z given X cannot be computed in general, since it is given by

$$z \mapsto p(z|x) = \frac{p(z)p(x|z)}{p(x)} \propto p(z)p(x|z)\,,$$

where $p(x) = \int p(z)p(x|z)\mathrm{d}z$ is an intractable integral. The conditional law of latent variables given observations is of utmost importance in many machine learning approaches. For instance, in the E-step of the EM algorithm, the intermediate quantity requires the computation of an expectation under such a distribution. In most situations, this distribution cannot be sampled from easily. Standard solutions to sample approximately from $p(z|x)$ include Markov chain Monte Carlo (Chapter 1) or sequential Monte Carlo (Chapter 3) methods. In this chapter, we focus on variational approaches where $p(z|x)$ is replaced by a simpler distribution obtained by solving an optimization problem.

In variational inference, we introduce a variational family i.e. a family of densities to approximate $z \mapsto p(z|x)$. Let $\mathcal{D}$ be such a family, where the densities $q(\cdot|x) \in \mathcal{D}$ satisfy the two following assumptions.

- For all $q(\cdot|x) \in \mathcal{D}$, $z \mapsto q(z|x)$ is easy to evaluate.
- For all $q(\cdot|x) \in \mathcal{D}$, it is easy to sample $Z \sim q(\cdot|x)$.

The notation $q(\cdot|x)$ only emphasizes that the approximation is designed for a given observation x; at this stage, $q(\cdot|x)$ is simply a probability density on $\mathbb{R}^d$.

Then, for all x such that $p(x) > 0$ and all $q(\cdot|x) \in \mathcal{D}$ such that the following quantities are finite, writing KL for the Kullback–Leibler divergence between two probability distributions,

$$\begin{aligned}\mathrm{KL}\left(q(\cdot|x)\|p(\cdot|x)\right) = \int q(z|x)\log\frac{q(z|x)}{p(z|x)}\mathrm{d}z &= \mathbb{E}_{q(\cdot|x)}[\log q(Z|x)] - \mathbb{E}_{q(\cdot|x)}[\log p(Z|x)]\,,\\ &= \mathbb{E}_{q(\cdot|x)}[\log q(Z|x)] - \mathbb{E}_{q(\cdot|x)}[\log p(Z,x)] + \log p(x)\,,\\ &= -\mathcal{L}(q(\cdot|x)) + \log p(x)\,,\end{aligned}$$

where

$$\mathcal{L}(q(\cdot|x)) = \mathbb{E}_{q(\cdot|x)}\left[\log\frac{p(Z,x)}{q(Z|x)}\right].$$

In other words,

$$\log p(x) = \mathcal{L}(q(\cdot|x)) + \mathrm{KL}\left(q(\cdot|x)\|p(\cdot|x)\right). \tag{2.1}$$

Using Jensen's inequality, we obtain $\mathrm{KL}\left(q(\cdot|x)\|p(\cdot|x)\right) \geq 0$, with equality if and only if $q(\cdot|x) = p(\cdot|x)$ almost everywhere, so that

$$\mathcal{L}(q(\cdot|x)) \leq \log p(x).$$

This inequality justifies the name evidence lower bound (ELBO) for $\mathcal{L}(q(\cdot|x))$, the evidence being the marginal likelihood $p(x)$. Since $\log p(x)$ does not depend on q, (2.1) shows that maximizing the ELBO is equivalent to minimizing $\mathrm{KL}(q(\cdot|x)\|p(\cdot|x))$, a quantity which cannot be computed, whereas the ELBO only involves the joint density $p(z,x)$, which is explicit. In variational inference, we then aim to approximate $p(\cdot|x)$ by $q_*(\cdot|x)$ where:

$$q_*(\cdot|x) \in \mathrm{argmax}_{q\in\mathcal{D}}\ \mathcal{L}(q(\cdot|x))\,.$$

Remark 2.1. i) *Reconstruction and regularization.* Since $p(z,x) = p(z)p(x|z)$,

$$\mathcal{L}(q(\cdot|x)) = \mathbb{E}_{q(\cdot|x)}[\log p(x|Z)] - \mathrm{KL}\left(q(\cdot|x)\|p\right), \tag{2.2}$$

where p denotes the prior. The first term favors distributions $q(\cdot|x)$ putting mass on latent values which explain the observation well, the second one keeps $q(\cdot|x)$ close to the prior. This decomposition is central for variational autoencoders (Section 2.4).

ii) *Energy and entropy.* $\mathcal{L}(q(\cdot|x)) = \mathbb{E}_{q(\cdot|x)}[\log p(Z,x)] + \mathrm{H}(q(\cdot|x))$, where $\mathrm{H}(q) = -\int q\log q$ is the (differential) entropy. This is the form used in statistical physics (the ELBO is the opposite of the variational free energy).

Remark 2.2. When the model depends on a parameter θ, $\mathcal{L}(q,\theta) = \mathbb{E}_q[\log(p_\theta(Z,x)/q(Z))]$ is a lower bound on $\log p_\theta(x)$ for all q and θ. Maximizing alternately in q *without restriction on the family* and in θ is exactly the EM algorithm: by (2.1), the maximum in q is attained at $q = p_\theta(\cdot|x)$ (E-step), and maximizing $\theta \mapsto \mathcal{L}(q,\theta) = \mathbb{E}_q[\log p_\theta(Z,x)] + \mathrm{H}(q)$ is the M-step [**?**]. Variational inference restricts q to a tractable family $\mathcal{D}$ when the E-step is not feasible; variational EM and variational autoencoders (Section 2.4) are built on this remark.

The direction of the divergence is dictated by computability: $\mathrm{KL}(q\|p(\cdot|x))$ only requires expectations under q, which can be sampled, while $\mathrm{KL}(p(\cdot|x)\|q)$ requires expectations under the intractable posterior. This choice has qualitative consequences. Since $q\log(q/p)$ is large where $q > 0$ and $p \approx 0$, minimizing $\mathrm{KL}(q\|p)$ forces q to vanish where p vanishes: it is *zero-forcing* (or mode-seeking) and tends to underestimate the spread of p. On the contrary, $\mathrm{KL}(p\|q)$ is *mass-covering*: it penalizes heavily the regions where $p > 0$ and $q \approx 0$. See Example 2.3 and Figure 2.1.

2.2 Coordinate ascent variational inference

The most straightforward approach to solve the optimization problem is to consider a mean-field variational family, i.e. to choose $\mathcal{D}$ such that

$$\mathcal{D} = \left\{ z \mapsto q(z|x) = \prod_{j=1}^{d} q_j(z_j|x) \text{ ; for all } 1 \leq j \leq d,\ q_j(\cdot|x) \text{ is a probability density on } \mathbb{R} \right\}.$$

Under $q \in \mathcal{D}$, the components of Z are independent, but no parametric form is imposed on the factors q_j. More generally, z can be split into blocks $z = (z_1, \ldots, z_J)$ of arbitrary dimensions; all the computations below hold verbatim. Even when considering a simple variational family such as $\mathcal{D}$, it is not possible to maximize the ELBO explicitly in general. Assume therefore that we want to optimize $\mathcal{L}(q(\cdot|x))$ with respect to $q_j(\cdot|x)$ only, for some $1 \leq j \leq d$, the other densities $(q_\ell(\cdot|x))_{\ell \neq j}$ being kept fixed. Write $z_{-j} = (z_\ell)_{1 \leq \ell \leq d; \ell \neq j}$ and, for all $q(\cdot|x) \in \mathcal{D}$, $q_{-j}(z_{-j}|x) = \prod_{1 \leq \ell \leq d; \ell \neq j} q_\ell(z_\ell|x)$. Then, since $p(z, x) = p(z_{-j}, x)p(z_j|z_{-j}, x)$,

$$\begin{aligned}
\mathcal{L}(q(\cdot|x)) &= \int \left\{ \prod_{\ell=1}^{d} q_\ell(z_\ell|x) \right\} \left\{ \log(p(z_{-j}, x)p(z_j|z_{-j}, x)) - \sum_{\ell=1}^{d} \log q_\ell(z_\ell|x) \right\} \mathrm{d}z_1 \ldots \mathrm{d}z_d \\
&= \int q_j(z_j|x) \left\{ \int \left\{ \prod_{\ell=1, \ell \neq j}^{d} q_\ell(z_\ell|x) \right\} \log p(z_j|z_{-j}, x)\, \mathrm{d}z_{-j} \right\} \mathrm{d}z_j \\
&\qquad + \int q_{-j}(z_{-j}|x) \log p(z_{-j}, x)\, \mathrm{d}z_{-j} - \sum_{\ell=1}^{d} \int q_\ell(z_\ell|x) \log q_\ell(z_\ell|x) \mathrm{d}z_\ell \\
&= \mathbb{E}_{q_j(\cdot|x)} \left[\mathbb{E}_{q_{-j}(\cdot|x)} \left[\log p(Z_j|Z_{-j}, x) \right] \right] - \mathbb{E}_{q_j(\cdot|x)} \left[\log q_j(Z_j|x) \right] + \mathrm{C},
\end{aligned}$$

where C does not depend on $q_j(\cdot|x)$. Consider the density $\tilde{q}_j(\cdot|x)$ such that

$$\tilde{q}_j(z_j|x) \propto \exp\left(\mathbb{E}_{q_{-j}(\cdot|x)} \left[\log p(z_j|Z_{-j}, x) \right] \right),$$

i.e. the density given by $\tilde{q}_j(z_j|x) = \exp(\mathbb{E}_{q_{-j}(\cdot|x)}[\log p(z_j|Z_{-j}, x)])/C_j$ where C_j does not depend on z_j (C_j is the normalizing constant to obtain a density, assumed to be finite). Therefore,

$$\mathcal{L}(q(\cdot|x)) = -\mathbb{E}_{q_j(\cdot|x)} \left[\log \frac{q_j(Z_j|x)}{\tilde{q}_j(Z_j|x)} \right] + \mathrm{C} + \log C_j = -\mathrm{KL}(q_j(\cdot|x) \| \tilde{q}_j(\cdot|x)) + \mathrm{C} + \log C_j.$$

Therefore, optimizing $\mathcal{L}(q(\cdot|x))$ with respect to $q_j(\cdot|x)$ only, the other densities $(q_\ell(\cdot|x))_{\ell \neq j}$ being kept fixed, yields a unique optimum given by $\tilde{q}_j(\cdot|x)$. Since $\log p(z_j|z_{-j}, x) = \log p(z_j, z_{-j}, x) - \log p(z_{-j}, x)$ and the second term does not depend on z_j, the update can equivalently be written

$$\tilde{q}_j(z_j|x) \propto \exp\left(\mathbb{E}_{q_{-j}(\cdot|x)} \left[\log p(z_j, Z_{-j}, x) \right] \right), \tag{2.3}$$

which is often more convenient since the joint density is explicit. The coordinate ascent variational inference (CAVI) algorithm therefore proposes to update sequentially $q_j(\cdot|x)$, $1 \leq j \leq d$, until a stopping criterion is met. In Algorithm 4, we propose a version of the algorithm where the variational distribution of only one component of Z, chosen at random, is updated at each iteration; of course many alternatives can be considered. A standard alternative is to update all the factors in turn at each iteration (systematic scan), as in Algorithm 5.

Data: Observation x, initial variational distribution $\{q_j^{(0)}(\cdot|x)\}_{1\le j\le d}$, maximum number of iterations N.
Result: A variational distribution for each coordinate of Z, $q_j^{(N)}(\cdot|x)$, $1 \le j \le d$.
for $k = 1 \to N$ **do**
  Draw $j \in \{1, \dots, d\}$ uniformly at random;
  Set $q_\ell^{(k)}(\cdot|x) = q_\ell^{(k-1)}(\cdot|x)$ for all $1 \le \ell \le d$, $\ell \ne j$, and $q_{-j}^{(k)}(\cdot|x) = \prod_{1\le \ell\le d, \ell\ne j} q_\ell^{(k)}(\cdot|x)$;
  Set $q_j^{(k)}(z_j|x) \propto \exp\left(\mathbb{E}_{q_{-j}^{(k)}(\cdot|x)}\left[\log p(z_j|Z_{-j}, x)\right]\right)$;
end

Algorithm 4: Coordinate ascent variational inference (random scan).

The CAVI algorithm has several key properties.

i) *Monotonicity.* Each update maximizes $\mathcal{L}$ with respect to one factor, so that $k \mapsto \mathcal{L}(q^{(k)}(\cdot|x))$ is nondecreasing. Since it is bounded by $\log p(x)$, the sequence of ELBO values converges. Monitoring the ELBO is therefore a natural stopping criterion, and a decrease of the ELBO reveals an error in the implementation (see Figure 2.3).
ii) *Local optima.* The ELBO is not concave in general: CAVI converges to a local maximum (or a stationary point) which depends on the initialization, and several initializations are usually compared through their final ELBO.
iii) *Conditionally conjugate models.* If the full conditional $p(z_j|z_{-j}, x)$ belongs to an exponential family, $p(z_j|z_{-j}, x) = h(z_j)\exp\{\eta_j(z_{-j}, x)^\top T(z_j) - A(\eta_j(z_{-j}, x))\}$, then $\tilde{q}_j(z_j|x) \propto h(z_j)\exp\{\mathbb{E}_{q_{-j}}[\eta_j(Z_{-j}, x)]^\top T(z_j)\}$: the optimal factor belongs to the same family, with natural parameter $\mathbb{E}_{q_{-j}}[\eta_j(Z_{-j}, x)]$ [Blei et al., 2017]. CAVI then reduces to fixed-point updates of finite-dimensional parameters.

Example 2.3 (Mean-field approximation of a Gaussian distribution). Let $p(\cdot|x) = \mathcal{N}(\mu, \Lambda^{-1})$ on $\mathbb{R}^d$, where the precision matrix Λ is positive definite. Since $\log p(z|x) = -\frac{1}{2}(z-\mu)^\top \Lambda(z-\mu) + \text{const}$, as a function of z_j,

$$\log p(z|x) = -\frac{\Lambda_{jj}}{2} z_j^2 + z_j\Big(\Lambda_{jj}\mu_j - \sum_{\ell\ne j}\Lambda_{j\ell}(z_\ell - \mu_\ell)\Big) + \text{const.}$$

By (2.3), the optimal factor is Gaussian, $q_j = \mathcal{N}(m_j, \Lambda_{jj}^{-1})$, with $m_j = \mu_j - \Lambda_{jj}^{-1}\sum_{\ell\ne j}\Lambda_{j\ell}(m_\ell - \mu_\ell)$. The unique fixed point is $m = \mu$ (the means are exact), but the variances are $\Lambda_{jj}^{-1} \le (\Lambda^{-1})_{jj}$ (the conditional variance is smaller than the marginal variance): *mean-field VI underestimates marginal variances.* For $d = 2$, unit variances and correlation ρ, $\Lambda_{jj}^{-1} = 1 - \rho^2$, which goes to 0 as $|\rho| \to 1$ while the marginal variances remain equal to 1. By contrast, minimizing $\mathrm{KL}(p\|q)$ over product densities gives $q_j = p_j$, the exact marginals: indeed, $\mathrm{KL}(p\|\prod_j q_j) = -\mathrm{H}(p) - \sum_j \mathbb{E}_p[\log q_j(Z_j)]$ and each term $-\mathbb{E}_{p_j}[\log q_j(Z_j)]$ is minimal for $q_j = p_j$. Both approximations are displayed in Figure 2.1, together with a bimodal example where the reverse KL locks onto one mode while the forward KL covers both modes with a single wide Gaussian.

Example 2.4 (Normal model with unknown mean and precision). Consider the model where $Y_1, \dots, Y_n$ are i.i.d. $\mathcal{N}(m, 1/\kappa)$ given (m, κ), with independent priors $m \sim \mathcal{N}(m_0, 1/\kappa_0)$ and $\kappa \sim \mathrm{Gamma}(a_0, b_0)$, and the mean-field family $q(m, \kappa) = q_1(m)q_2(\kappa)$. Up to additive constants,

$$\log p(m, \kappa, y_{1:n}) = -\frac{\kappa_0}{2}(m - m_0)^2 + \Big(a_0 + \frac{n}{2} - 1\Big)\log\kappa - b_0\kappa - \frac{\kappa}{2}\sum_{i=1}^n (y_i - m)^2.$$

Taking expectations with respect to q_2 (resp. q_1) in (2.3) yields $q_1 = \mathcal{N}(m_N, \lambda_N^{-1})$ and $q_2 = \mathrm{Gamma}(a_N, b_N)$ with

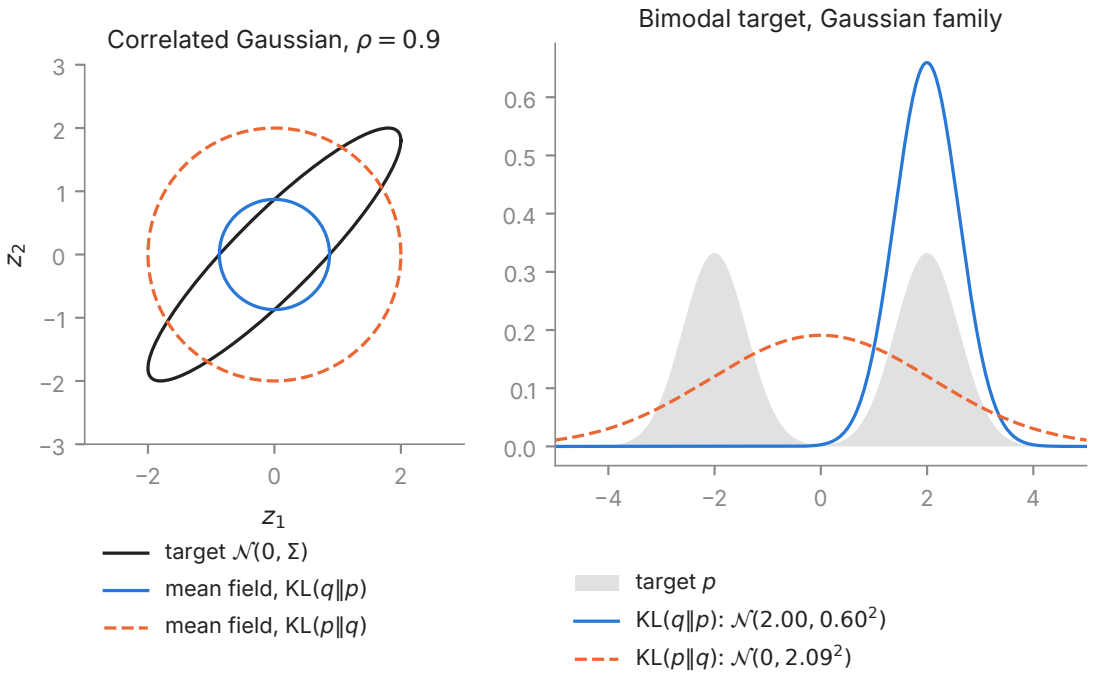


Fig. 2.1 Left: level sets (2 standard deviations) of a correlated Gaussian target and of its best mean-field approximations for $\mathrm{KL}(q\|p)$ (variances $1-\rho^2$) and $\mathrm{KL}(p\|q)$ (exact marginals). Right: bimodal target $0.5\mathcal{N}(-2, 0.6^2) + 0.5\mathcal{N}(2, 0.6^2)$ and its best Gaussian approximations; minimizing $\mathrm{KL}(q\|p)$ (computed by quadrature) selects one mode, minimizing $\mathrm{KL}(p\|q)$ matches the mean and the variance of p.

$$\lambda_N = \kappa_0 + n\,\mathbb{E}_{q_2}[\kappa], \quad m_N = \frac{\kappa_0 m_0 + \mathbb{E}_{q_2}[\kappa]\sum_{i=1}^n y_i}{\lambda_N},$$

$$a_N = a_0 + \frac{n}{2}, \quad b_N = b_0 + \frac{1}{2}\Big(\sum_{i=1}^n (y_i - m_N)^2 + \frac{n}{\lambda_N}\Big),$$

where $\mathbb{E}_{q_2}[\kappa] = a_N/b_N$ and we used $\mathbb{E}_{q_1}[(y_i - m)^2] = (y_i - m_N)^2 + \lambda_N^{-1}$. Figure 2.2 compares the result with the exact posterior: the mean-field approximation is accurate around the mode but cannot reproduce the dependence between m and κ (the posterior of m is wider when κ is small) and is too concentrated.

2.3 Application to a mixture of Gaussian distributions

This example can be found in [Blei et al., 2017]. Consider a mixture of K Gaussian distributions with means $\mu = (\mu_k)_{1 \leqslant k \leqslant K}$ and variance 1. The variables $\mu_1, \ldots, \mu_K$ are i.i.d. Gaussian with mean 0 and variance σ^2. For all $1 \leq i \leq n$, we denote by $c_i \in \{1, \ldots, K\}$ the group the i-th observation belongs to. The variables μ and c are not observed. The random variables $c = (c_i)_{1 \leq i \leq n}$ are independent of μ, and are i.i.d. with categorical distribution with parameters $(\omega_1, \ldots, \omega_K)$, i.e. $\mathbb{P}(c_i = k) = \omega_k$, where $\sum_{k=1}^K \omega_k = 1$ and the weights ω_k are known.

Conditionally on μ and c, the observations $(X_i)_{1 \leqslant i \leqslant n}$ are independent and X_i has a Gaussian distribution with mean μ_{c_i} and variance 1. Marginalizing over c, conditionally on μ, the observations $(X_i)_{1 \leqslant i \leqslant n}$ are i.i.d. and the conditional probability density of X_1 is

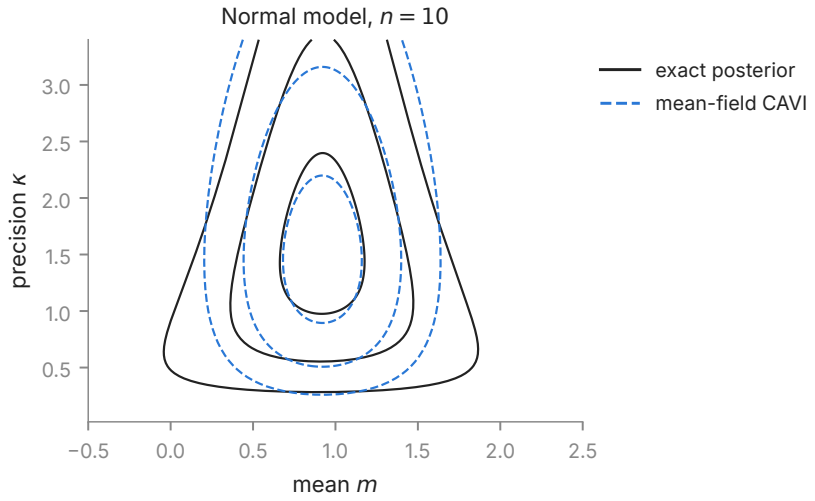


Fig. 2.2 Exact posterior (computed on a grid) and mean-field CAVI approximation for the normal model of Example 2.4, with $n = 10$ observations drawn from $\mathcal{N}(1, 1)$, $m_0 = 0$, $\kappa_0 = 0.1$, $a_0 = b_0 = 1$. The contours correspond to the same density levels relative to the mode.

$$x \mapsto p(x|\mu) = \sum_{k=1}^{K} \omega_k \varphi_{\mu_k,1}(x),$$

where φ_{μ_k,η^2} is the Gaussian probability density function with mean μ_k and variance η^2. The marginal likelihood (evidence) is therefore

$$p(x_{1:n}) = \int p(x_{1:n}|\mu)p(\mu)\mathrm{d}\mu = \int \prod_{i=1}^{n} p(x_i|\mu)p(\mu)\mathrm{d}\mu = \int \prod_{i=1}^{n} \left(\sum_{k=1}^{K} \omega_k \varphi_{\mu_k,1}(x_i) \right) p(\mu)\mathrm{d}\mu.$$

Expanding the product yields a sum of K^n Gaussian integrals, one for each configuration of c, which is not computable as soon as n is moderately large. Writing $z = (\mu, c)$ and $x = x_{1:n}$, our objective is to approximate $p(\mu, c|x)$, where $c = (c_1, \cdots, c_n)$ are the labels of the observations. Consider the following mean-field variational family:

$$q(\mu, c|x) = \prod_{k=1}^{K} \varphi_{m_k,s_k}(\mu_k) \prod_{i=1}^{n} \mathrm{Cat}_{\phi_i}(c_i),$$

which means that under $q(\cdot|x)$:

- μ and c are independent.
- $(\mu_k)_{1 \leqslant k \leqslant K}$ are independent Gaussian random variables with means $(m_k)_{1 \leqslant k \leqslant K}$ and variances $(s_k)_{1 \leqslant k \leqslant K}$.
- $(c_i)_{1 \leqslant i \leqslant n}$ are independent with categorical distributions with parameters $(\phi_i)_{1 \leqslant i \leqslant n}$: $q(c_i = k|x) = \phi_i(k)$ for $1 \leqslant k \leqslant K$.

The Gaussian form of the factors is not an assumption: by (2.3), it is the form of the optimal factors in the mean-field family $q(\mu, c|x) = \prod_k q_k(\mu_k) \prod_i q_i(c_i)$, as shown by the computations below. As in Section 2.1,

$$\mathrm{KL}\left(q(\cdot|x) \| p(\cdot|x)\right) = -\mathcal{L}(q(\cdot|x)) + \log p(x),$$

where

$$\mathcal{L}(q(\cdot|x)) = \mathbb{E}_{q(\cdot|x)}[\log p(\mu, c, x)] - \mathbb{E}_{q(\cdot|x)}[\log q(\mu, c|x)].$$

The CAVI algorithm computes iteratively, for $1 \leqslant k \leqslant K$,

$$q(\mu_k|x) \propto \exp\left(\mathbb{E}_{\tilde{q}_{\mu_k}(\cdot|x)}[\log p(\mu_k|x, c, \mu_{-k})]\right)$$

and, for all $1 \leqslant i \leqslant n$,

$$q(c_i|x) \propto \exp\left(\mathbb{E}_{\tilde{q}_{c_i}(\cdot|x)}[\log p(c_i|x, c_{-i}, \mu)]\right),$$

where $\mu_{-k} = (\mu_j)_{1\leq j\leq K, j\neq k}$, $c_{-i} = (c_j)_{1\leq j\leq n, j\neq i}$, and $\mathbb{E}_{\tilde{q}_z(\cdot|x)}$ is the expectation under the variational distribution of all variables except z.

Update of the variational distribution of c_i, $1 \leq i \leq n$

Note that

$$p(c_i|x, c_{-i}, \mu) \propto p(c_i)p(x_i|c_i, \mu) \propto p(c_i) \prod_{k=1}^{K} \left(\varphi_{\mu_k,1}(x_i)\right)^{1_{c_i=k}} .$$

Then,

$$\mathbb{E}_{\tilde{q}_{c_i}(\cdot|x)}[\log p(c_i|x, c_{-i}, \mu)] = \log p(c_i) + \sum_{k=1}^{K} 1_{c_i=k} \mathbb{E}_{\tilde{q}_{c_i}(\cdot|x)}[\log \varphi_{\mu_k,1}(x_i)] + \text{const},$$

and

$$\begin{aligned}\exp\left(\mathbb{E}_{\tilde{q}_{c_i}(\cdot|x)}[\log p(c_i|x, c_{-i}, \mu)]\right) &\propto p(c_i)\exp\left(\sum_{k=1}^{K} 1_{c_i=k} \mathbb{E}_{\tilde{q}_{c_i}(\cdot|x)}[\log \varphi_{\mu_k,1}(x_i)]\right) \\ &\propto p(c_i)\exp\left(\sum_{k=1}^{K} 1_{c_i=k} \mathbb{E}_{\tilde{q}_{c_i}(\cdot|x)}[-(x_i - \mu_k)^2/2]\right) .\end{aligned}$$

Since $\mathbb{E}_{\tilde{q}_{c_i}(\cdot|x)}[(x_i - \mu_k)^2] = x_i^2 - 2x_i m_k + m_k^2 + s_k$ and x_i^2 does not depend on k, the update writes:

$$\phi_i(k) \propto \omega_k \exp\left(m_k x_i - \frac{m_k^2 + s_k}{2}\right).$$

Update of the variational distribution of μ_k, $1 \leq k \leq K$

On the other hand,

$$p(\mu_k|x, c, \mu_{-k}) \propto p(\mu_k) \prod_{i=1}^{n} p(x_i|c_i, \mu) .$$

Then,

$$\mathbb{E}_{\tilde{q}_{\mu_k}(\cdot|x)}[\log p(\mu_k|x, c, \mu_{-k})] = \log p(\mu_k) + \sum_{i=1}^{n} \mathbb{E}_{\tilde{q}_{\mu_k}(\cdot|x)}[\log p(x_i|\mu, c_i)] + \text{const},$$

and, since the terms with $\ell \neq k$ do not depend on μ_k and $\mathbb{E}_{\tilde{q}_{\mu_k}(\cdot|x)}[\mathbf{1}_{c_i=k}] = \phi_i(k)$,

$$\begin{aligned}\exp\Big(\mathbb{E}_{\tilde{q}_{\mu_k}(\cdot|x)}[\log p(\mu_k|x,c,\mu_{-k})]\Big) &\propto p(\mu_k)\exp\left(\sum_{i=1}^n\sum_{\ell=1}^K \mathbb{E}_{\tilde{q}_{\mu_k}(\cdot|x)}[1_{c_i=\ell}\log\varphi_{\mu_\ell,1}(x_i)]\right)\\ &\propto p(\mu_k)\exp\left(\sum_{i=1}^n \phi_i(k)\mathbb{E}_{\tilde{q}_{\mu_k}(\cdot|x)}[\log\varphi_{\mu_k,1}(x_i)]\right)\\ &\propto \exp\left(-\frac{\mu_k^2}{2\sigma^2}-\frac{1}{2}\sum_{i=1}^n\phi_i(k)(x_i-\mu_k)^2\right),\\ &\propto \exp\left(-\frac{\mu_k^2}{2\sigma^2}+\sum_{i=1}^n\phi_i(k)x_i\mu_k-\frac{1}{2}\sum_{i=1}^n\phi_i(k)\mu_k^2\right).\end{aligned}$$

This is a Gaussian density, and the update writes therefore

$$m_k=\frac{\sum_{i=1}^n\phi_i(k)x_i}{1/\sigma^2+\sum_{i=1}^n\phi_i(k)}\quad\text{and}\quad s_k=\frac{1}{1/\sigma^2+\sum_{i=1}^n\phi_i(k)}.$$

The mean m_k is a weighted average of the observations, with weights given by the probabilities $\phi_i(k)$ that observation i belongs to group k, shrunk towards the prior mean 0.

Data: Observations $x=(x_1,\dots,x_n)$, initial values of $\phi_i(k)$, m_k and s_k, $1\le k\le K$, $1\le i\le n$, maximum number of iteration N
Result: A variational distribution for each coordinate of μ_k and c_i, $1\le k\le K$, $1\le i\le n$.
for $p=1\to N$ **do**
  for $i=1\to n$ **do**
    Set
    $$\phi_i(k)\propto\omega_k\exp\left(m_kx_i-\frac{m_k^2+s_k}{2}\right).$$
    ;
  end
  for $k=1\to K$ **do**
    Set
    $$m_k=\frac{\sum_{i=1}^n\phi_i(k)x_i}{1/\sigma^2+\sum_{i=1}^n\phi_i(k)}\quad\text{and}\quad s_k=\frac{1}{1/\sigma^2+\sum_{i=1}^n\phi_i(k)}.$$
    ;
  end
end

Algorithm 5: A version of CAVI algorithm for a Bayesian mixture of Gaussian distributions.

To monitor the convergence, the ELBO is available in closed form: using the expectations above and the entropies of the Gaussian and categorical distributions,

$$\begin{aligned}\mathcal{L}(q(\cdot|x)) = \sum_{k=1}^K\Big\{-\frac{1}{2}\log(2\pi\sigma^2)-\frac{m_k^2+s_k}{2\sigma^2}+\frac{1}{2}\log(2\pi\mathrm{e}\,s_k)\Big\}\\ +\sum_{i=1}^n\sum_{k=1}^K\phi_i(k)\Big\{\log\omega_k-\frac{1}{2}\log(2\pi)-\frac{(x_i-m_k)^2+s_k}{2}-\log\phi_i(k)\Big\}.\end{aligned}$$

Figure 2.3 displays the result of Algorithm 5 on simulated data and the evolution of the ELBO, which is nondecreasing as expected.

Label switching.

The posterior distribution is invariant by any permutation of the labels $\{1,\dots,K\}$ (when the weights ω_k are equal), so that it has (at least) $K!$ symmetric modes and the posterior means of

the μ_k are all equal. Mean-field VI, being mode-seeking, selects one of these modes depending on the initialization: different initializations in Figure 2.3 reach the same ELBO with permuted components. This is harmless for clustering, but shows that the variational approximation describes one mode of the posterior and not the posterior itself. An MCMC sampler would face the opposite difficulty: it should visit all the modes, which rarely happens in practice.

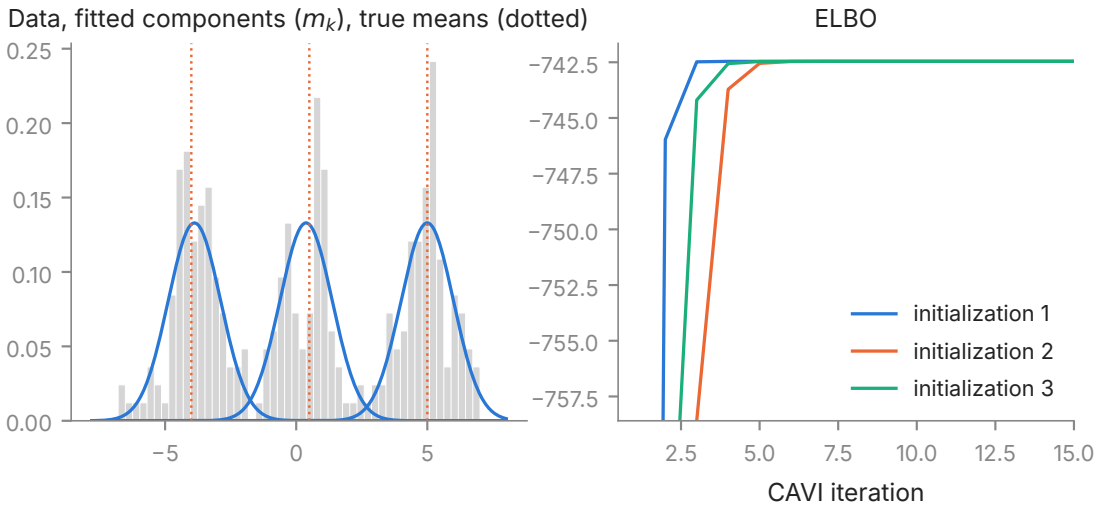


Fig. 2.3 CAVI for a mixture of $K = 3$ unit-variance Gaussian distributions ($n = 300$, true means -4, 0.5 and 5, $\sigma^2 = 10$, equal weights). Left: data and fitted components. Right: ELBO along the iterations for three random initializations, which converge (in a few iterations) to the same ELBO up to a permutation of the labels.

2.4 Variational autoencoders

Variational autoencoders (VAE) [Kingma and Welling, 2013, Rezende et al., 2014] are very popular approaches to introduce approximations of a target conditional distribution in the context of latent data models. Assume that $(X_1, \ldots, X_n)$ are i.i.d. random variables in X with unknown probability distribution π_{data}. We consider a family of joint probability densities $\{(z, x) \mapsto p_\theta(z, x)\}_{\theta \in \Theta}$ on $(\mathsf{Z} \times \mathsf{X}, \mathcal{Z} \otimes \mathcal{X})$ where Z is a latent variable and X is the observation. In this setting, we often write, for all $\theta \in \Theta$, $x \in \mathsf{X}$, $z \in \mathsf{Z}$,

$$p_\theta(z, x) = p_\theta(z) p_\theta(x|z).$$

The latent variable generative model defines a joint density $(z, x) \mapsto p_\theta(z, x)$ on $(\mathsf{Z} \times \mathsf{X}, \mathcal{Z} \otimes \mathcal{X})$ by specifying a prior $z \mapsto p_\theta(z)$ over the latent variable Z and a conditional density $x \mapsto p_\theta(x|z)$, also referred to as the decoder. Typically, $p_\theta(z)$ is a fixed standard Gaussian density and $p_\theta(x|z)$ is a Gaussian (or Bernoulli) density whose parameters are the output of a neural network with input z and weights θ. New data are generated by drawing $Z \sim p_\theta(\cdot)$ and then $X \sim p_\theta(\cdot|Z)$. The normalized log-likelihood is given by

$$\ell_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \log p_\theta(X_i) = \frac{1}{n} \sum_{i=1}^{n} \log \int p_\theta(z) p_\theta(X_i|z) \mathrm{d}z,$$

and the conditional distribution of the latent variable is $p_\theta(z|x) \propto p_\theta(z)p_\theta(x|z)$. In most cases, maximizing the average marginal log-likelihood of the data is not possible, as the marginal likelihood functions $p_\theta(X_i)$, $1 \leqslant i \leqslant n$, are not available explicitly since the integral used to marginalize the latent variable is intractable. Since a maximum likelihood estimator cannot be computed simply, VAEs introduce a variational approach which aims at simultaneously providing a parameter estimate and an approximation of the conditional distribution of the latent variable given the observation. Consider a family of conditional probability density functions $\{(z,x) \mapsto q_\varphi(z|x)\}_{\varphi\in\Phi}$, referred to as the encoder. Then, we can write, for all $\varphi \in \Phi, \theta \in \Theta$, $x \in \mathsf{X}$,

$$\begin{aligned}\log p_\theta(x) = \int \log p_\theta(x) q_\varphi(z|x) \mathrm{d}z &= \mathbb{E}_{q_\varphi(\cdot|x)}\left[\log p_\theta(x)\right] \\ &= \mathbb{E}_{q_\varphi(\cdot|x)}\left[\log \frac{p_\theta(Z,x)}{p_\theta(Z|x)}\right] \\ &= \mathbb{E}_{q_\varphi(\cdot|x)}\left[\log \frac{q_\varphi(Z|x)}{p_\theta(Z|x)}\right] + \mathbb{E}_{q_\varphi(\cdot|x)}\left[\log \frac{p_\theta(Z,x)}{q_\varphi(Z|x)}\right].\end{aligned}$$

The first term of the right-hand side is the Kullback–Leibler divergence between $q_\varphi(\cdot|x)$ and $p_\theta(\cdot|x)$, so that $\log p_\theta(x) \geq \mathcal{L}(\theta,\varphi;x)$, where

$$\mathcal{L}(\theta,\varphi;x) = \mathbb{E}_{q_\varphi(\cdot|x)}\left[\log \frac{p_\theta(Z,x)}{q_\varphi(Z|x)}\right]$$

is the evidence lower bound (ELBO).

In Section 2.2, a variational distribution is optimized separately for each observation (or dataset) x. In a VAE, the map $x \mapsto q_\varphi(\cdot|x)$ is a neural network shared by all observations: the cost of inference is *amortized*, since approximating $p_\theta(\cdot|x)$ for a new x only requires a forward pass of the encoder. The price to pay is an additional approximation error, the *amortization gap* [?]: $\mathcal{L}(\theta,\varphi;x)$ may be smaller than the best ELBO achievable by a distribution of the same parametric family optimized for x only. Figure 2.5 summarizes a Gaussian VAE architecture.

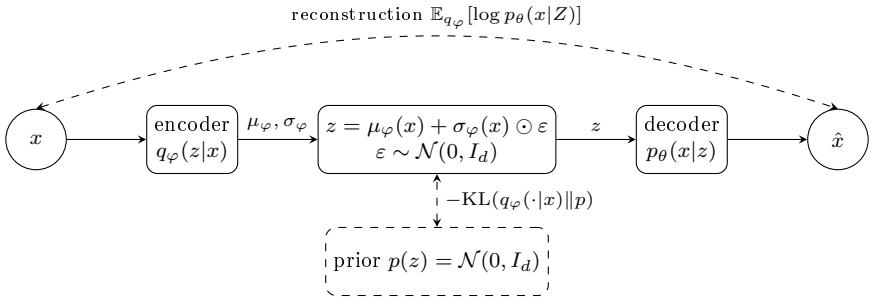


Fig. 2.4 Architecture of a Gaussian VAE. The encoder outputs the parameters of $q_\varphi(\cdot|x)$, a latent sample is obtained with the reparameterization $z = \mu_\varphi(x) + \sigma_\varphi(x) \odot \varepsilon$, and the decoder outputs the parameters of $p_\theta(\cdot|z)$. The two terms of the ELBO (2.2) are indicated with dashed arrows.

Example 2.5 (Gaussian VAE). Let $p(z) = \mathcal{N}(0, I_d)$, $p_\theta(x|z) = \mathcal{N}(\mu_\theta(z), \eta^2 I_m)$ and $q_\varphi(z|x) = \mathcal{N}(\mu_\varphi(x), \mathrm{diag}(\sigma_\varphi^2(x)))$, where μ_θ, μ_φ and σ_φ are neural networks. By (2.2),

$$\mathcal{L}(\theta,\varphi;x) = \mathbb{E}_{q_\varphi(\cdot|x)}[\log p_\theta(x|Z)] - \mathrm{KL}(q_\varphi(\cdot|x)\|\mathcal{N}(0, I_d)),$$

where the KL term is explicit:

$$\mathrm{KL}\big(\mathcal{N}(\mu, \mathrm{diag}(\sigma^2)) \| \mathcal{N}(0, I_d)\big) = \frac{1}{2}\sum_{j=1}^{d}\big(\mu_j^2 + \sigma_j^2 - 1 - \log\sigma_j^2\big),$$

and the reconstruction term is

$$\mathbb{E}_{q_\varphi(\cdot|x)}[\log p_\theta(x|Z)] = -\frac{1}{2\eta^2}\mathbb{E}_{q_\varphi(\cdot|x)}\big[\|x - \mu_\theta(Z)\|^2\big] - \frac{m}{2}\log(2\pi\eta^2).$$

Only the reconstruction term requires a Monte Carlo approximation. With one sample per observation, the negative ELBO is a squared reconstruction error penalized by the KL term; the ratio between the two terms is governed by η^2. For binary data, a Bernoulli decoder replaces the squared error by a cross-entropy.

Example 2.6 (Linear Gaussian model: probabilistic PCA). Let $p(z) = \mathcal{N}(0, I_d)$ and $p_\theta(x|z) = \mathcal{N}(Wz + b, \eta^2 I_m)$ with $\theta = (W, b, \eta^2)$, $W \in \mathbb{R}^{m\times d}$. Everything is explicit in this model [**?**]: $p_\theta(x) = \mathcal{N}(b, WW^\top + \eta^2 I_m)$, and, with $M = W^\top W + \eta^2 I_d$,

$$p_\theta(z|x) = \mathcal{N}\big(M^{-1}W^\top(x - b),\, \eta^2 M^{-1}\big).$$

Hence, for a Gaussian encoder whose mean is an affine function of x and whose covariance does not depend on x, the variational family contains the exact posterior, the ELBO is tight at the optimum, and maximizing it in (θ, φ) yields the maximum likelihood estimator, whose columns of W span the principal subspace of the empirical covariance matrix. A VAE can thus be seen as a nonlinear generalization of PCA, and this model is a useful test case to check an implementation, since $\log p_\theta(x)$ can be compared with the ELBO.

This motivates the introduction of the following loss function, the negative expected ELBO:

$$\mathcal{J}(\theta, \varphi) = \mathbb{E}_{\pi_{\mathrm{data}}}[-\mathcal{L}(\theta, \varphi; X)] = \mathbb{E}_{\pi_{\mathrm{data}}}\left[\mathbb{E}_{q_\varphi(\cdot|X)}\left[\log\frac{q_\varphi(Z|X)}{p_\theta(Z, X)}\right]\right].$$

The empirical loss is then given by

$$\mathcal{J}_n(\theta, \varphi) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}_{q_\varphi(\cdot|X_i)}\left[\log\frac{q_\varphi(Z|X_i)}{p_\theta(Z, X_i)}\right],$$

where $(X_1, \ldots, X_n)$ are i.i.d. with distribution π_{data}, and we aim at solving the optimization problem:

$$(\widehat{\theta}_n, \widehat{\varphi}_n) \in \mathrm{Argmin}_{\theta\in\Theta, \varphi\in\Phi}\mathcal{J}_n(\theta, \varphi). \tag{2.4}$$

Since $\mathcal{J}_n(\theta, \varphi) \geq -\ell_n(\theta)$, minimizing $\mathcal{J}_n$ amounts to maximizing a lower bound of the log-likelihood; the gap $n^{-1}\sum_i \mathrm{KL}(q_\varphi(\cdot|X_i)\|p_\theta(\cdot|X_i))$ also depends on θ, so that $\widehat{\theta}_n$ differs from the maximum likelihood estimator in general (the model is biased towards parameters for which the posterior is easy to approximate). The joint optimization of θ and φ is a complex problem both for practical and theoretical reasons, and many research works have been devoted to this problem in the past few years. In most cases, $\mathcal{J}_n(\theta, \varphi)$ cannot be computed explicitly since expectations under the variational distribution are not explicit. Therefore, $\mathcal{J}_n(\theta, \varphi)$ is replaced by a Monte Carlo estimate $\widehat{\mathcal{J}}_n(\theta, \varphi)$:

$$\widehat{\mathcal{J}}_n(\theta, \varphi) = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{M}\sum_{j=1}^{M}\log\frac{q_\varphi(Z_{i,j}|X_i)}{p_\theta(Z_{i,j}, X_i)},$$

where for all $1 \leqslant i \leqslant n$, $Z_{i,1}, \ldots, Z_{i,M}$ are i.i.d. with distribution $q_\varphi(\cdot|X_i)$.

The ELBO can be tightened with importance sampling (see Chapter 3). Let $w_{\theta,\varphi}(x, z) = p_\theta(z, x)/q_\varphi(z|x)$, so that $\mathbb{E}_{q_\varphi(\cdot|x)}[w_{\theta,\varphi}(x, Z)] = p_\theta(x)$. For $K \geq 1$, the importance weighted ELBO

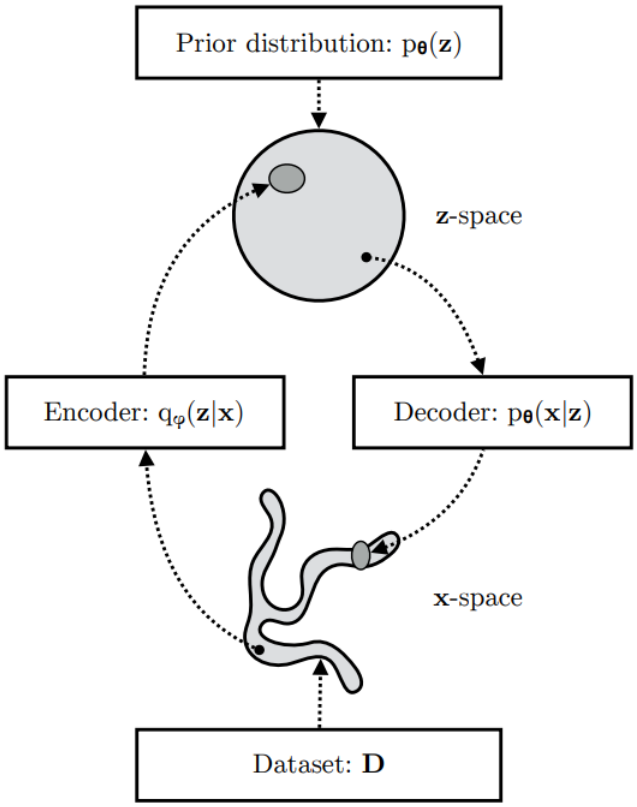


Fig. 2.5 An illustration of a VAE. From "An Introduction to Variational Autoencoders", Kingma et al., 2019.

[?] is

$$\mathcal{L}_K(\theta,\varphi;x) = \mathbb{E}\left[\log \frac{1}{K}\sum_{k=1}^{K} w_{\theta,\varphi}(x,Z_k)\right], \qquad Z_1,\dots,Z_K \overset{\text{i.i.d}}{\sim} q_\varphi(\cdot|x).$$

Proposition 2.7. *$\mathcal{L}_1(\theta,\varphi;x) = \mathcal{L}(\theta,\varphi;x)$ and, for all $K \geq 1$, $\mathcal{L}_K(\theta,\varphi;x) \leq \mathcal{L}_{K+1}(\theta,\varphi;x) \leq \log p_\theta(x)$.*

Proof. The first statement is the definition of the ELBO and the upper bound follows from Jensen's inequality, since the empirical mean of the weights is an unbiased estimator of $p_\theta(x)$. Write $w_k = w_{\theta,\varphi}(x,Z_k)$ and let I be uniformly distributed over the subsets of $\{1,\dots,K+1\}$ with K elements, independent of (Z_k). Since each index belongs to I with probability $K/(K+1)$,

$$\frac{1}{K+1}\sum_{k=1}^{K+1} w_k = \mathbb{E}_I\left[\frac{1}{K}\sum_{k\in I} w_k\right],$$

where $\mathbb{E}_I$ is the expectation with respect to I only. By concavity of the logarithm,

$$\log \frac{1}{K+1}\sum_{k=1}^{K+1} w_k \geq \mathbb{E}_I\left[\log \frac{1}{K}\sum_{k\in I} w_k\right],$$

and taking expectations, since $(w_k)_{k\in I}$ has the same distribution as $(w_1,\dots,w_K)$, $\mathcal{L}_{K+1} \geq \mathcal{L}_K$. □

If the weights are bounded, the strong law of large numbers and dominated convergence show that $\mathcal{L}_K(\theta,\varphi;x) \to \log p_\theta(x)$ as $K \to \infty$. The importance weighted autoencoder (IWAE) replaces

$\mathcal{L}$ by $\mathcal{L}_K$ in the loss; the encoder then plays the role of an importance (proposal) distribution rather than of an approximation of the posterior.

2.5 Gradient-based optimization

In order to solve (2.4), we need to compute the gradient of the loss function. Under classical regularity assumptions allowing to differentiate under the integral sign:

$$\nabla_\theta \mathcal{J}(\theta, \varphi) = -\mathbb{E}_{\pi_{\text{data}}}\left[\mathbb{E}_{q_\varphi(\cdot|X)}\left[\nabla_\theta \log p_\theta(Z, X)\right]\right].$$

This gradient can be estimated using samples from π_{data}. Given a mini-batch of i.i.d. observations $(X_1, \dots, X_B)$ with distribution π_{data} (in practice, drawn from the training set), an unbiased estimator of $\nabla_\theta \mathcal{J}(\theta, \varphi)$ can be computed as follows:

$$S_\theta(\theta, \varphi; \{X_i\}_{i=1}^B) = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{M}\sum_{j=1}^{M}\nabla_\theta \log p_\theta(Z_{i,j}, X_i),$$

where for all $1 \leqslant i \leqslant B$, $Z_{i,1}, \dots, Z_{i,M}$ are i.i.d. with distribution $q_\varphi(\cdot|X_i)$. Computing the gradient with respect to the variational parameter φ is more challenging since the distribution of the integration variable depends on φ: the gradient of an expectation under q_φ is not the expectation of the gradient. There are two common methods for computing this gradient.

2.5.1 Pathwise gradient

The reparameterization trick consists in expressing the variational distribution through a deterministic transform $g(\varepsilon, \varphi; x)$, where ε is an auxiliary random variable, independent of φ, drawn from a known probability density function p_ε: if $\varepsilon \sim p_\varepsilon$, then $g(\varepsilon, \varphi; x) \sim q_\varphi(\cdot|x)$. To simplify notation, we write $g(\varepsilon, \varphi)$ when x is fixed. Using this trick, the ELBO can be expressed as:

$$\mathcal{L}(\theta, \varphi; x) = \mathbb{E}_{p_\varepsilon}\left[\log w_{\theta,\varphi}(x, g(\varepsilon, \varphi))\right],$$

where $w_{\theta,\varphi}(x, z) = p_\theta(z, x)/q_\varphi(z|x)$ are the unnormalized importance weights and $\mathbb{E}_{p_\varepsilon}$ is the expectation when $\varepsilon \sim p_\varepsilon$. Now the integration variable ε does not depend on φ, and the chain rule yields the pathwise gradient [Kingma and Welling, 2013, Rezende et al., 2014] of the ELBO:

$$\nabla_\varphi \mathcal{L}(\theta, \varphi; x) = \mathbb{E}_{p_\varepsilon}\left[J_\varphi g(\varepsilon, \varphi)^\top \nabla_z \log w_{\theta,\varphi}(x, z)_{|z=g(\varepsilon,\varphi)}\right] - \mathbb{E}_{p_\varepsilon}\left[\nabla_\varphi \log q_\varphi(z|x)_{|z=g(\varepsilon,\varphi)}\right], \quad (2.5)$$

where $J_\varphi g$ is the Jacobian matrix of $\varphi \mapsto g(\varepsilon, \varphi)$ and, in the second term, the gradient is taken with respect to φ at fixed z (this term accounts for the explicit dependence of $w_{\theta,\varphi}$ on φ). The gradient with respect to φ of the loss can be estimated using samples from the dataset. Let $(X_1, \dots, X_B)$ be i.i.d. with distribution π_{data}. Then, an estimator of $\nabla_\varphi \mathcal{J}(\theta, \varphi)$ can be computed as follows:

$$\begin{aligned} &S_\varphi(\theta, \varphi; \{X_i\}_{i=1}^B) \\ &\quad = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{M}\sum_{j=1}^{M}\left\{J_\varphi g(\varepsilon_{i,j}, \varphi; X_i)^\top \nabla_z \log \frac{p_\theta(z, X_i)}{q_\varphi(z|X_i)}_{|z=Z_{i,j}} - \nabla_\varphi \log q_\varphi(z|X_i)_{|z=Z_{i,j}}\right\}, \end{aligned} \quad (2.6)$$

where, for all $1 \leq i \leq B$, $\varepsilon_{i,1}, \dots, \varepsilon_{i,M}$ are independent samples from p_ε and $Z_{i,j} = g(\varepsilon_{i,j}, \varphi; X_i)$. In practice, (2.6) is not coded by hand: it is obtained by automatic differentiation of the Monte Carlo estimate of the loss written as a function of (θ, φ) with the $\varepsilon_{i,j}$ held fixed.

Example 2.8 (Gaussian encoder). In a context of deep Gaussian models, $q_\varphi(\cdot|x)$ is a Gaussian probability density function with mean $\mu_\varphi(x) \in \mathbb{R}^d$ and covariance matrix $\mathrm{diag}(\sigma^2_\varphi(x)) \in \mathbb{R}^{d\times d}$ where $(\mu_\varphi(x), \sigma^2_\varphi(x))$ is the output of a neural network with input x. Then, if $z = \mu_\varphi(x) + \mathrm{diag}(\sigma_\varphi(x))\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_d)$, $z \sim q_\varphi(\cdot|x)$. Writing $g(\varepsilon, \varphi; x) = \mu_\varphi(x) + \sigma_\varphi(x) \odot \varepsilon$, where $\odot$ is the componentwise product, the Jacobian matrix of g with respect to ε is

$$J_\varepsilon g(\varepsilon, \varphi; x) = \mathrm{diag}(\sigma_\varphi(x)).$$

Therefore, by the change of variables formula, with $z = g(\varepsilon, \varphi; x)$,

$$\log q_\varphi(g(\varepsilon, \varphi; x)|x) = \log p_\varepsilon(\varepsilon) - \sum_{i=1}^{d} \log \sigma_{\varphi,i}(x).$$

The reparameterization trick applies more generally to location-scale families and to distributions which can be sampled by inverting their cumulative distribution function; it does not apply directly to discrete latent variables.

2.5.2 Score function gradient

lternatively, the score function gradient, also known as the REINFORCE gradient or likelihood ratio gradient [Glynn, 1990, Williams, 1992, Paisley et al., 2012], can be used. Unlike the pathwise gradient, this method does not require a reparameterization nor the differentiability of $z \mapsto \log p_\theta(z, x)$, and is applicable to a wider range of variational distributions, including discrete ones.

Proposition 2.9. *Under regularity conditions allowing to differentiate under the integral sign, for all $\theta \in \Theta$, $\varphi \in \Phi$, we have:*

$$\nabla_\varphi \mathcal{J}(\theta, \varphi) = -\mathbb{E}_{\pi_{\mathrm{data}}}\left[\mathbb{E}_{q_\varphi(\cdot|X)}\left[\log \frac{p_\theta(Z, X)}{q_\varphi(Z|X)} \nabla_\varphi \log q_\varphi(Z|X)\right]\right].$$

Proof. For all $x \in \mathsf{X}$, the gradient of the ELBO with respect to φ is given by:

$$\begin{aligned}
\nabla_\varphi \mathcal{L}(\theta, \varphi; x) &= \nabla_\varphi \mathbb{E}_{q_\varphi(\cdot|x)}\left[\log p_\theta(Z, x) - \log q_\varphi(Z|x)\right] \\
&= \nabla_\varphi \int (\log p_\theta(z, x) - \log q_\varphi(z|x)) q_\varphi(z|x)\, \mathrm{d}z \\
&= \int \nabla_\varphi \left[(\log p_\theta(z, x) - \log q_\varphi(z|x)) q_\varphi(z|x)\right] \mathrm{d}z \\
&= \mathbb{E}_{q_\varphi(\cdot|x)}\left[\nabla_\varphi \log q_\varphi(Z|x)(\log p_\theta(Z, x) - \log q_\varphi(Z|x))\right] - \mathbb{E}_{q_\varphi(\cdot|x)}\left[\nabla_\varphi \log q_\varphi(Z|x)\right],
\end{aligned}$$

where we used $\nabla_\varphi q_\varphi(z|x) = q_\varphi(z|x)\nabla_\varphi \log q_\varphi(z|x)$. Using the fact that $\mathbb{E}_{q_\varphi(\cdot|x)}\left[\nabla_\varphi \log q_\varphi(Z|x)\right] = \int \nabla_\varphi q_\varphi(z|x)\mathrm{d}z = \nabla_\varphi \int q_\varphi(z|x)\mathrm{d}z = 0$ under regularity conditions on $q_\varphi(z|x)$ yields

$$\begin{aligned}
\nabla_\varphi \mathcal{L}(\theta, \varphi; x) &= \mathbb{E}_{q_\varphi(\cdot|x)}\left[\nabla_\varphi \log q_\varphi(Z|x)(\log p_\theta(Z, x) - \log q_\varphi(Z|x))\right] \\
&= \mathbb{E}_{q_\varphi(\cdot|x)}\left[\log \frac{p_\theta(Z, x)}{q_\varphi(Z|x)} \nabla_\varphi \log q_\varphi(Z|x)\right].
\end{aligned}$$

The result follows by taking the expectation with respect to $X \sim \pi_{\mathrm{data}}$ and changing the sign. □

The gradient with respect to φ of the loss can be estimated using samples from the dataset. Let $(X_1, \ldots, X_B)$ be i.i.d. with distribution π_{data}. Then, an estimator of $\nabla_\varphi \mathcal{J}(\theta, \varphi)$ can be computed as follows:

$$S_\varphi(\theta, \varphi; \{X_i\}_{i=1}^B) = -\frac{1}{B}\sum_{i=1}^{B}\frac{1}{M}\sum_{j=1}^{M}\log\frac{p_\theta(Z_{i,j}, X_i)}{q_\varphi(Z_{i,j}|X_i)}\nabla_\varphi \log q_\varphi(Z_{i,j}|X_i), \tag{2.7}$$

where, for all $1 \leq i \leq B$, $Z_{i,1}, \ldots, Z_{i,M}$ are independent samples from $q_\varphi(\cdot|X_i)$.

The pathwise gradient estimator often yields lower-variance estimates than the score function estimator [Miller et al., 2017, Buchholz et al., 2018], but its variance can sometimes exceed that of the score function estimator, especially when the score function correlates with other components of the pathwise estimator. Several methods have been proposed to further reduce variance, such as the Rao-Blackwellization estimator [Ranganath et al., 2014], Control Variates [Liévin et al., 2020], Stop Gradient estimator [Roeder et al., 2017], Quasi-Monte Carlo VAE [Buchholz et al., 2018], and Multi-Level Monte Carlo estimator [Fujisawa and Sato, 2021, He et al., 2022]. While our analysis focuses on the convergence rate of score function and pathwise gradient estimators, our convergence results also apply to most of these other methods.

Chapter 3

Sequential Monte Carlo Methods

The use of Monte Carlo methods for non-linear filtering can be traced back to the pioneering work of [Handschin and Mayne, 1969]. These early approaches relied on sequential versions of importance sampling, where samples are generated from an instrumental (proposal) distribution and the target distribution is approximated by assigning appropriate importance weights to these samples. In the context of non-linear filtering, importance sampling can be implemented sequentially by carefully defining a sequence of proposal distributions. This makes it unnecessary to regenerate the entire set of samples whenever a new observation becomes available. This approach is known as sequential importance sampling (SIS).

Although SIS was already known in the early 1970s, its application to non-linear filtering problems remained limited for many years. This was partly due to insufficient computational power at the time, but also to a more fundamental issue that was not fully understood until later: *weight degeneracy*. As the number of iterations increases, most samples tend to receive very small normalized importance weights and therefore contribute negligibly to the approximation of the target distribution.

A major breakthrough came with the work of [Gordon et al., 1993], who proposed addressing this problem through resampling. Their approach rejuvenates the particle population by duplicating samples with high importance weights while eliminating those with low weights. This led to the development of the particle filter, which represented the first successful application of sequential Monte Carlo (SMC) methods to non-linear filtering. Resampling introduces its own difficulty, the loss of diversity of the particle population (*sample impoverishment*).

Since then, SMC methods have been widely adopted across many fields, including computer vision, signal processing, control, econometrics, robotics, and statistics. The purpose of this chapter is to review the fundamental components required to implement sequential Monte Carlo methods.

3.1 Importance sampling

Throughout this section, μ will denote a probability measure of interest on a measurable space $(\mathsf{X}, \mathcal{X})$, which we shall refer to as the target distribution, and ν an instrumental probability measure (note that the roles of these letters differ from Chapter 1). The aim is to approximate integrals of the form

$$\mu(f) = \int_{\mathsf{X}} f(x)\, \mu(\mathrm{d}x),$$

for real-valued measurable functions f. The Monte Carlo approach presented in the first part of the course consists in drawing an i.i.d. sample $(\xi_1, \ldots, \xi_N)$ from the probability measure μ and

then evaluating the sample mean $\sum_{i=1}^N f(\xi_i)/N$. Of course, this technique is applicable only when it is possible (and reasonably simple) to sample from the target distribution μ.

Importance sampling is based on the idea that it may be simpler or computationally cheaper to sample from an instrumental distribution ν, and then to apply a change of measure. Assume that the target probability measure μ is absolutely continuous with respect to an instrumental probability measure ν from which sampling is easily feasible, and write

$$w = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}$$

for the *importance weight function* (if μ and ν have densities, still denoted μ and ν, with respect to a common measure, $w = \mu/\nu$). Then, for any μ-integrable function f,

$$\mu(f) = \int f(x)\,\mu(\mathrm{d}x) = \int f(x)\,w(x)\,\nu(\mathrm{d}x) = \nu(fw).$$

In particular, if $(\xi_1, \dots, \xi_N)$ is an i.i.d. sample from ν, we can introduce the following (unnormalized) importance sampling estimator of $\mu(f)$:

$$\tilde{\mu}_N^{\mathrm{IS}}(f) = \frac{1}{N}\sum_{i=1}^N f(\xi_i)\,w(\xi_i),$$

which is an unbiased estimator of $\mu(f)$. In many practical situations, the target probability measure μ is only known up to a normalizing constant. This situation arises in particular when applying importance sampling techniques to hidden Markov models and, more generally, in Bayesian statistics. In such cases, the Radon–Nikodym derivative is only available up to a multiplicative constant: we can compute $\tilde{w} = c\,w$ for an unknown constant $c > 0$. Nevertheless, the importance sampling methodology can still be applied by resorting to the self-normalized importance sampling estimator, defined as

$$\hat{\mu}_N^{\mathrm{IS}}(f) = \frac{\sum_{i=1}^N f(\xi_i)\,w(\xi_i)}{\sum_{i=1}^N w(\xi_i)} = \frac{\sum_{i=1}^N f(\xi_i)\,\tilde{w}(\xi_i)}{\sum_{i=1}^N \tilde{w}(\xi_i)},$$

which does not depend on c. It is a weighted empirical mean with normalized weights $\tilde{w}(\xi_i)/\sum_j \tilde{w}(\xi_j)$, i.e. the integral of f with respect to the weighted empirical measure $\sum_{i=1}^N \{\tilde{w}(\xi_i)/\sum_j \tilde{w}(\xi_j)\}\delta_{\xi_i}$. As a by-product, $N^{-1}\sum_{i=1}^N \tilde{w}(\xi_i)$ is an unbiased estimator of the unknown constant $c = \nu(\tilde{w})$, which is the normalizing constant of the target when $\tilde{w}$ is computed from an unnormalized density.

Remark 3.1. Importance sampling has already appeared in this course: the independent Metropolis–Hastings sampler of Chapter 1 uses the same weight function $w = \pi/g$, the pseudo-marginal algorithm relies on unbiased importance sampling estimates of a marginal likelihood, and the importance weighted ELBO is the expectation of the logarithm of $\tilde{\mu}_N^{\mathrm{IS}}(1)$ computed with $\tilde{w} = p_\theta(\cdot, x)/q_\varphi(\cdot|x)$.

3.1.1 Convergence guarantees for the IS estimator

Since the importance sampling estimator can be written as a sample mean of independent random variables, a wide range of theoretical results is available to quantify the accuracy of $\tilde{\mu}_N^{\mathrm{IS}}(f)$ as an estimator of $\mu(f)$. Some of these results are asymptotic in nature, such as the law of large numbers (LLN) and the central limit theorem (CLT). It is also possible to derive non-asymptotic guarantees, including Berry–Esseen type bounds, bounds on the moments of the error $\mathbb{E}\big[\big|\tilde{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\big|^p\big]$ for $p > 0$, or bounds on tail probabilities.

Theorem 3.2. *Assume that $\mu \ll \nu$, let f be a real-valued measurable function such that $\mu(|f|) < \infty$, and let $(\xi_i)_{i\geq 1}$ be i.i.d. with distribution ν. Then,*

$$\lim_{N\to\infty} \tilde{\mu}_N^{\mathrm{IS}}(f) = \mu(f), \qquad \textit{almost surely.}$$

Assume in addition that

$$\nu(f^2w^2) = \int f^2(x)w^2(x)\nu(\mathrm{d}x) = \int f^2(x)w(x)\mu(\mathrm{d}x) < \infty. \tag{3.1}$$

Then,

$$\sqrt{N}\big(\tilde{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\big) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \mathrm{Var}_\nu(fw)), \qquad \textit{as } N \to \infty,$$

where

$$\mathrm{Var}_\nu(fw) = \int \big(f(x)w(x) - \mu(f)\big)^2 \nu(\mathrm{d}x) = \nu(f^2w^2) - \mu(f)^2.$$

Proof. The random variables $f(\xi_i)w(\xi_i)$, $i \geq 1$, are i.i.d. with $\nu(|f|w) = \mu(|f|) < \infty$ and mean $\nu(fw) = \mu(f)$. The two statements are the strong law of large numbers and the central limit theorem for these variables. □

It is enough to assume that μ is absolutely continuous with respect to ν on the set $\{f \neq 0\}$, i.e. that ν puts mass wherever $f\mu$ does. Although the importance sampling estimator is generic, its efficiency depends crucially on the interplay between the target distribution μ, the instrumental distribution ν, and the function f. In particular, for a given function f, it is possible to define an instrumental distribution ν that leads to a significantly lower variance than that obtained with the standard Monte Carlo approach, corresponding to the choice $\nu = \mu$.

Proposition 3.3 (Optimal instrumental distribution). *Let f be such that $0 < \mu(|f|) < \infty$ and define $\nu^\star(\mathrm{d}x) = |f(x)|\mu(\mathrm{d}x)/\mu(|f|)$. Then, for all instrumental distributions ν such that $\mu \ll \nu$,*

$$\mathrm{Var}_{\nu^\star}(fw^\star) = \mu(|f|)^2 - \mu(f)^2 \leq \mathrm{Var}_\nu(fw),$$

where $w^\star = \mathrm{d}\mu/\mathrm{d}\nu^\star$ on $\{f \neq 0\}$.

Proof. On $\{f \neq 0\}$, $fw^\star = \mathrm{sign}(f)\mu(|f|)$, so that $\nu^\star(f^2(w^\star)^2) = \mu(|f|)^2$. For any ν, by the Cauchy–Schwarz inequality, $\mu(|f|)^2 = \nu(|f|w)^2 \leq \nu(f^2w^2)$, and the result follows by subtracting $\mu(f)^2$ on both sides. □

The optimal distribution is not usable in practice ($\nu^\star$ involves the unknown quantity $\mu(|f|)$), but it provides a guideline: a good instrumental distribution should put mass where $|f|\mu$ is large, i.e. sample *more often* in the important regions. In particular, importance sampling can be much more efficient than crude Monte Carlo for rare events.

Example 3.4 (Gaussian tail probability). Let $\mu = \mathcal{N}(0,1)$ and $f = \mathbf{1}_{(a,\infty)}$, so that $\mu(f) = \mathbb{P}(X > a) = \bar{\Phi}(a)$, where $\bar{\Phi}$ is the Gaussian tail function. For $a = 4$, $\bar{\Phi}(4) \approx 3.17 \cdot 10^{-5}$: with crude Monte Carlo, the relative variance per sample is $(1 - \bar{\Phi}(a))/\bar{\Phi}(a) \approx 3.2 \cdot 10^4$, and about 10^7 samples are needed to get a relative error of 5%. With $\nu = \mathcal{N}(a,1)$, $w(x) = \mathrm{e}^{-ax+a^2/2}$ and

$$\nu(f^2w^2) = \int_a^\infty \mathrm{e}^{-ax+a^2/2}\varphi(x)\,\mathrm{d}x = \mathrm{e}^{a^2}\bar{\Phi}(2a),$$

where φ is the standard Gaussian density, so that the relative variance per sample is $\mathrm{e}^{a^2}\bar{\Phi}(2a)/\bar{\Phi}(a)^2 - 1 \approx 4.5$: importance sampling divides the number of samples needed by a factor of about 7 000. The instrumental distribution is centered at the boundary of the rare event, which mimics the shape of $\nu^\star \propto \mathbf{1}_{(a,\infty)}\varphi$.

We may also use inequalities on the control of moments of sums of independent random variables to provide non-asymptotic bound for the importance sampling estimator.

Theorem 3.5 (Marcinkiewicz–Zygmund inequality). *Let $(X_1, \dots, X_n)$ be independent random variables and let $p \geq 2$. Then,*

$$\mathbb{E}\left[\left|\sum_{i=1}^{n}\big(X_i - \mathbb{E}[X_i]\big)\right|^p\right] \leq C_p\, n^{p/2-1} \sum_{i=1}^{n} \mathbb{E}[|X_i - \mathbb{E}[X_i]|^p], \tag{3.2}$$

for some positive constant C_p depending only on p.

Controlling tail probabilities is often of utmost interest. This topic has also been extensively studied, and a canonical result in this area is Hoeffding's inequality.

Theorem 3.6 (Hoeffding inequality). *Let $(X_1, \dots, X_n)$ be independent bounded random variables such that $\mathbb{P}(a_i \leq X_i \leq b_i) = 1$ for all $i = 1, \dots, n$. Then, for any $t \geq 0$,*

$$\mathbb{P}\left(\sum_{i=1}^{n}\big(X_i - \mathbb{E}[X_i]\big) \geq t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right),$$

and

$$\mathbb{P}\left(\sum_{i=1}^{n}\big(X_i - \mathbb{E}[X_i]\big) \leq -t\right) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

These inequalities make it possible to derive non-asymptotic bounds on both moments and tail probabilities of importance sampling estimators.

Theorem 3.7. *Let $p \geq 2$, $N \geq 1$, and let f be such that $\nu(|fw|^p) < \infty$. Then,*

$$\mathbb{E}\ \left[\left|\tilde{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\right|^p\right]\ \leq\ C_p\, N^{-p/2}\, \nu\big(|fw - \mu(f)|^p\big),$$

where $C_p < \infty$ depends only on p. Moreover, if $\|fw\|_\infty < \infty$, for any $t \geq 0$,

$$\mathbb{P}\Big(\left|\tilde{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\right| \geq t\Big)\ \leq\ 2\exp\left(-\frac{Nt^2}{2\|fw\|_\infty^2}\right).$$

Proof. Apply Theorem 3.5 to $X_i = f(\xi_i)w(\xi_i)$, which are i.i.d. with mean $\mu(f)$: $\mathbb{E}[|\sum_{i=1}^{N}(X_i - \mu(f))|^p] \leq C_p N^{p/2}\nu(|fw - \mu(f)|^p)$, and divide by N^p. For the tail bound, apply Theorem 3.6 with $a_i = -\|fw\|_\infty$, $b_i = \|fw\|_\infty$ and t replaced by Nt, to both deviations. □

3.1.2 Convergence guarantees for the self-normalized IS estimator

Theorem 3.8. *Let f be a measurable function such that $\mu(|f|) < \infty$, and assume that $\mu \ll \nu$. Let $(\xi_i)_{i\geq 1}$ be i.i.d. with distribution ν. Then,*

$$\hat{\mu}_N^{\mathrm{IS}}(f)\ \xrightarrow{a.s.}\ \mu(f), \qquad \text{as } N \to \infty.$$

Assume in addition that

$$\int \big[1 + f(x)^2\big] w(x)^2 \nu(\mathrm{d}x) < \infty. \tag{3.3}$$

Then,

$$\sqrt{N}\left(\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\right)\ \xrightarrow{\mathcal{D}}\ \mathcal{N}\big(0, \sigma^2(\nu, f)\big), \qquad \text{as } N \to \infty,$$

where

$$\sigma^2(\nu, f) = \int w(x)^2 \big(f(x) - \mu(f)\big)^2 \nu(\mathrm{d}x). \tag{3.4}$$

Proof. Note that

$$\hat{\mu}_N^{\mathrm{IS}}(f) = \frac{N^{-1}\sum_{i=1}^N f(\xi_i)\, w(\xi_i)}{N^{-1}\sum_{i=1}^N w(\xi_i)}.$$

Therefore, the first result is a direct consequence of the strong law of large numbers:

$$\frac{1}{N}\sum_{i=1}^N f(\xi_i)\, w(\xi_i) \xrightarrow{\text{a.s.}} \mu(f), \qquad \frac{1}{N}\sum_{i=1}^N w(\xi_i) \xrightarrow{\text{a.s.}} \nu(w) = 1.$$

For the central limit theorem, note that

$$\sqrt{N}\left(\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\right) = \frac{N^{-1/2}\sum_{i=1}^N w(\xi_i)\big(f(\xi_i) - \mu(f)\big)}{N^{-1}\sum_{i=1}^N w(\xi_i)}.$$

The random variables $w(\xi_i)(f(\xi_i)-\mu(f))$ are i.i.d., centered (since $\nu(w(f-\mu(f))) = \mu(f)-\mu(f) = 0$) and have variance $\sigma^2(\nu, f) < \infty$ by (3.3). By the central limit theorem, the numerator converges in distribution to $\mathcal{N}(0, \sigma^2(\nu, f))$, while the denominator converges almost surely to 1. The proof is then completed with Slutsky's lemma. □

Note that $\sigma^2(\nu, f)$ is the variance of the unnormalized estimator applied to the centered function $f - \mu(f)$: self-normalization removes the error due to the fluctuations of $N^{-1}\sum_i w(\xi_i)$ around 1 only to first order. Unlike $\tilde{\mu}_N^{\mathrm{IS}}(f)$, $\hat{\mu}_N^{\mathrm{IS}}(f)$ is biased; a second-order expansion (delta method) shows that, under moment conditions, $\mathbb{E}[\hat{\mu}_N^{\mathrm{IS}}(f)] - \mu(f) = -N^{-1}\nu\big(w^2(f-\mu(f))\big) + o(N^{-1})$, which is negligible compared with the standard deviation of order $N^{-1/2}$.

Theorem 3.9. *Assume that $\nu(w^p) < \infty$ for some $p \geq 2$. Then, there exists a constant $C_p < \infty$ depending only on p such that, for any $N \geq 1$ and any bounded measurable function f,*

$$\mathbb{E}\left[\left|\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\right|^p\right] \;\leq\; C_p\, \|f\|_\infty^p\, \nu(w^p)\, N^{-p/2}. \tag{3.5}$$

In particular, for $p = 2$, $\mathbb{E}[|\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)|^2] \leq 4\|f\|_\infty^2 \nu(w^2)/N$. Moreover, if $\|w\|_\infty < \infty$, for all $t \geq 0$,

$$\mathbb{P}\Big(\left|\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)\right| \geq t\Big) \;\leq\; 4\exp\left(-\frac{Nt^2}{8\|f\|_\infty^2\|w\|_\infty^2}\right). \tag{3.6}$$

Proof. Write $A_N = N^{-1}\sum_{i=1}^N f(\xi_i)w(\xi_i)$ and $B_N = N^{-1}\sum_{i=1}^N w(\xi_i)$, so that $\hat{\mu}_N^{\mathrm{IS}}(f) = A_N/B_N$ and $|A_N/B_N| \leq \|f\|_\infty$. Then

$$\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f) = \frac{A_N}{B_N}(1-B_N) + (A_N - \mu(f)), \quad \text{so that} \quad |\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)| \leq \|f\|_\infty |B_N - 1| + |A_N - \mu(f)|.$$

For $p = 2$, $(a+b)^2 \leq 2a^2 + 2b^2$ yields $\mathbb{E}[|\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)|^2] \leq 2\|f\|_\infty^2 \mathrm{Var}_\nu(w)/N + 2\mathrm{Var}_\nu(fw)/N \leq 4\|f\|_\infty^2\nu(w^2)/N$. For general p, the same decomposition, $(a+b)^p \leq 2^{p-1}(a^p + b^p)$ and Theorem 3.7 applied to 1 and f give the result, since $\nu(|w-1|^p) \leq 2^p\nu(w^p)$ and $\nu(|fw - \mu(f)|^p) \leq 2^p\|f\|_\infty^p\nu(w^p)$ (we used $\nu(w^p) \geq \nu(w)^p = 1$). For the tail bound, $\mathbb{P}(|\hat{\mu}_N^{\mathrm{IS}}(f) - \mu(f)| \geq t) \leq \mathbb{P}(|B_N - 1| \geq t/(2\|f\|_\infty)) + \mathbb{P}(|A_N - \mu(f)| \geq t/2)$, and both terms are bounded with Hoeffding's inequality, since $w(\xi_i) \in [0, \|w\|_\infty]$ and $f(\xi_i)w(\xi_i) \in [-\|f\|_\infty\|w\|_\infty, \|f\|_\infty\|w\|_\infty]$. □

3.2 Sequential importance sampling

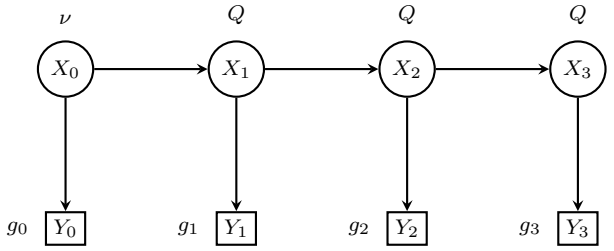
3.2.1 Hidden Markov models

A hidden Markov model (HMM) is defined as a bivariate process $\{(X_k, Y_k)\}_{k\geq 0}$ such that:

- $\{X_k\}_{k\geq 0}$ is a Markov chain with transition kernel Q and initial distribution ν;
- conditionally on the state process $\{X_k\}_{k\geq 0}$, the observations $\{Y_k\}_{k\geq 0}$ are independent, and for each n, the conditional distribution of Y_n depends only on X_n.

An alternative definition can be given as follows: an HMM is a bivariate Markov chain that is only partially observed, and whose transition kernel possesses a special structure: it is such that both the joint process $\{(X_k, Y_k)\}_{k\geq 0}$ and the marginal, hidden chain $\{X_k\}_{k\geq 0}$ are Markovian. A graphical representation of an HMM is given in Figure 3.2.1.

Fig. 3.1 A graphical model describing the joint distribution of a hidden Markov model.



Definition 3.10 (Hidden Markov Model). Let $(\mathsf{X}, \mathcal{X})$ and $(\mathsf{Y}, \mathcal{Y})$ be two measurable spaces, and let Q and G denote, respectively, a Markov transition kernel on $(\mathsf{X}, \mathcal{X})$ and a Markov kernel from $(\mathsf{X}, \mathcal{X})$ to $(\mathsf{Y}, \mathcal{Y})$. Define a Markov transition kernel T on the product space $(\mathsf{X}\times\mathsf{Y}, \mathcal{X}\otimes\mathcal{Y})$ as follows: for all bounded measurable functions f on $\mathsf{X}\times\mathsf{Y}$, and all $(x, y) \in \mathsf{X}\times\mathsf{Y}$,

$$Tf(x,y) \;:=\; \int Q(x, \mathrm{d}x')\, G(x', \mathrm{d}y') f(x', y'). \tag{3.7}$$

Then, a Markov chain $\{(X_k, Y_k)\}_{k\geq 0}$ with transition kernel T and initial distribution $\nu(\mathrm{d}x)G(x, \mathrm{d}y)$ is called a *hidden Markov model*.

Note that $Tf(x, y)$ does not depend on y: the next state only depends on the current state. We assume that there exists a σ-finite measure λ on $(\mathsf{Y}, \mathcal{Y})$ such that, for all $x \in \mathsf{X}$, $G(x, \cdot)$ is absolutely continuous with respect to λ, i.e., $G(x, \cdot) \ll \lambda(\cdot)$, with transition density function $g(x, \cdot)$. We also assume that there exists a σ-finite measure μ on $(\mathsf{X}, \mathcal{X})$ such that for all $x \in \mathsf{X}$, $Q(x, \cdot) \ll \mu(\cdot)$ with transition density function $q(x, \cdot)$ (the model is then said to be *fully dominated*). The joint Markov transition kernel T is then dominated by the product measure $\mu \otimes \lambda$ and admits the transition density function

$$t((x, y), (x', y')) \;:=\; q(x, x')\, g(x', y').$$

Proposition 3.11. *Let $\{(X_k, Y_k)\}_{k\geq 0}$ be a Markov chain on the product space $\mathsf{X}\times\mathsf{Y}$ with transition kernel T defined by (3.7) and initial distribution $\nu(\mathrm{d}x)G(x, \mathrm{d}y)$. Then, for any integer p, any ordered set of indices $k_1 < \cdots < k_p$, and any bounded measurable functions $f_1, \ldots, f_p$,*

$$\mathbb{E}\left[\prod_{j=1}^{p} f_j(Y_{k_j}) \,\middle|\, X_{k_1}, \ldots, X_{k_p}\right] \;=\; \prod_{j=1}^{p} \int_{\mathsf{Y}} f_j(y)\, G(X_{k_j}, \mathrm{d}y). \tag{3.8}$$

Proof. For all bounded measurable functions h, writing $n = k_p$, we have, by Proposition 1.6 of Chapter 1 applied to the chain $\{(X_k, Y_k)\}_{k\geq 0}$,

$$\begin{aligned}\mathbb{E}\left[\prod_{j=1}^{p} f_j(Y_{k_j})\, h(X_{k_1},\dots,X_{k_p})\right] &= \int \nu(\mathrm{d}x_0)G(x_0,\mathrm{d}y_0)\prod_{i=1}^{n} Q(x_{i-1},\mathrm{d}x_i)G(x_i,\mathrm{d}y_i)\\ &\quad\times\left[\prod_{j=1}^{p} f_j(y_{k_j})\right] h(x_{k_1},\dots,x_{k_p})\\ &= \int \nu(\mathrm{d}x_0)\prod_{i=1}^{n} Q(x_{i-1},\mathrm{d}x_i)\, h(x_{k_1},\dots,x_{k_p})\prod_{j=1}^{p}\int_{\mathsf{Y}} f_j(y)G(x_{k_j},\mathrm{d}y)\\ &= \mathbb{E}\left[h(X_{k_1},\dots,X_{k_p})\prod_{j=1}^{p}\int_{\mathsf{Y}} f_j(y)G(X_{k_j},\mathrm{d}y)\right],\end{aligned}$$

where we integrated with respect to the y_i (using $\int G(x_i,\mathrm{d}y_i)=1$ for $i\notin\{k_1,\dots,k_p\}$). Since the product in the last expectation is a measurable function of $(X_{k_1},\dots,X_{k_p})$, this identity, valid for all h, characterizes the conditional expectation, which completes the proof. □

The joint probability of the unobservable states and the observations up to index n is given as follows. For any bounded measurable function f,

$$\begin{aligned}\mathbb{E}\big[f(X_0,Y_0,\dots,X_n,Y_n)\big] &= \int f(x_0,y_0,\dots,x_n,y_n)\,\nu(\mathrm{d}x_0)g(x_0,y_0)\\ &\quad\times\prod_{k=1}^{n} Q(x_{k-1},\mathrm{d}x_k)g(x_k,y_k)\,\lambda^{\otimes(n+1)}(\mathrm{d}y_0,\dots,\mathrm{d}y_n),\end{aligned}\tag{3.9}$$

where $\lambda^{\otimes(n+1)}$ denotes the $(n+1)$-fold product measure on $(\mathsf{Y}^{n+1},\mathcal{Y}^{\otimes(n+1)})$. Marginalizing over the unobservable states $X_0,\dots,X_n$ yields the distribution of the observations alone:

$$\mathbb{E}\big[f(Y_0,\dots,Y_n)\big] = \int f(y_0,\dots,y_n)\,L_n(y_0,\dots,y_n)\,\lambda^{\otimes(n+1)}(\mathrm{d}y_0,\dots,\mathrm{d}y_n),\tag{3.10}$$

where L_n is defined below and plays a central role in likelihood-based inference.

Definition 3.12 (Likelihood). The *likelihood* of the observations $(Y_0,\dots,Y_n)$ is the probability density function with respect to $\lambda^{\otimes(n+1)}$, given, for all $(y_0,\dots,y_n)\in\mathsf{Y}^{n+1}$, by

$$L_n(y_0,\dots,y_n) := \int \nu(\mathrm{d}x_0)g(x_0,y_0)\prod_{k=1}^{n} Q(x_{k-1},\mathrm{d}x_k)g(x_k,y_k).\tag{3.11}$$

The *log-likelihood function* is defined, dropping the dependency on $(y_0,\dots,y_n)$, as

$$\ell_n \;:=\; \log L_n.\tag{3.12}$$

Example 3.13 (Three classical HMMs).

i) *Finite state space.* If $\mathsf{X}=\{1,\dots,K\}$, the integral (3.11) is a sum over K^{n+1} paths, but it can be computed exactly with $\mathcal{O}(nK^2)$ operations with the forward recursion below, where integrals become finite sums. This is the setting of speech recognition or of the segmentation of genomic sequences.

ii) *Linear Gaussian state space model.* Let $\mathsf{X}=\mathsf{Y}=\mathbb{R}$ and

$$X_k = aX_{k-1}+\sigma_x U_k,\qquad Y_k = X_k+\sigma_y V_k,$$

where $(U_k)_{k\ge1}$ and $(V_k)_{k\ge0}$ are independent i.i.d. $\mathcal{N}(0,1)$ sequences, independent of $X_0\sim\mathcal{N}(0,\sigma_x^2/(1-a^2))$, $|a|<1$. All the conditional distributions are Gaussian (we denote by φ_{m,s^2} the Gaussian density with mean m and variance s^2) and the filtering distributions

$\phi_k = \mathcal{N}(m_k, P_k)$ are given by the *Kalman filter*: with $m^\star_{-1} = 0$, $P^\star_{-1} = \sigma_x^2/(1-a^2)$ (prediction at time 0) and, for $k \geq 1$, $m^\star_{k-1} = a m_{k-1}$, $P^\star_{k-1} = a^2 P_{k-1} + \sigma_x^2$,

$$K_k = \frac{P^\star_{k-1}}{P^\star_{k-1} + \sigma_y^2}, \qquad m_k = m^\star_{k-1} + K_k(Y_k - m^\star_{k-1}), \qquad P_k = (1 - K_k)P^\star_{k-1}.$$

Moreover, the predictive distribution of Y_k given $Y_{0:k-1}$ is $\mathcal{N}(m^\star_{k-1}, P^\star_{k-1} + \sigma_y^2)$, so that the likelihood is explicit: $\ell_n = \sum_{k=0}^n \log \varphi_{m^\star_{k-1}, P^\star_{k-1}+\sigma_y^2}(Y_k)$. This model is used below as a benchmark for particle methods.

The likelihood function given in Definition 3.12 is not available in closed form in general. Therefore, in a setting where the law of the HMM depends on an unknown parameter $\theta \in \Theta$, and in the fully dominated case, the log-likelihood

$$\ell_n(\theta) := \log \int \nu_\theta(x_0) g_\theta(x_0, y_0) \prod_{k=1}^n q_\theta(x_{k-1}, x_k) g_\theta(x_k, y_k) \mu^{\otimes(n+1)}(\mathrm{d}x_{0:n}),$$

where ν_θ also denotes the density of the initial distribution with respect to μ, cannot be optimized easily. A natural solution is to estimate θ using the EM algorithm. Therefore, at iteration $p \geq 0$ with a current parameter estimate $\widehat{\theta}_p$, it is required to compute the intermediate quantity:

$$\begin{aligned}\mathcal{Q}(\theta, \widehat{\theta}_p) &= \mathbb{E}_{\widehat{\theta}_p}\left[\log p_\theta(X_{0:n}, Y_{0:n}) | Y_{0:n}\right] \\ &= \mathbb{E}_{\widehat{\theta}_p}\left[\log\{\nu_\theta(X_0) g_\theta(X_0, Y_0)\} | Y_{0:n}\right] + \sum_{k=1}^n \mathbb{E}_{\widehat{\theta}_p}\left[\log\{q_\theta(X_{k-1}, X_k) g_\theta(X_k, Y_k)\} | Y_{0:n}\right],\end{aligned}$$

where p_θ denotes the joint density of $(X_{0:n}, Y_{0:n})$ (we write $\mathcal{Q}$ to avoid any confusion with the kernel Q). This means that we need to sample (at least approximately) from the conditional distributions of (X_{k-1}, X_k) given $Y_{0:n}$ for a given parameter value $\widehat{\theta}_p$. This is in general not possible exactly and an appealing solution is to use sequential Monte Carlo methods. In the next subsection we present these sequential methods, omitting the dependency on $\widehat{\theta}_p$ for simplicity.

3.2.2 Sequential Monte Carlo methods for HMMs

We now specialize the sampling techniques introduced above to hidden Markov models. As in the previous section, we consider the HMM where Q denotes the Markov transition kernel of the hidden chain, ν is the distribution of the initial state X_0, and $g(x, y)$, for $x \in \mathsf{X}$ and $y \in \mathsf{Y}$, denotes the conditional density of the observation given the state, with respect to the measure λ on $(\mathsf{Y}, \mathcal{Y})$. The observations $Y_{0:n}$ are fixed, and, to simplify the notation, we will also use the shorthand

$$g_k(\cdot) := g(\cdot, Y_k).$$

In the following, we consider the following distributions.

- For all $0 \leq k \leq n$, ϕ_k is the conditional distribution of X_k given $Y_{0:k}$, and is referred to as the filtering distribution at time k.
- For all $0 \leq k \leq \ell \leq n$, $\phi_{k:\ell|n}$ is the conditional distribution of $X_{k:\ell}$ given $Y_{0:n}$, and is referred to as the joint smoothing distribution of $X_{k:\ell}$ given $Y_{0:n}$.
- For all $0 \leq k \leq n$, $\phi_{k|n} = \phi_{k:k|n}$ is the marginal smoothing distribution at time k.

With a slight abuse of notation, we use the same letters for these distributions and for their densities with respect to μ or $\mu^{\otimes(\ell-k+1)}$; ν also denotes the density of the initial distribution with respect to μ. Note that for all bounded measurable functions f,

$$\phi_0(f) = \frac{\int f(x_0)\, g_0(x_0)\, \nu(\mathrm{d}x_0)}{\int g_0(x_0)\, \nu(\mathrm{d}x_0)}.$$

Therefore, $\phi_0(f)$ can be estimated using self-normalized importance sampling. Sample $(\xi_0^1, \ldots, \xi_0^N)$ i.i.d. with probability density χ with respect to μ, associate each ξ_0^i with the importance weight $\omega_0^i = \nu(\xi_0^i) g_0(\xi_0^i)/\chi(\xi_0^i)$ and compute

$$\phi_0^N(f) = \sum_{i=1}^N \frac{\omega_0^i}{\sum_{j=1}^N \omega_0^j} f(\xi_0^i).$$

If $\chi = \nu$, the weights are simply $\omega_0^i = g_0(\xi_0^i)$. In addition, for all $n \geq 0$ and all bounded measurable functions f,

$$\phi_{0:n|n}(f) = \frac{\int \nu(x_0) g_0(x_0) \prod_{k=1}^n q(x_{k-1}, x_k) g_k(x_k) f(x_{0:n}) \mu^{\otimes(n+1)}(\mathrm{d}x_{0:n})}{\int \nu(x_0) g_0(x_0) \prod_{k=1}^n q(x_{k-1}, x_k) g_k(x_k) \mu^{\otimes(n+1)}(\mathrm{d}x_{0:n})}.$$

Note that the denominator is the likelihood $L_n(Y_{0:n})$. Therefore, it is straightforward to note that for all $0 \leq k \leq n-1$,

$$\phi_{0:k+1|k+1}(x_{0:k+1}) \propto \phi_{0:k|k}(x_{0:k}) q(x_k, x_{k+1}) g_{k+1}(x_{k+1}).$$

Integrating with respect to $x_{0:k-1}$, and since ϕ_k is the marginal of $\phi_{0:k|k}$ at time k, for all bounded measurable functions f,

$$\phi_{k+1}(f) = \frac{\int \phi_k(x_k) q(x_k, x_{k+1}) g_{k+1}(x_{k+1}) f(x_{k+1}) \mu(\mathrm{d}x_k) \mu(\mathrm{d}x_{k+1})}{\int \phi_k(x_k) q(x_k, x_{k+1}) g_{k+1}(x_{k+1}) \mu(\mathrm{d}x_k) \mu(\mathrm{d}x_{k+1})}. \tag{3.13}$$

This is the *filtering recursion*: a prediction step $\phi_k \mapsto \phi_k Q$ (the distribution of X_{k+1} given $Y_{0:k}$), followed by a correction (Bayes) step which multiplies by the likelihood g_{k+1} of the new observation and normalizes. In a finite state space, it can be computed exactly; for the linear Gaussian model, it yields the Kalman filter of Example 3.13. As these integrals are intractable in general, if at time k we have access to a weighted sample $\{(\omega_k^i, \xi_k^i)\}_{1 \leq i \leq N}$ targeting ϕ_k, we may estimate ϕ_{k+1} by

$$\tilde{\phi}_{k+1}^N(x_{k+1}) \propto \sum_{i=1}^N \omega_k^i q(\xi_k^i, x_{k+1}) g_{k+1}(x_{k+1}).$$

This approximation is a mixture distribution, which suggests that an importance sampling estimator of $\phi_{k+1}(f)$ can be obtained as follows.

- For all $1 \leq i \leq N$, sample I_{k+1}^i in $\{1, \ldots, N\}$ with probabilities proportional to $\{\omega_k^j\}_{1 \leq j \leq N}$.
- For all $1 \leq i \leq N$, sample $\xi_{k+1}^i \sim P(\xi_k^{I_{k+1}^i}, \cdot)$ where P is a proposal kernel with density p with respect to μ.
- For all $1 \leq i \leq N$, compute

$$\omega_{k+1}^i = \frac{q(\xi_k^{I_{k+1}^i}, \xi_{k+1}^i) g_{k+1}(\xi_{k+1}^i)}{p(\xi_k^{I_{k+1}^i}, \xi_{k+1}^i)}.$$

Then, $\phi_{k+1}(f)$ is estimated by

$$\phi_{k+1}^N(f) = \sum_{i=1}^N \frac{\omega_{k+1}^i}{\sum_{j=1}^N \omega_{k+1}^j} f(\xi_{k+1}^i).$$

The first step is a *resampling* (or selection) step, the second one a *mutation* (or propagation) step and the third one a *weighting* step. The variables I_{k+1}^i are the indices of the *ancestors* of the particles at time $k+1$. This algorithm is known as sequential importance sampling with resampling (SISR), or particle filter; when $P = Q$, the weights reduce to $\omega_{k+1}^i = g_{k+1}(\xi_{k+1}^i)$ and we obtain the *bootstrap filter* of [Gordon et al., 1993].

Data: Observations $Y_{0:n}$, number of particles N, initial instrumental density χ, proposal kernel P with density p.

Result: Weighted samples $\{(\omega_k^i, \xi_k^i)\}_{1 \le i \le N}$ targeting ϕ_k, $0 \le k \le n$, and an estimate $\widehat{L}_n$ of the likelihood.

For $1 \le i \le N$, draw $\xi_0^i \sim \chi$ and set $\omega_0^i = \nu(\xi_0^i) g_0(\xi_0^i)/\chi(\xi_0^i)$;

Set $\widehat{L}_0 = N^{-1} \sum_{i=1}^N \omega_0^i$;

for $k = 0 \to n-1$ **do**

Resampling: for $1 \le i \le N$, draw I_{k+1}^i with $\mathbb{P}(I_{k+1}^i = j) = \omega_k^j / \sum_{\ell=1}^N \omega_k^\ell$;

Mutation: for $1 \le i \le N$, draw $\xi_{k+1}^i \sim P(\xi_k^{I_{k+1}^i}, \cdot)$;

Weighting: for $1 \le i \le N$, set $\omega_{k+1}^i = q(\xi_k^{I_{k+1}^i}, \xi_{k+1}^i) g_{k+1}(\xi_{k+1}^i) / p(\xi_k^{I_{k+1}^i}, \xi_{k+1}^i)$;

Set $\widehat{L}_{k+1} = \widehat{L}_k \times N^{-1} \sum_{i=1}^N \omega_{k+1}^i$;

end

Algorithm 6: Particle filter (sequential importance sampling with resampling).

Without the resampling step, i.e. setting $I_{k+1}^i = i$ and multiplying the weights, $\omega_{k+1}^i = \omega_k^i \, q(\xi_k^i, \xi_{k+1}^i) g_{k+1}(\xi_{k+1}^i) / p(\xi_k^i, \xi_{k+1}^i)$, the algorithm is plain importance sampling on the path space X^{n+1} targeting $\phi_{0:n|n}$, with instrumental density $\chi(x_0) \prod_{k=1}^n p(x_{k-1}, x_k)$: this is *sequential importance sampling* (SIS). The weights of the paths are products of $n+1$ factors, and their variance typically grows exponentially with n: after a few iterations, one particle carries almost all the weight. Figure 3.2 illustrates this phenomenon on the linear Gaussian model of Example 3.13: the ESS of SIS drops to 1 within 20 time steps and its estimate of the filtering mean departs from the exact Kalman filter, whereas the ESS of the bootstrap filter remains of the order of N. Resampling discards the particles with negligible weights and duplicates the promising ones, so that the computational effort is focused on the relevant regions of the state space at each time step.

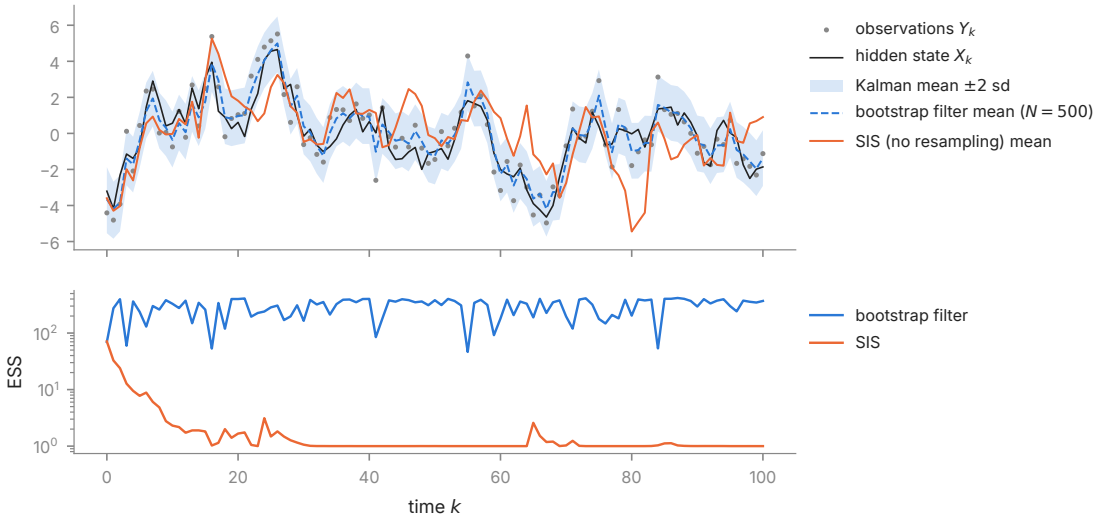


Fig. 3.2 Linear Gaussian state space model of Example 3.13 with $a = 0.9$, $\sigma_x = \sigma_y = 1$. Top: hidden states, observations, exact filtering mean ±2 standard deviations (Kalman filter), and estimates of the filtering mean by the bootstrap filter and by SIS with $N = 500$ particles. Bottom: effective sample size along time (log scale).

Appendix A

M-estimation Z-estimation, maximum likelihood

A.1 Method of moments

Consider a measurable space $(\Omega, \mathcal{F})$ and i.i.d. random variables $(X_1, \dots, X_n)$ taking values in a measurable space $(\mathsf{X}, \mathcal{X})$. We assume that we have access to probabilities $(\mathbb{P}_\theta)_{\theta \in \Theta}$, where $\Theta \subset \mathbb{R}^d$. For all $\theta \in \Theta$, we write $\mathbb{E}_\theta$ the expectation under $\mathbb{P}_\theta$ and $\mathbb{V}_\theta$ the variance. The objective is to estimate the unknown parameter $\theta \in \Theta$. The method of moments consists in choosing d functions $T_j : \mathsf{X} \to \mathbb{R}$, $1 \leq j \leq d$, such that $\mathbb{E}_\theta[|T_j(X_1)|] < \infty$. Then, write for all $1 \leq j \leq d$, $\theta \in \Theta$,

$$e_j(\theta) = \mathbb{E}_\theta[T_j(X_1)].$$

As the quantities $e_j(\theta)$, $1 \leq j \leq d$, $\theta \in \Theta$, are usually unknown, they may be estimated by using empirical estimates. Assuming that for $1 \leq j \leq d$, $\mathbb{E}_\theta[|T_j(X_1)|^2] < \infty$, the Bienayme-Tchebychev inequality allows to quantify the empirical estimation error: for all $\varepsilon > 0$,

$$\mathbb{P}_\theta\left(\left|\frac{1}{n}\sum_{i=1}^n T_j(X_i) - e_j(\theta)\right| \geq \varepsilon\right) \leq \frac{\mathbb{V}_\theta[T_j(X_1)]}{n\varepsilon^2}.$$

In order to estimate the unknown parameter θ we may consider the system of equations:

$$\forall j \in \{1, \dots, d\}, \quad e_j(\theta) = \frac{1}{n}\sum_{i=1}^n T_j(X_i).$$

Assuming that this system has a unique solution $\widehat{\theta}_n$, $\widehat{\theta}_n$ is referred to as the moment estimator associated with $\{T_j\}_{1 \leq j \leq d}$.

Example A.1. Let $(X_1, \dots, X_n)$ be i.i.d. random variables with exponential distribution with parameter $\theta > 0$. Using $T_1 : x \mapsto x$ and $T_2 : x \mapsto x^2$ we have for all $\theta > 0$,

$$e_1(\theta) = \theta^{-1} \quad \text{and} \quad e_2(\theta) = 2\theta^{-2}.$$

The moment estimator associated with T_1 is

$$\widehat{\theta}_{n,1} = \frac{n}{\sum_{i=1}^n X_i}.$$

The moment estimator associated with T_2 is

$$\widehat{\theta}_{n,2} = \left(\frac{2n}{\sum_{i=1}^{n} X_i^2} \right)^{1/2}.$$

A.2 Z-estimation

The moment estimator associated with $\{T_j\}_{1 \leq j \leq d}$ is a solution to a system of equations of the form

$$\frac{1}{n} \sum_{i=1}^{n} \psi(\theta, X_i) = 0,$$

where for all $\theta \in \Theta$, $x \in \mathsf{X}$,

$$\psi(\theta, x) = \begin{pmatrix} T_1(x) - \mathbb{E}_\theta[T_1(X_1)] \\ \vdots \\ T_d(x) - \mathbb{E}_\theta[T_d(X_1)] \end{pmatrix}.$$

Consider now arbitrary functions ψ_j, $1 \leq j \leq d$, such that for all $\theta_* \in \Theta$, $1 \leq j \leq d$, $\mathbb{E}_{\theta_*}[|\psi_j(\theta_*, X_1)|] < \infty$. A Z-estimator associated with $\psi = (\psi_1, \ldots, \psi_d)^\top$ is any solution $\widehat{\theta}_n$ satisfying

$$\psi_n(\widehat{\theta}_n) = 0,$$

where for all $\theta \in \Theta$,

$$\psi_n(\theta) = \frac{1}{n} \sum_{i=1}^{n} \psi(\theta, X_i).$$

Example A.2. Let F be a distribution function on $\mathbb{R}$ such that for all $x \in \mathbb{R}$, $F(x) = 1 - F(-x)$. Let $(X_1, \ldots, X_n)$ be i.i.d with distribution function F_{θ_*} where for all $\theta \in \mathbb{R}$, $x \in \mathbb{R}$, $F_\theta(x) = F(x - \theta)$. In this setting,

$$\mathbb{E}_\theta[X_1] = \theta,$$

which suggests to choose $\psi(\theta, x) = x - \theta$. In this case, the Z-estimator associated with ψ is given by $\widehat{\theta}_n = n^{-1} \sum_{i=1}^{n} X_i$.

A.3 Maximum likelihood

Definition A.3. Let $(\mathsf{X}, \mathcal{X})$ be a measurable space equipped with a sigma-finite measure μ. Let $(f_\theta)_{\theta \in \Theta}$ be a family of probability densities with respect to μ and $(X_i)_{1 \leq i \leq n}$ be i.i.d. random variables with probability density f_{θ_*}, $\theta_* \in \Theta$. The likelihood of $(X_i)_{1 \leq i \leq n}$ is the function

$$L_n : \theta \mapsto \prod_{i=1}^{n} f_\theta(X_i).$$

A maximum likelihood estimator associated with L_n is any estimator solution to the following optimization problem

$$\widehat{\theta}_n \in \mathrm{Argmax}_{\theta \in \Theta} L_n(\theta).$$

Example A.4. Let $(X_i)_{1 \leq i \leq n}$ be i.i.d. Bernoulli random variables with parameter $\theta_* \in (0, 1)$. For all $\theta \in (0, 1)$,

$$L_n(\theta) = \prod_{i=1}^{n} \theta^{X_i} (1 - \theta)^{1 - X_i}$$

and

$$\ell_n(\theta) = \log L_n(\theta) = \left(\sum_{i=1}^n X_i\right)\log\theta + \left(\sum_{i=1}^n (1-X_i)\right)\log(1-\theta).$$

The function ℓ_n/n is stricly concave on $(0,1)$ with $\lim_{\theta\to 0}\ell_n(\theta)/n = -\infty$ and $\lim_{\theta\to 1}\ell_n(\theta)/n = -\infty$. This function has therefore a unique maximum given by

$$\widehat{\theta}_n = \frac{1}{n}\sum_{i=1}^n X_i.$$

A.4 M-estimation

Maximum likelihood estimators are defined as solutions to optimization problems. This is the case of many estimation procedures. Consider for instance a function $m : \Theta \times \mathsf{X} \to \mathbb{R}$, $(\theta, x) \mapsto m_\theta(x)$, such that for all $\theta, \theta_* \in \Theta$, $\mathbb{E}_{\theta_*}[|m_\theta(X_1)|] < \infty$ and consider also $M_n : \theta \mapsto n^{-1}\sum_{i=1}^n m_\theta(X_i)$. For all $\delta > 0$,

$$\mathbb{P}_{\theta_*}\left(|M_n(\theta) - M_{\theta_*}(\theta)| \geq \delta\right) \leq \frac{\mathbb{V}_{\theta_*}[m_\theta(X_1)]}{n\delta^2},$$

where

$$M_{\theta_*}(\theta) = \mathbb{E}_{\theta_*}[m_\theta(X_1)].$$

A M-estimator is any solution to the following optimization problem:

$$\widehat{\theta}_n \in \mathrm{Argmax}_{\theta\in\Theta} M_n(\theta).$$

Example A.5. For all $1 \leq k \leq n$, let $x_k \in \mathbb{R}^d$ and consider $(\xi_k)_{1\leq k\leq n}$ i.i.d. random variables with distribution $\mathcal{N}(0,1)$ and the linear regression model:

$$Y_k = \sum_{\ell=0}^p \beta_\ell \varphi_\ell(x_k) + \sigma\varepsilon_k,$$

where $\theta = (\sigma, \beta) \in \mathbb{R}_+^* \times \mathbb{R}^{p+1}$. The joint density of the observations is:

$$f_n : \theta \mapsto (2\pi\sigma^2)^{-n/2}\exp\left(-\frac{1}{2\sigma^2}\sum_{k=1}^n\left(Y_k - \sum_{\ell=0}^p \beta_\ell\varphi_\ell(x_k)\right)^2\right).$$

The maximum likelihood estimator of β coincides with the mean squared error estimator

$$\widehat{\beta}_n \in \mathrm{Argmin}_{\beta\in\mathbb{R}^{p+1}} \sum_{k=1}^n\left(Y_k - \sum_{\ell=0}^p \beta_\ell\varphi_\ell(x_k)\right)^2.$$

Consider the matrix Φ in $\mathbb{R}^{n\times(p+1)}$ such that for all $1 \leq i \leq n$, $1 \leq j \leq p+1$, $\Phi_{i,j} = \varphi_{j-1}(x_i)$. Then, $\widehat{\beta}_n$ is solution to

$$\Phi\Phi^\top\widehat{\beta}_n = \Phi Y,$$

where $Y = (Y_1, \ldots, Y_n)^\top$.

A.5 Consistency

When for all θ, θ_*, $\mathbb{E}_{\theta_*}[|m_\theta(X_1)|] < \infty$, by the law of large numbers, in $\mathbb{P}_{\theta_*}$-probability,

$$M_n(\theta) = \frac{1}{n}\sum_{i=1}^{n} m_\theta(X_i) \to_{n\to\infty} M_{\theta_*}(\theta) = \mathbb{E}_{\theta_*}[m_\theta(X_1)].$$

We also assume that θ_* is a maximum of M_{θ_*}.

Theorem A.6. *Consider the following assumptions.*

- *For all $\theta_* \in \Theta$, in $\mathbb{P}_{\theta_*}$-probability, $\sup_{\theta\in\Theta}|M_n(\theta) - M_{\theta_*}(\theta)| \to_{n\to\infty} 0$.*
- *For all $\theta_* \in \Theta$ and $\varepsilon > 0$,*

$$\sup_{\theta\in\Theta;|\theta-\theta_*|>\varepsilon} M_{\theta_*}(\theta) < M_{\theta_*}(\theta_*).$$

- *$(\widehat{\theta}_n)_{n\geq 0}$ is such that there exists $(\rho_n)_{n\geq 0}$ satisfying for all $\theta_* \in \Theta$, in $\mathbb{P}_{\theta_*}$-probability, $\rho_n \to_{n\to\infty} 0$ and*

$$\liminf_{n\to\infty} \mathbb{P}_{\theta_*}\left(M_n(\widehat{\theta}_n) \geq M_n(\theta_*) - \rho_n\right) = 1.$$

Then, for all $\theta_* \in \Theta$, in $\mathbb{P}_{\theta_*}$-probability, $\widehat{\theta}_n \to \theta_*$.

Proof. For all $\theta_* \in \Theta$, since θ_* is a maximum of M_{θ_*},

$$\begin{aligned} 0 \leq M_{\theta_*}(\theta_*) - M_{\theta_*}(\widehat{\theta}_n) &\leq M_{\theta_*}(\theta_*) - M_n(\theta_*) + M_n(\theta_*) - M_n(\widehat{\theta}_n) + M_n(\widehat{\theta}_n) - M_{\theta_*}(\widehat{\theta}_n) \\ &\leq 2\sup_{\theta\in\Theta}|M_n(\theta) - M_{\theta_*}(\theta)| + \rho_n \\ &\quad + \left\{M_n(\theta_*) - M_n(\widehat{\theta}_n) - \rho_n\right\} \mathbb{1}_{M_n(\theta_*)-\rho_n > M_n(\widehat{\theta}_n)}. \end{aligned}$$

Let $\varepsilon > 0$. There exists $\eta > 0$ such that $M_{\theta_*}(\theta) \leq M_{\theta_*}(\theta_*) - \eta$ for all $\theta \in \Theta$ such that $|\theta - \theta_*| \geq \varepsilon$. Therefore, $\{|\widehat{\theta}_n - \theta_*| \geq \varepsilon\} \subset \{M_{\theta_*}(\widehat{\theta}_n) \leq M_{\theta_*}(\theta_*) - \eta\}$. This yields

$$\mathbb{P}_{\theta_*}\left(|\widehat{\theta}_n - \theta_*| \geq \varepsilon\right) \leq \mathbb{P}_{\theta_*}\left(M_{\theta_*}(\widehat{\theta}_n) \leq M_{\theta_*}(\theta_*) - \eta\right) \leq \mathbb{P}_{\theta_*}\left(M_{\theta_*}(\theta_*) - M_{\theta_*}(\widehat{\theta}_n) > \eta\right),$$

which concludes the proof. □

Remark A.7. If Θ is compact in $\mathbb{R}^d$, M_{θ_*} is continuous, and for all $\theta \neq \theta_*$, $M_{\theta_*}(\theta) < M_{\theta_*}(\theta_*)$, the second assumption is satisfied.

Exponential models

Let $(X_1, \ldots, X_n)$ be i.i.d. random variables with density p_{θ_*} with respect to a reference measure μ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. The family $\{p_\theta\}_{\theta\in\Theta}$ is said to be in the exponential family if there exist $\eta : \theta \to \mathbb{R}^d$, $T : \mathsf{X} \to \mathbb{R}^d$, $h : \mathsf{X} \to \mathbb{R}_+$, $B : \Theta \to \mathbb{R}$ such that for all $x \in \mathsf{X}$,

$$p_\theta(x) = h(x)\exp\left(\langle\eta(\theta); T(x)\rangle - B(\theta)\right).$$

Example A.8.

- The density of a Poisson distribution with parameter $\theta > 0$ is given by

$$p_\theta : x \mapsto \frac{\theta^x}{x!}\mathrm{e}^{-\theta},$$

  so that $h(x) = (x!)^{-1}$, $T(x) = x$, $\eta(\theta) = \log\theta$, $B(\theta) = -\theta$.
- If $\theta = (\mu, \sigma^2) \in \mathbb{R} \times \mathbb{R}_+^*$ and p_θ is the Gaussian probability density with mean μ and variance σ^2, $h(x) = 1$, $T(x) = (x, x^2)^\top$, $\eta(\theta) = (\mu/\sigma^2, -1/(2\sigma^2))^\top$, $B(\theta) = \log(2\pi\sigma^2)/2 + \mu/(2\sigma^2)$.

The canonical exponential family is given, for all $x \in \mathsf{X}$, by

$$p_\eta(x) = h(x)\exp\left(\langle\eta; T(x)\rangle - A(\eta)\right),$$

where

$$A(\eta) = \log\left(\int h(x) \exp\left(\langle \eta; T(x) \rangle\right) \mu(\mathrm{d}x)\right).$$

Appendix B

Expectation Maximization algorithm

The Expectation Maximization (EM) algorithm [Dempster et al., 1977] is a general iterative method for maximum likelihood estimation in statistical models that involve latent (hidden) variables or missing data. When the likelihood function is difficult or impossible to optimize directly because part of the information is hidden, the EM algorithm provides a systematic way to alternate between estimating the missing information and optimizing the parameters. This algorithm is widely used in statistics, machine learning, and signal processing, especially for mixture models, clustering, density estimation, and probabilistic inference.

B.1 Introduction

Let $(\mathsf{X}, \mathcal{X})$ be a measurable space and μ be a measure on $(\mathsf{X}, \mathcal{X})$. Consider also a family $\{f_\theta\}_{\theta\in\Theta}$ of μ-integrable and positive functions. Define

$$L(\theta) = \int f_\theta(x)\mu(\mathrm{d}x).$$

We aim at solving

$$\widehat{\theta} \in \mathrm{Argmax}_{\theta\in\Theta} L(\theta).$$

When L is positive, the problem is often written:

$$\widehat{\theta} \in \mathrm{Argmax}_{\theta\in\Theta}\, \ell(\theta) = \log L(\theta).$$

B.2 Algorithm

In the following, we write $q_\theta : x \mapsto f_\theta(x)/L(\theta)$. Solving the optimization problem is not possible in general frameworks. The Expectation Maximization (EM) algorithm computes sequentially $\{\theta_k\}_{k\geq 0}$ to estimate $\widehat{\theta}$. For all $\theta, \theta' \in \Theta$, we introduce the following quantity:

$$Q(\theta, \theta') = \int \log f_\theta(x) q_{\theta'}(x)\mu(\mathrm{d}x) = \mathbb{E}_{\theta'}[\log f_\theta(X)],$$

where $\mathbb{E}_\theta$ is a notation for the expectation under the density q_θ. Then, we can write for all $\theta, \theta' \in \Theta$

$$Q(\theta, \theta') = \int \log(L(\theta) q_\theta(x)) q_{\theta'}(x) \mu(\mathrm{d}x) = \ell(\theta) - H(\theta, \theta'),$$

where $H(\theta, \theta') = -\int \log q_\theta(x) q_{\theta'}(x) \mu(\mathrm{d}x)$.

Lemma B.1. *For all $\theta, \theta' \in \Theta$,*

$$\ell(\theta) - \ell(\theta') \geq Q(\theta, \theta') - Q(\theta', \theta').$$

Proof. By definition, for all $\theta, \theta' \in \Theta$,

$$\begin{aligned} Q(\theta, \theta') - Q(\theta', \theta') &= \ell(\theta) - \ell(\theta') + H(\theta', \theta') - H(\theta, \theta') \\ &= \ell(\theta) - \ell(\theta') + \int \log\left(\frac{q_\theta(x)}{q_{\theta'}(x)}\right) q_{\theta'}(x) \mu(\mathrm{d}x). \end{aligned}$$

As log is concave, by Jensen's inequality,

$$\int \log\left(\frac{q_\theta(x)}{q_{\theta'}(x)}\right) q_{\theta'}(x) \mu(\mathrm{d}x) \leq \log \int \frac{q_\theta(x)}{q_{\theta'}(x)} q_{\theta'}(x) \mu(\mathrm{d}x) = 0,$$

which concludes the proof. The inequality $\int \log(q_\theta(x)) q_{\theta'}(x) \mu(\mathrm{d}x) \leq \int \log(q_{\theta'}(x)) q_{\theta'}(x) \mu(\mathrm{d}x)$ is known as Gibbs' inequality. □

By Lemma B.1, starting from a parameter estimate θ_k, $k \geq 0$, a direct solution to obtain a parameter θ such that $\ell(\theta) \geq \ell(\theta_k)$ is to choose θ such that $Q(\theta, \theta_k) \geq Q(\theta_k, \theta_k)$. This result motivates the Expectation Maximization (EM) algorithm given in Algorithm 7 and introduced in [Dempster et al., 1977].

Data: Initial parameter estimate θ_0.
Result: A sequence of parameter estimate $\{\theta_k\}_{k \geq 0}$.
for $k \geq 0$ **do**
  Compute the E-step: $\theta \mapsto Q(\theta, \theta_k)$.
  Compute the M-step: $\theta_{k+1} \in \mathrm{Argmax}_{\theta \in \Theta} Q(\theta, \theta_k)$.
end

Algorithm 7: A generic EM algorithm

B.3 Convergence properties

The first theoretical guarantees for the EM algorithm were provided in [Wu, 1983]. Given $k \geq 0$ and $\theta_k \in \Theta$, the EM update is

$$\theta_{k+1} \in M(\theta_k), \quad \text{where} \quad M(\theta') = \arg\max_{\theta \in \Theta} Q(\theta, \theta').$$

Therefore, M is a map from points of Θ to subsets of Θ and is referred to as a point-to-set map on Θ. This map is said to be closed at $\eta_* \in \Theta$ if for all sequence $\{\eta_k\}_{k \geq 0}$ such that η_k converges to η_* as $k \to \infty$, if for all $k \geq 0$, $y_k \in M(\eta_k)$ and $\{y_k\}_{k \geq 0}$ converges to $y_* \in \Theta$ as $k \to \infty$, then $y_* \in M(\eta_*)$. Convergence of the EM algorithm relies on a more general convergence theorem for point-to-set map.

Theorem B.2. *Assume that $\Theta \subset \mathbb{R}^m$ and let M be a point-to-set map defined on Θ. Consider a sequence $\{\theta_k\}_{k \geq 0}$ such that for all $k \geq 1$,*

$$\theta_{k+1} \in M(\theta_k).$$

and let $\mathcal{S} \subset \Theta$ denote the set of solution points. Assume that the sequence $\{\theta_k\}_{k\geq 0}$ is contained in a compact subset of Θ and that M is closed at all points $\theta \in \Theta \setminus \mathcal{S}$. Assume also that there exists $\alpha : \Theta \to \mathbb{R}$ a continuous function, such that for all $\theta \in \Theta$, if $\theta \in \Theta \setminus \mathcal{S}$ then for all $y \in M(\theta)$, $\alpha(y) > \alpha(\theta)$ and if $\theta \in \mathcal{S}$, then $\alpha(y) \geq \alpha(\theta)$ for all $y \in M(\theta)$. Then, every accumulation point of the sequence $\{\theta_k\}_{k\geq 0}$ belongs to the solution set $\mathcal{S}$.

Proof. The proof can be found in [Zangwill, 1969]. □

H1 The level set

$$\{\theta \in \Theta : \ell(\theta) \geq \ell(\theta_0)\}.$$

is compact.

By Lemma B.1, the sequence $\{\ell(\theta_k)\}_{k\geq 0}$ is non-decreasing. Therefore, by H1 this sequence converges to some limit point ℓ_*. However, there is no guarantee that ℓ_* is the global maximum of ℓ. Assumptions H1 ensures that $\{\theta_k\}_{k\geq 0}$ is contained in a compact subset of Θ.

H2 The functions $(\theta, \theta') \mapsto Q(\theta, \theta')$ and $\theta \mapsto \ell(\theta)$ are continuously differentiable, and differentiation under the integral sign is valid.

Convergence of the EM algorithm is a specific application of Theorem B.2 as stated in Theorem B.3

Theorem B.3. *Assume that H1 holds. Let $\{\theta_p\}_{p\geq 0}$ be a sequence generated by the EM algorithm:*

$$\theta_{p+1} \in M(\theta_p),$$

and suppose that there exists a set S such that:

(i) M is a closed point-to-set mapping over the complement of S;
(ii) for all $p \geq 0$, $L(\theta_{p+1}) > L(\theta_p)$ if $\theta_p \notin S$.

Then, all limit points of $\{\theta_p\}_{p\geq 0}$ are stationary points (local maxima) of L, and

$$\lim_{p\to\infty} L(\theta_p) = L(\theta^*) \quad \text{for some } \theta^* \in S.$$

By assumption H2, we can define the set of stationary points of the log-likelihood:

$$\mathcal{S} = \{\theta \in \Theta\,,\, \nabla\ell(\theta_*)\}\,.$$

Theorem B.4. *Assume that H1-2 hold. Then, all limit points of any EM sequence $\{\theta_p\}_{p\geq 0}$ are stationary points of L, and*

$$\lim_{p\to\infty} L(\theta_p) = L(\theta^*) \quad \text{for some stationary point } \theta^*.$$

Proof. The proof is a consequence of Theorem B.3. By H2, M is closed at all points $\theta \in \theta \in \Theta \setminus \mathcal{S}$, and for all limit point θ^* of $\{\theta_k\}_{k\geq 0}$, $\theta^* \in M(\theta^*)$. By Lemma B.1, and the definition of M, for all $\theta \in \Theta$, if $\theta \in \mathcal{S}$ then for all $y \in M(\theta)$, $\alpha(y) \geq \alpha(\theta)$. Let $p \geq 0$ and $\theta_p \in \Theta \setminus \mathcal{S}$. By definition of θ_p,

$$\nabla\ell(\theta_p) = \nabla_\theta Q(\theta, \theta_p)\big|_{\theta=\theta_p}.$$

Therefore, if $\theta_p \notin \mathcal{S}$, $\nabla\ell(\theta_p) \neq 0$ and θ_p is not a local maximum of $\theta \mapsto Q(\theta, \theta_p)$ which implies that $Q(\theta_{p+1}, \theta_p) > Q(\theta_p, \theta_p)$ and $L(\theta_{p+1}) > L(\theta_p)$. This concludes the frist part of the proof. Since θ^* maximizes $\theta \mapsto Q(\theta, \theta^*)$,

$$\nabla_\theta Q(\theta, \theta^*)\big|_{\theta=\theta^*} = 0.$$

Under the differentiability assumption,

$$\nabla_\theta Q(\theta, \theta^*)\big|_{\theta=\theta^*} = \nabla \ell(\theta^*),$$

which proves the result. □

B.4 Application to latent data models

Let (X, Y) in $(\mathsf{X} \times \mathsf{Y}, \mathcal{X} \otimes \mathcal{Y})$ be two random variables where Y is observed and X is not observed. The measurable space $(\mathsf{X} \times \mathsf{Y}, \mathcal{X} \otimes \mathcal{Y})$ is endowed with the product measure $\mu \otimes \lambda$. A very popular setting for the EM algorithm is when $f_\theta : x \mapsto p_\theta(x, Y)$ where for all $\theta \in \Theta$, p_θ is a joint probability density function with respect to $\mu \otimes \lambda$ used to model the joint distribution of (X, Y). Assuming that the random variable Y is observed and that X is not observed, we consider the likelihood function

$$L(\theta) = \int f_\theta(x, Y)\mu(\mathrm{d}x),$$

which is a random variable depending on Y. This is the marginal density of Y when the parameter is θ. In this setting, $f_\theta(X, Y)/L(\theta)$ is the probability density of the conditional distribution of X given Y.

Solving $\widehat{\theta} \in \mathrm{Argmax}_{\theta \in \Theta} L(\theta)$ amounts to solving the maximum likelihood estimation problem. However, in this setting, as in many other settings, the integral is intractable and the optimization problem cannot be solved directly. We have

$$Q(\theta, \theta') = \int \log p_\theta(x, Y) p_{\theta'}(x|Y)\mu(\mathrm{d}x) = \mathbb{E}_{\theta'}[\log p_\theta(X, Y)|Y].$$

Therefore, the E-step of the EM algorithm amounts to computing the conditional expectation given Y of the complete data (joint) loglikelihood.

If $\{(X_i, Y_i)\}_{1 \le i \le n}$ are i.i.d we write $Y = (Y_1, \ldots, Y_n)$ and $X = (X_1, \ldots, X_n)$. In this case, the intermediate quantity of the EM algorithm is

$$Q(\theta, \theta') = \sum_{i=1}^{n} \mathbb{E}_{\theta'}[\log p_\theta(X_i, Y_i)|Y_i].$$

Curved exponential families

We may further restrict the analysis to the case where the complete-data likelihood p_θ belongs to the class of curved exponential family densities. This setting covers a wide range of practical models. We now introduce a set of assumptions that are satisfied in many scenarios. Let $\phi : \Theta \to \mathbb{R}$, $\psi : \Theta \to \mathbb{R}^q$, $h : \mathsf{X} \times \mathsf{Y} \to \mathbb{R}_+^*$ and $S : \mathsf{X} \times \mathsf{Y} \to \mathbb{R}^q$. Assume then that the complete-data likelihood may be written:

$$p_\theta(x, y) = h(x, y) \exp\left\{\phi(\theta) + \langle S(x, y); \psi(\theta)\rangle\right\}.$$

Therefore, the intermediate quantity of the EM algorithm is given by

$$Q(\theta, \theta') = \int \log p_\theta(x, Y) p_{\theta'}(x|Y)\mu(\mathrm{d}x) = \mathbb{E}_{\theta'}[\log h(X, Y)|Y] + \phi(\theta) + \mathbb{E}_{\theta'}[\langle S(X, Y); \psi(\theta)\rangle|Y].$$

If $p \ge 0$ and θ_p is the current parameter estimate, θ_{p+1} may be computed by solving

$$\theta_{p+1} \in \arg\max_{\theta \in \Theta} \left\{\phi(\theta) + \left\langle \mathbb{E}_{\theta_p}[S(X, Y)|Y]; \psi(\theta)\right\rangle\right\}.$$

B.5 Example: mixture of Gaussian distributions

In this example, we assume that the joint distribution of (X, Y) belongs to a family of distributions parametrized by a vector θ with real components. For $k \in \{1, \dots, M\}$, write $\pi_k = \mathbb{P}_\theta(X = k)$. Assume that conditionally on the event $\{X = k\}$, Z has a Gaussian distribution with mean $\mu_k \in \mathbb{R}^d$ and covariance matrix $\Sigma \in \mathbb{R}^{d\times d}$. The probability density of this conditional distribution is written g_k^θ, where the parameter $\theta = (\{\pi_k\}_{1\le k\le K}, \{\mu_k\}_{1\le k\le K}, \Sigma)$ belongs to the set $\Theta = \mathbb{S}_K \times (\mathbb{R}^d)^K \times \mathbb{R}^{d\times d}$ with $\mathbb{S}_K = \{(\pi_1, \dots, \pi_K) \in [0,1]^K \; ; \; \sum_{k=1}^K \pi_k = 1\}$. For all $1 \le k \le K$, the explicit computation of $\mathbb{P}_\theta(X = k|Y)$ writes

$$\mathbb{P}_\theta\left(X = k|Y\right) = \frac{\pi_k g_k^\theta(Y)}{\sum_{\ell=1}^K \pi_\ell g_\ell^\theta(Y)}.$$

Assume that $\{(X_i, Y_i)\}_{1\le i\le n}$ are i.i.d. with this distribution parameterized by $\theta \in \Theta$. Then, the complete-data loglikelihood writes

$$\log p_\theta(X_{1:n}, Y_{1:n}) = \sum_{i=1}^n \sum_{k=1}^K \mathbb{1}_{X_i=k} \left\{ \log(\pi_k) - \frac{1}{2}\log\det(2\pi\Sigma) - \frac{1}{2}(Y_i - \mu_k)^\top \Sigma^{-1}(Y_i - \mu_k) \right\}.$$

Therefore, the intermediate quantity of the EM algorithm is given, for all $\theta, \theta' \in \Theta$, by

$$\begin{aligned} Q(\theta, \theta') &= \mathbb{E}_{\theta'}\left[\log p_\theta(X_{1:n}, Y_{1:n})|X_{1:n}\right], \\ &= \sum_{i=1}^n \sum_{k=1}^K \left\{ \mathbb{P}_{\theta'}(X_i = k|Y_i) \left(\log(\pi_k) - \frac{1}{2}\log\det(2\pi\Sigma) - \frac{1}{2}(Y_i - \mu_k)^\top \Sigma^{-1}(Y_i - \mu_k) \right) \right\}. \end{aligned}$$

The algorithm is initialized by choosing $\theta^{(0)}$ randomly. Then, for each iteration $p \ge 0$, the current parameter estimate is written

$$\theta^{(p)} = \left\{ \{\pi_k^{(p)}\}_{1\le k\le K}, \{\mu_k^{(p)}\}_{1\le k\le K}, \Sigma^{(p)} \right\},$$

and the update is decomposed into two steps.

1. Compute $\mathbb{P}_{\theta^{(p)}}(X_i = k|Y_i)$ for all $1 \le i \le n$, $1 \le k \le K$:

$$\mathbb{P}_{\theta^{(p)}}(X_i = k|Y_i) = \frac{\pi_k^{(p)} g_k^{\theta^{(p)}}(Y)}{\sum_{\ell=1}^K \pi_\ell^{(p)} g_\ell^{\theta^{(p)}}(Y)} = \omega_{i,k}^{(p)}.$$

2. Update the parameter estimate by computing:

$$\theta^{(p+1)} \in \mathrm{Argmax}_{\theta\in\Theta} Q(\theta, \theta^{(p)}).$$

The intermediate quantity is given, for all θ, by:

$$Q(\theta, \theta^{(p)}) = \sum_{i=1}^n \sum_{k=1}^K \left\{ \omega_{i,k}^{(p)} \left(\log(\pi_k) - \frac{1}{2}\log\det(2\pi\Sigma) - \frac{1}{2}(Y_i - \mu_k)^\top \Sigma^{-1}(Y_i - \mu_k) \right) \right\}.$$

By Lemma B.6, and using that $\pi_K = 1 - \sum_{k=1}^{K-1} \pi_k$, for all $1 \le k \le K-1$,

$$\frac{\partial Q(\theta, \theta^{(p)})}{\partial \pi_k} = \left(\sum_{i=1}^n \omega_{i,k}^{(p)} \right) \frac{1}{\pi_k} - \left(\sum_{i=1}^n \omega_{i,K}^{(p)} \right) \frac{1}{\pi_K}.$$

In addition, for all $1 \le k \le K$,

$$\frac{\partial Q(\theta,\theta^{(p)})}{\partial \mu_k} = \sum_{i=1}^{n} \omega_{i,k}^{(p)} \left(2\Sigma^{-1}Y_i - 2\Sigma^{-1}\mu_k\right),$$

and

$$\frac{\partial Q(\theta,\theta^{(p)})}{\partial \Sigma^{-1}} = \frac{n}{2}\Sigma - \frac{1}{2}\sum_{i=1}^{n}\sum_{k=1}^{K} \omega_{i,k}^{(p)} \left(Y_i-\mu_k\right)\left(Y_i-\mu_k\right)^\top.$$

The maximum likelihood estimator is defined as the only parameter $\widehat{\theta}^{(p+1)}$ such that all these equations are set to 0. We can note that there exists $c > 0$ such that for all $k \in \{1,\dots,K\}$, $\pi_k = c\sum_{i=1}^{n} \omega_{i,k}^{(p)}$. Computing the sum for $k = 1$ to $k = K$ yields $c = n$. For $k \in \{1,\dots,K\}$, it is given by

$$\begin{aligned}
\pi_k^{(p+1)} &= \frac{1}{n}\sum_{i=1}^{n} \omega_{i,k}^{(p)},\\
\mu_k^{(p+1)} &= \frac{1}{\sum_{i=1}^{n} \omega_{i,k}^{(p)}} \sum_{i=1}^{n} \omega_{i,k}^{(p)} Y_i,\\
\Sigma^{(p+1)} &= \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} \omega_{i,k}^{(p)} \left(Y_i-\mu_k^{(p+1)}\right)\left(Y_i-\mu_k^{(p+1)}\right)^\top.
\end{aligned}$$

Remark B.5. Note that the computation of $\mathbb{P}_{\theta^{(p)}}(X_i = k|Y_i)$ is explicit. The fact that the conditional expectation (and therefore $Q(\theta,\theta^{(p)})$) can be computed explicitly is a consequence of the fact that X_i is a discrete random variable. In other cases, $Q(\theta,\theta^{(p)})$ is likely to be unavailable explicitly and is often replaced by Monte Carlo estimators. In the setting of Gaussian mixtures, the computation of $\theta^{(p+1)}$ is also explicit. The M-step is often replaced by simply choosing an estimator $\theta^{(p+1)}$ such that $Q(\theta^{(p+1)},\theta^{(p)}) > Q(\theta^{(p)},\theta^{(p)})$ which is tractable in many cases and yields the Generalized EM algorithm.

Lemma B.6. *Let Σ be a symmetric and invertible matrix in $\mathbb{R}^{d\times d}$.*

(i) The derivative of the real valued function $\Sigma \mapsto \log\det(\Sigma)$ defined on $\mathbb{R}^{d\times d}$ is given by:

$$\partial_\Sigma\{\log\det(\Sigma)\} = \Sigma^{-1},$$

where, for all real valued function f defined on $\mathbb{R}^{d\times d}$, $\partial_\Sigma f(\Sigma)$ denotes the $\mathbb{R}^{d\times d}$ matrix such that for all $1 \leqslant i,j \leqslant d$, $\{\partial_\Sigma f(\Sigma)\}_{i,j}$ is the partial derivative of f with respect to $\Sigma_{i,j}$.

(ii) The derivative of the real valued function $x \mapsto x^\top\Sigma x$ defined on $\mathbb{R}^d$ is given by:

$$\partial_x\{x^\top\Sigma x\} = 2\Sigma x.$$

Proof. (i) Recall that for all $i \in \{1,\dots,d\}$ we have $\det(\Sigma) = \sum_{k=1}^{d} \Sigma_{i,k}\Delta_{i,k}$ where $\Delta_{i,j}$ is the (i,j)-cofactor associated with Σ. For any fixed i,j, the component $\Sigma_{i,j}$ does not appear in anywhere in the decomposition $\sum_{k=1}^{d} \Sigma_{i,k}\Delta_{i,k}$, except for the term $k = j$. This implies

$$\frac{\partial \log\det(\Sigma)}{\partial \Sigma_{i,j}} = \frac{1}{\det\Sigma}\frac{\partial\det(\Sigma)}{\partial\Sigma_{i,j}} = \frac{\Delta_{i,j}}{\det\Sigma}$$

Recalling the identity $\Sigma\,[\Delta_{j,i}]_{1\leq i,j\leq d} = (\det\Sigma)\,I_d$ so that $\Sigma^{-1} = \frac{[\Delta_{j,i}]_{1\leq i,j\leq d}^\top}{\det\Sigma}$, we finally get

$$\left[\frac{\partial \log\det(\Sigma)}{\partial\Sigma_{i,j}}\right]_{1\leq i,j\leq d} = (\Sigma^{-1})^\top = \Sigma^{-1}$$

where the last equality follows from the fact that Σ is symmetric.

(ii) Define $\varphi(x) = x^\top \Sigma x$. Then, by straightforward algebra, $\varphi(x+h) = \varphi(x) + 2h^\top \Sigma x + \varphi(h) = \varphi(x) + 2h^\top \Sigma x + o(\|h\|)$, which concludes the proof.

□

B.6 Monte Carlo EM in the curved exponential family setting

Assume that the complete-data log-likelihood belongs to a curved exponential family, i.e.,

$$\log p_\theta(X,Y) = \log h(x,y) + \phi(\theta) + \langle S(X,Y), \psi(\theta)\rangle,$$

where $S(X,Y) \in \mathbb{R}_+^*$, $S(X,Y) \in \mathbb{R}^q$, $\phi : \Theta \to \mathbb{R}$, $\psi : \Theta \to \mathbb{R}^q$. In this case,

$$Q(\theta, \theta^{(k)}) = \phi(\theta) + \mathbb{E}_{\theta^{(k)}}[\log h(X,Y) \mid Y] + \langle \mathbb{E}_{\theta^{(k)}}[S(X,Y) \mid Y],\ \psi(\theta)\rangle\,.$$

Hence, the E-step reduces to computing the conditional expectation of the sufficient statistics, and the M-step consists in solving

$$\theta^{(k+1)} = \arg\max_{\theta\in\Theta} \left\{ \phi(\theta) + \left\langle \bar{S}(\theta^{(k)}),\ \psi(\theta)\right\rangle \right\}, \qquad \bar{S}(\theta^{(k)}) := \mathbb{E}_{\theta^{(k)}}[S(X,Y) \mid Y].$$

When $\bar{S}(\theta^{(k)})$ is not available in closed form, the Monte Carlo EM (MCEM) algorithm estimates it with a simulation-based approximation. At iteration k, let

$$X_{k,1}, \ldots, X_{k,M_k} \sim p_{\theta^{(k)}}(X|Y)$$

be (approximately) independent draws from the conditional distribution of X given Y under $\theta^{(k)}$. Define the Monte Carlo estimate

$$\widehat{S}^{(k)}(\theta^{(k)}) := \frac{1}{M_k} \sum_{m=1}^{M_k} S(X_{k,m}, Y)\,.$$

The corresponding update is obtained by the approximate M-step

$$\theta^{(k+1)} \in \arg\max_{\theta\in\Theta} \left\{ \phi(\theta) + \left\langle \widehat{S}^{(k)}(\theta^{(k)}),\ \psi(\theta)\right\rangle \right\}.$$

Under standard regularity conditions, $\widehat{S}^{(k)}$ converges almost surely to $\mathbb{E}_{\theta^{(k)}}[S(X,Y) \mid Y]$ as $M_k \to \infty$, and convergence to a stationary point of the observed-data log-likelihood can be ensured when the number of Monte Carlo samples M_k increases sufficiently with k. Convergence rates of the MCEM algorithm can be obtained when nonasymptotic controls of the Monte Carlo estimation $\widehat{S}^{(k)}$ can be derived, see for instance [Fort and Moulines, 2003].

In many models, direct sampling from $p_{\theta^{(k)}}(X \mid Y)$ is not feasible but $p_{\theta^{(k)}}(X \mid Y)$ is known up to a normalizing constant. A common alternative is to approximate the conditional expectation in the E-step using a Markov chain Monte Carlo (MCMC) method targeting $p_{\theta^{(k)}}(X \mid Y)$. Specifically, at iteration k, let $\{X_{k,t}\}_{p\geq 1}$ be a Markov chain with invariant distribution $p_{\theta^{(k)}}(X \mid Y)$ (e.g., a Metropolis–Hastings or Gibbs sampler). After a burn-in period B_k, an ergodic average yields

$$\widehat{S}^{(k)}_{\mathrm{MCMC}}(\theta^{(k)}) := \frac{1}{M_k} \sum_{p=B_k+1}^{B_k+M_k} S(X_{k,p}, Y)\,,$$

which replaces $\bar{S}(\theta^{(k)})$ in the E-step. The resulting MCMC-EM update is

$$\theta^{(k+1)} \in \arg\max_{\theta\in\Theta} \left\{ \phi(\theta) + \left\langle \widehat{S}^{(k)}_{\mathrm{MCMC}}(\theta^{(k)}),\ \psi(\theta)\right\rangle \right\}.$$

Compared to MCEM with independent samples, the estimator $\widehat{S}^{(k)}_{\mathrm{MCMC}}$ is typically correlated due to the Markov dependence, but consistency follows from the ergodic theorem under standard assumptions (irreducibility, aperiodicity, and geometric ergodicity). In practice, the accuracy of the E-step approximation can be controlled by the burn-in length B_k, the chain length M_k.

Example B.7. Assume that we observe independent data $Y = (Y_1, \dots, Y_n)$ where for all $1 \leq i \leq n$,

$$Y_i \sim \mathrm{Poisson}(\lambda_i)$$

and

$$\lambda_i = \exp(\beta + X_i),$$

with independent latent variables $X = (X_1, \dots, X_n)$ such that for all $1 \leq i \leq n$,

$$X_i \sim \mathcal{N}(0, \sigma^2).$$

The unknown parameter is $\theta = (\beta, \sigma^2)$. The complete-data likelihood is given by

$$p_\theta(X, Y) = \prod_{i=1}^{n} p_\theta(Y_i|X_i)\, p_\theta(X_i).$$

The conditional distribution of the observations is

$$p_\theta(Y_i|X_i) = \frac{\exp(-\exp(\beta + X_i))\exp(Y_i(\beta + X_i))}{Y_i!}$$

and the prior distribution of the latent variable is

$$p_\theta(X_i) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{X_i^2}{2\sigma^2}\right).$$

Therefore, the complete-data loglikelihood is

$$\log p_\theta(X, Y) = \sum_{i=1}^{n} \Big[Y_i(\beta + X_i) - \exp(\beta + X_i)\Big] - \sum_{i=1}^{n} \frac{X_i^2}{2\sigma^2} - \frac{n}{2}\log\sigma^2 + C,$$

where C is a constant that does not depend on θ. Then, at iteration $k \geq 0$ of the algorithm, the intermediate quantity of the EM algorithm is

$$Q(\theta, \theta^{(k)}) = \sum_{i=1}^{n} \mathbb{E}_{\theta^{(k)}}\left[Y_i(\beta + X_i) - \exp(\beta + X_i) - \frac{X_i^2}{2\sigma^2}\middle| Y_i\right] - \frac{n}{2}\log\sigma^2 + C.$$

However the conditional distribution $p_{\theta^{(k)}}(X_i|Y_i)$ does not have a closed form. However, we can write

$$p_{\theta^{(k)}}(X_i|Y_i) \propto p_{\theta^{(k)}}(Y_i|X_i) p_{\theta^{(k)}}(X_i).$$

Therefore, the intermediate quantity cannot be computed analytically. Since direct sampling from $p_{\theta^{(k)}}(X_i|Y_i)$ is not possible but this distribution is known up to a multiplicative constant, we may use any MCMC method (Random Walk Metropolis Hastings, MALA, HMC) to approximate the intermediate quantity. Let $\{X_p^{(k)}\}_{p\geq 0}$ be the Markov chain generated by the chosen MCMC algorithm with target distribution $p_{\theta^{(k)}}(X|Y)$. The parameters are updated by maximizing the Monte Carlo approximation

$$\theta^{(k+1)} = \arg\max_\theta \frac{1}{M_k} \sum_{p=B_k+1}^{B_k+M_k} \log p_\theta(X_p^{(k)}, Y).$$

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